Assessing Mediation in Multiple Regression

PSYC 7015
Whitton
(Guest Lecture by Sierra Corbin)
3/22/2022

Class Outline

1. Introduction to Mediation and the Mediation Model
2. Estimating Mediation: The Baron and Kenny Method
3. Testing Significance of Mediation/Indirect Effects
   a. Sobel Test
   b. Process (SPSS) vs. SEM Mediation (JASP)
4. Example
5. In Class Practice Problem
6. Final Notes

1. Introduction to Mediators and Mediation Models

- **Definition of Mediator:** a variable that accounts for the association between the IV and DV
  - Also called:
    - the **indirect effect**
    - an *intervening* or *process* variable
- **Mediation Models:**
  - Causal models in which the mediator is the mechanism through which X affects Y
    - Note: To differentiate, I think of a **moderator** as a steering wheel/gas pedal in a car (can explain the direction/strength of the effect of X on Y using the moderator) while the **mediator** is the road itself (the path between X and Y goes through the mediator)

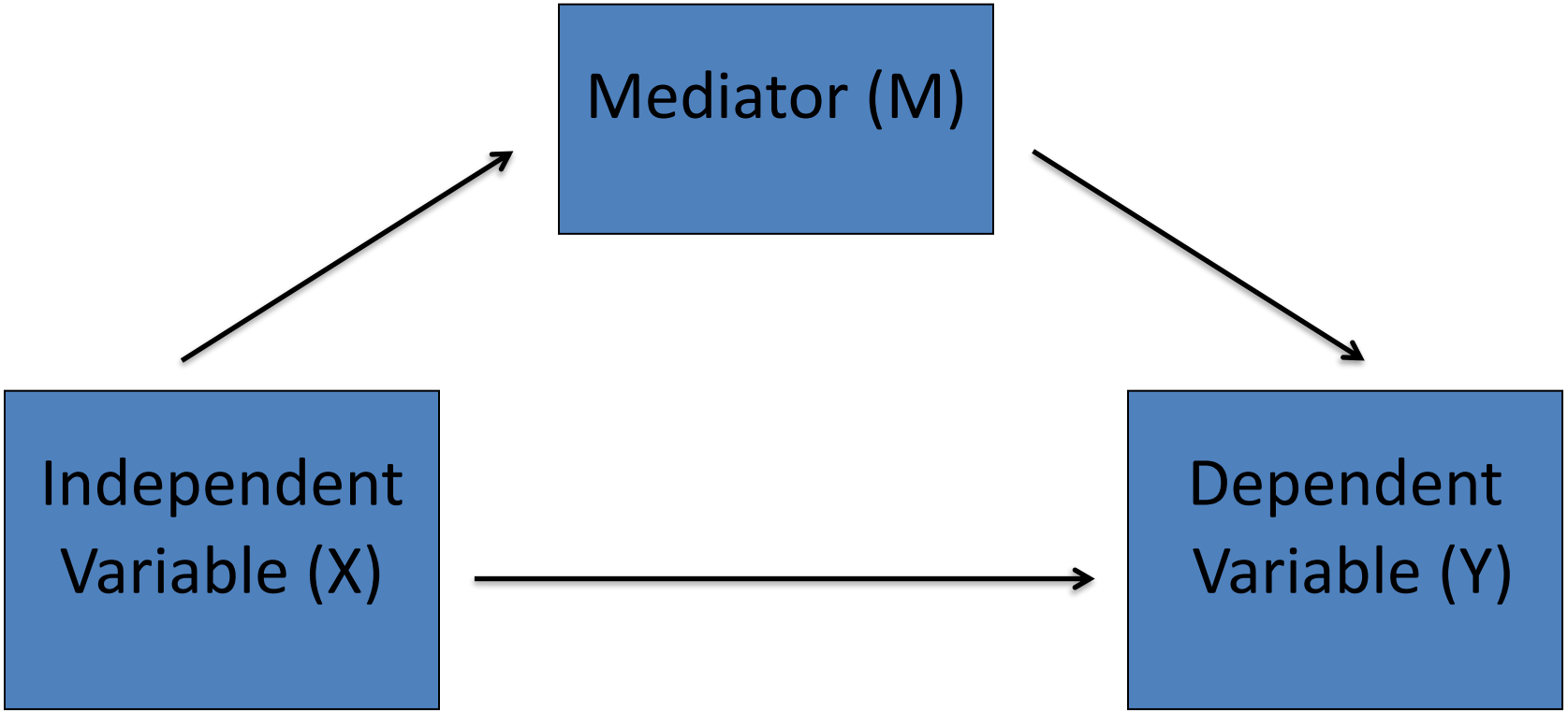
Examples of Mediation Models

- Through what mechanisms does TV viewing raise risk for obesity in children?
  - Sedentary lifestyle/lack of physical activity
  - Unhealthy snacking during TV viewing
- Through what mechanisms does couple relationship distress affect partner's mental health?
  - Reduced social support, hopelessness

The Unmediated Model

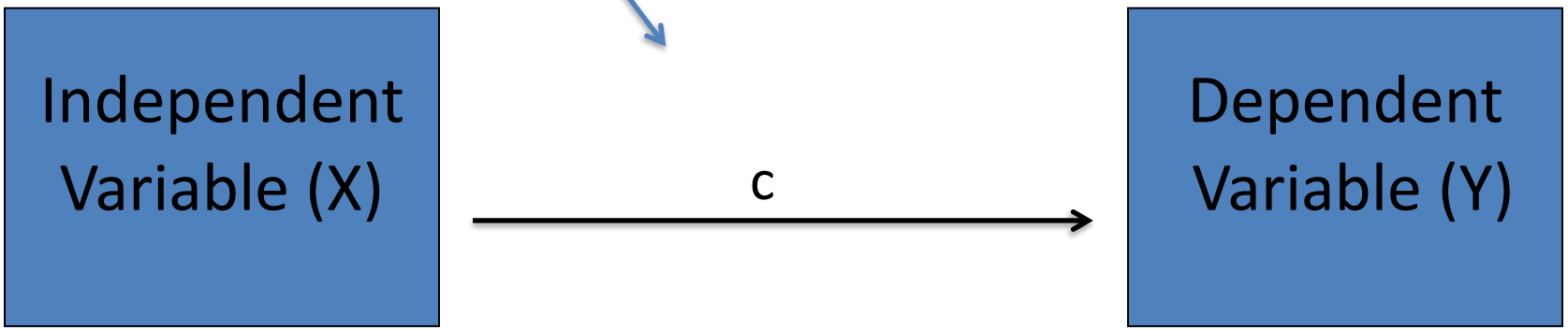


The Mediated Model

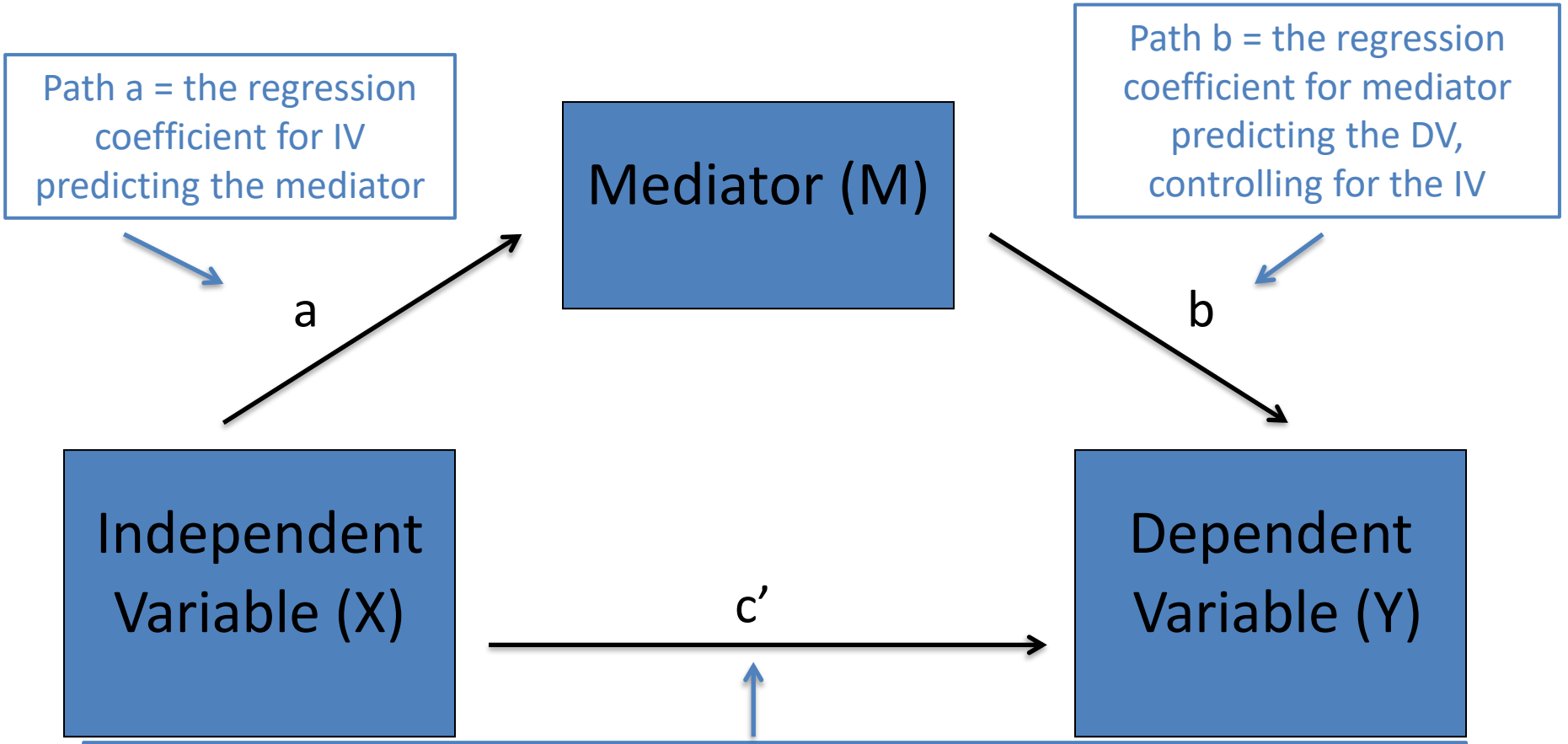


The Unmediated Model

Path c reflects the association between the IV and DV (or predictor and criterion) before controlling for the potential mediator. Called the ***total effect***



The Mediated Model



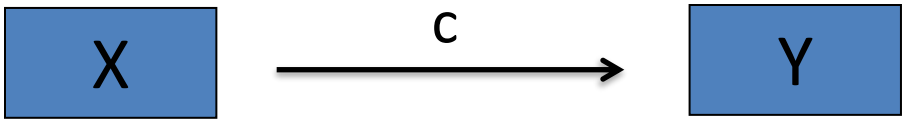
Path c' represents the ***direct effect***. Reflects the association between the IV and DV (or predictor and criterion) when controlling for the potential mediator. If it is *n.s.*, then there may be full mediation. If c' < c but still significant, then there may be partial mediation

Estimating Mediation using MR:
The Baron and Kenny (1986) Method

There are 4 steps to this method.

1) Test whether the independent variable is correlated with the dependent variable.

   -- estimate and test *path c (total effect)*



To do so, regress Y on X in a bivariate regression.

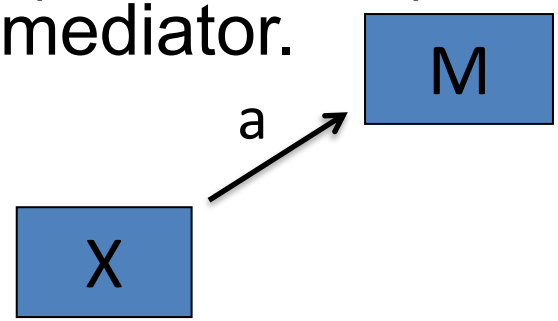
- If the coefficient for X is significant, it shows that there is an effect (or association) between the variables that may be mediated by M.

**Note- Can include controls, but if do so must keep them consistent in all models.

Estimating Mediation using MR:
The Baron and Kenny (1986) Method

2) Test whether the independent (or predictor) variable is correlated with the mediator.

-- estimate and test *path a*

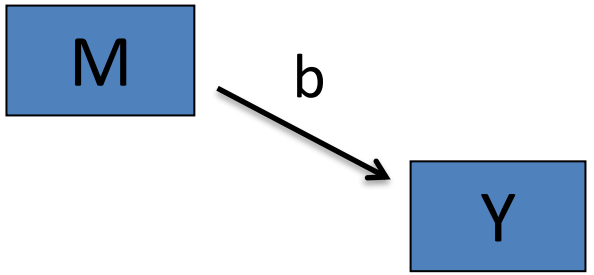


To do so, regress M on X in a bivariate regression.

- If the coefficient for X is significant, it tells you that X does have an association with M. This indicates that the effect of X on Y might occur indirectly through M.

Estimating Mediation using MR: The Baron and Kenny (1986) Method

3) Test whether the mediator affects the outcome variable.
-- estimate and test *path b*



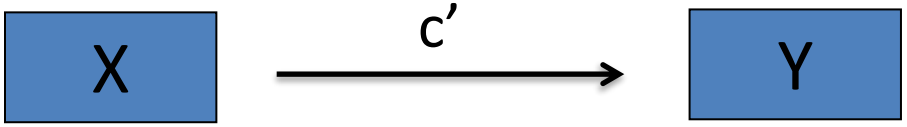
To do so, <u>regress Y on both X and M simultaneously</u>

- You must control for X: an observed zero-order correlation between M and Y could occur simply because they are both related to (predicted by) X.
- If the coefficient for M in this regression model is significant, it shows that M is associated with Y, controlling for X.

Estimating Paths using Multiple Regression: The Baron and Kenny (1986) Method

4) Test whether the effect of X on Y controlling for M is equal to zero (is it significant or non-significant).

-- i.e., estimate and test *path c' (direct effect)*



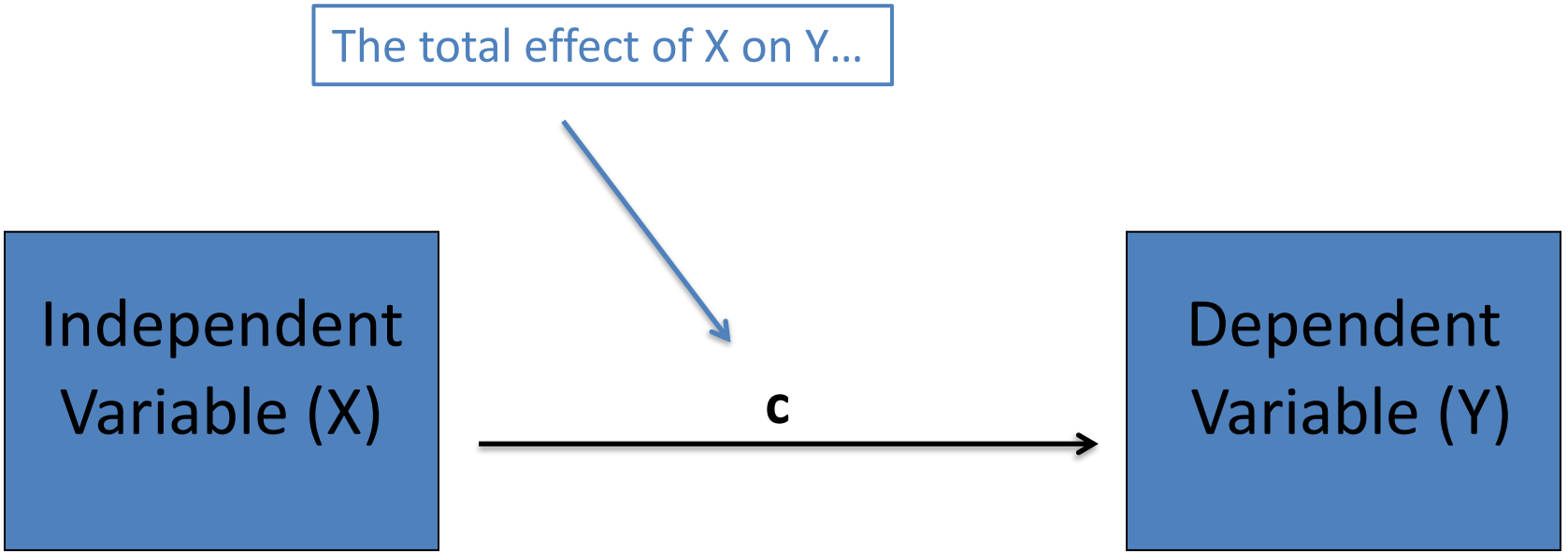
To do so, use the same equation as Step #3.

- If the coefficient associated with X = 0, this indicates full/complete mediation.
  - a non-significant beta for X means that we can't reject H_0 that beta = 0 in presence of M.
- If the coefficient is reduced but is still significant, this indicates that there may be *partial mediation*.

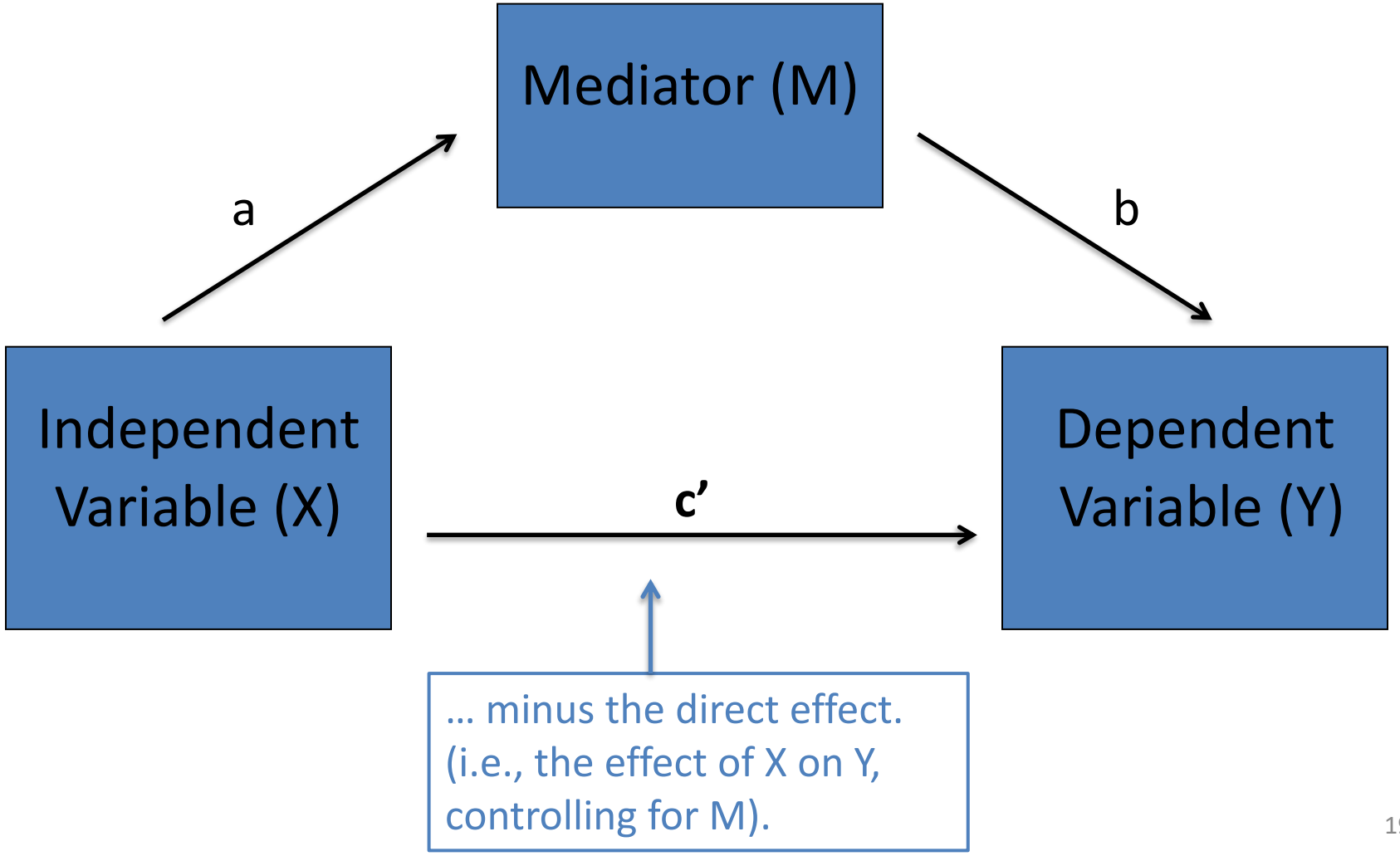
Estimating Mediation/ The Indirect Effect
Baron and Kenny (1986) Method:

- Calculate the *indirect effect:* this is the amount of mediation that occurs in the model.
- Defined as the reduction of the effect of X on Y when M is included.
  - Represented by c - c'.
    - Total effect (c) – the direct effect (c') = indirect effect
  - *In the figure…*

The Indirect Effect



The Indirect Effect



Beware of Missing Data

- If you are missing data from participants on M that is not missing on X or Y, then your model to test the total effect is run on a different sample than your mediated model!

- So, differences in C and C' may be due to differences in the samples rather than due to the mediator.

3. Testing Significance of Mediation/Indirect Effects

- In the past, it was considered acceptable to state that mediation occurred if:
  - C was significant
  - C' was not significant

- But, this is problematic because:
  - Does not test if the indirect effect is different from zero
  - Lacks power
  - The amount of reduction in the effect of X on Y is not equivalent to R^2 change, change in *F,* or its *p* value.

Testing the significance of the mediating (indirect) effect

- Assumes that c – c’ is theoretically equal to the product of the effect of X on M times the effect of M on Y

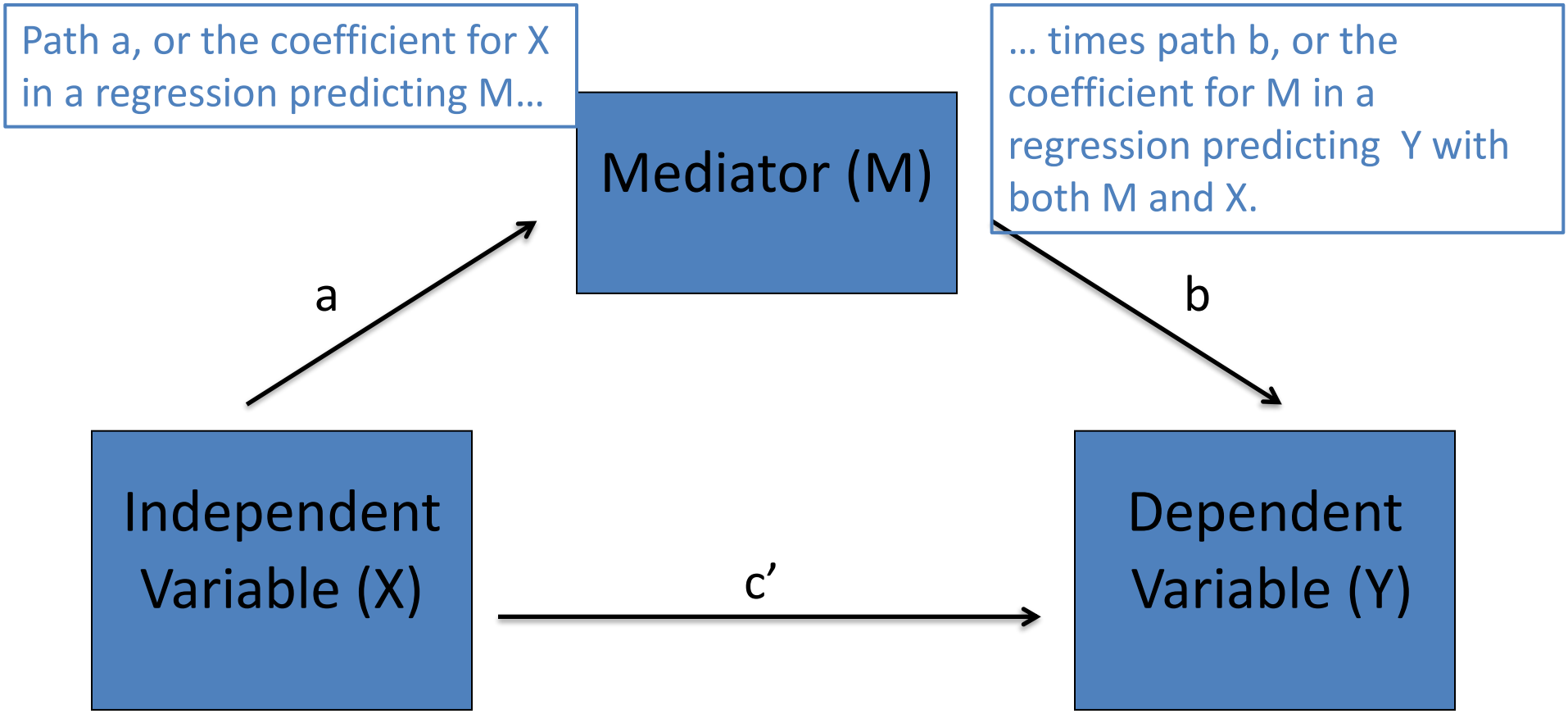
- Path a multiplied by path b, or a*b
  - The two are exactly equal when:

    a) multiple regression is used,

    b) there are no missing data,

    c) and the same covariates are in all of the equations used in steps 1-4.

The Indirect Effect



Interpreting a*b

- Recall that a regression coefficient tells you how many units the DV increases when the IV increases by one.
- So:
  - if the unstandardized beta for X predicting M is .5, then if X increases by 1, M increases by .5
  - Then, if the unstandardized beta for M predicting Y is .8, that increase in M of .5 means that Y increases by .40
  - .5*.8 = .40

Testing the significance of the mediating (indirect) effect

- Typically done by evaluating whether a*b is significantly different from zero.
  - Rather than focusing on c- c’
- This requires dividing a*b by an estimate of the standard error of a*b
  - There are different approaches to estimating this standard error

3a. The Sobel Test

- The most commonly used procedure is the Sobel test:

$$Z = \frac{a * b}{\sqrt{b^2 s_a^2 + a^2 s_b^2}}$$

Note: the denominator of the formula in D&H chapter is incorrect!

- numerator = the product of the effect of X on M times the effect of M on Y
- The denominator is 1 estimate of the standard error of a*b .
  - a, b, s_a and s_b are included in standard regression output
- The equation results in a ratio that can be interpreted as a Z test (i.e., Z scores > 1.96 are significant at alpha = .05).

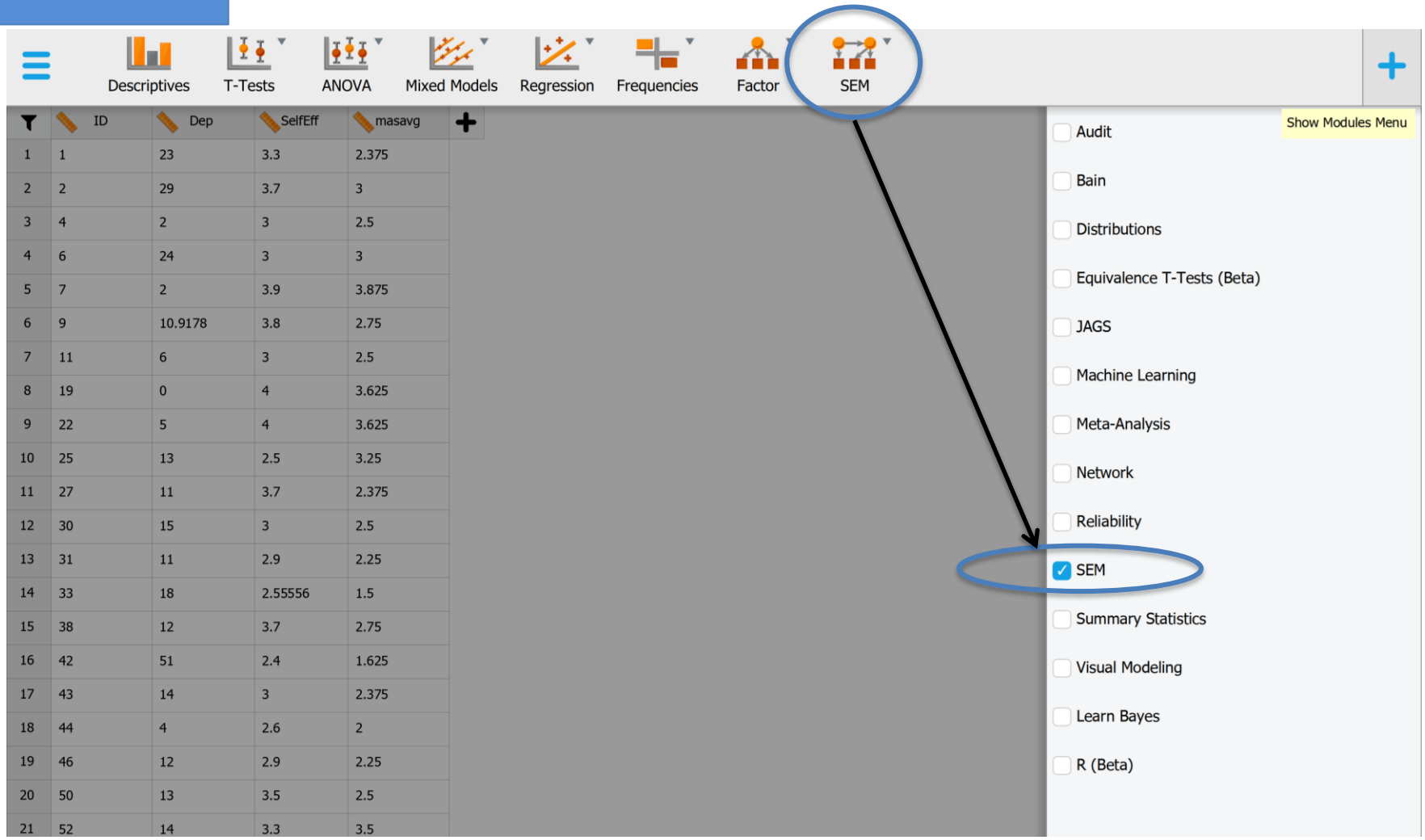
- Sobel calculator: http://quantpsy.org/sobel/sobel.htm

3b. Bootstrapping

- In small samples (n<400 or <200), Sobel lacks power, because the sampling distribution of a*b is not normally distributed.
- Bootstrapping is the preferred method:
  - Draw 100+ artificial samples of size n by sampling with replacement from the original n observations.
  - Calculate a*b in each bootstrap sample, save these values
  - Estimate a confidence interval from percentiles of the bootstrapped distribution of a*b coefficients
  - Then, reject H0 of no mediation if the CI does not contain zero

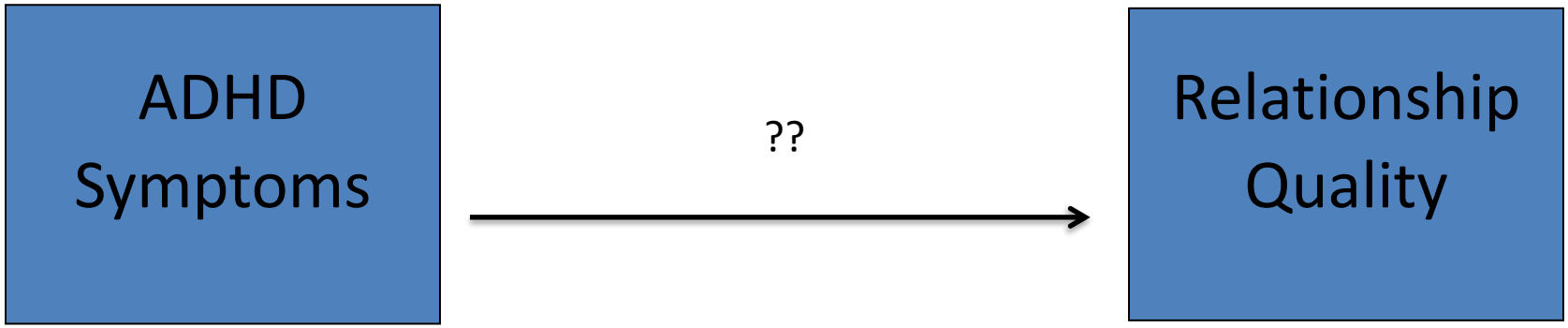
- Options for Mediation with Bootstrapping:
  - **SPSS:** PROCESS Macro (http://www.afhayes.com/introduction-to-mediation-moderation-and-conditional-process-analysis.html)
  - **JASP:** SEM Menu

In JASP:



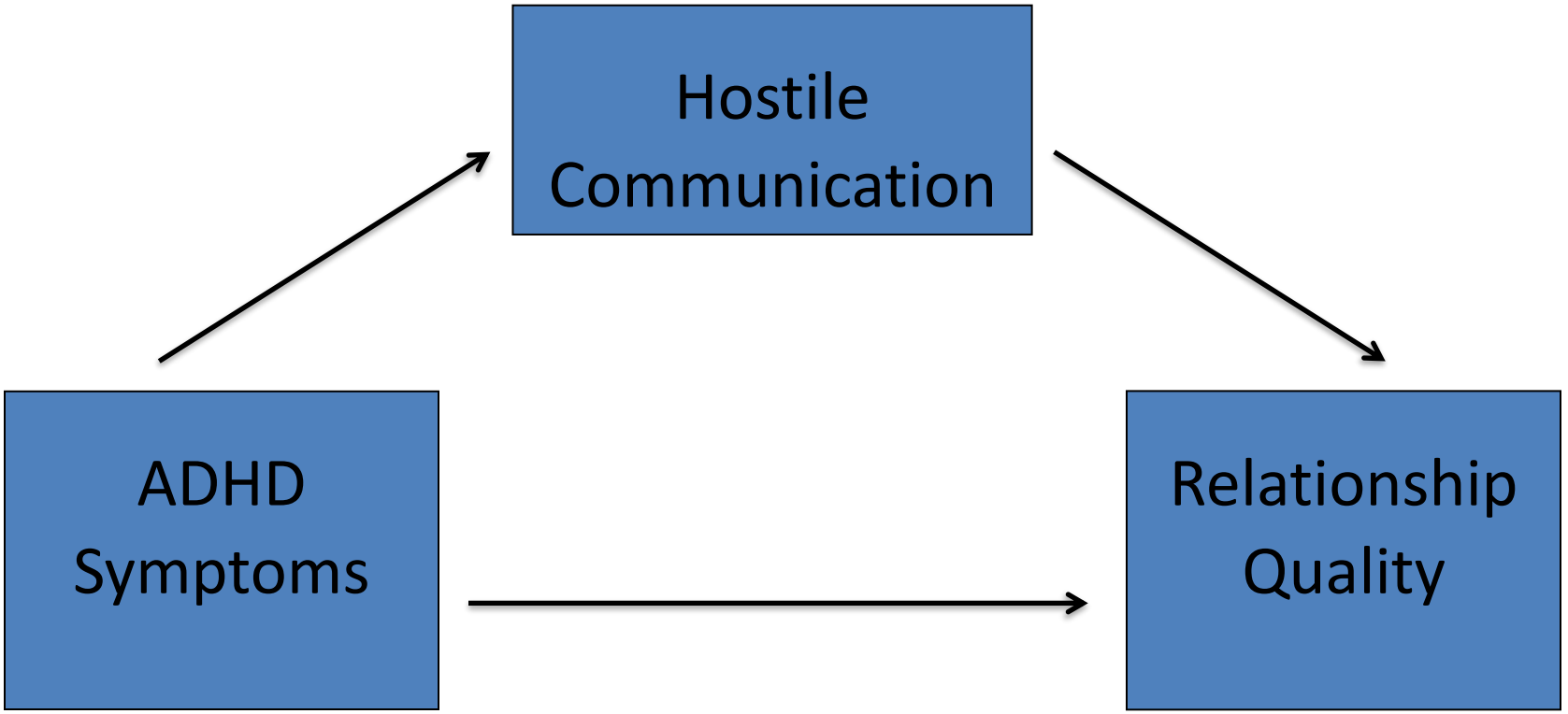
	ID	Dep	SelfEff	masavg
1	1	23	3.3	2.375
2	2	29	3.7	3
3	4	2	3	2.5
4	6	24	3	3
5	7	2	3.9	3.875
6	9	10.9178	3.8	2.75
7	11	6	3	2.5
8	19	0	4	3.625
9	22	5	4	3.625
10	25	13	2.5	3.25
11	27	11	3.7	2.375
12	30	15	3	2.5
13	31	11	2.9	2.25
14	33	18	2.55556	1.5
15	38	12	3.7	2.75
16	42	51	2.4	1.625
17	43	14	3	2.375
18	44	4	2.6	2
19	46	12	2.9	2.25
20	50	13	3.5	2.5
21	52	14	3.3	3.5

4. Example



Are ADHD symptoms associated with poorer relationship quality?

Mediation Model: Hypothesis



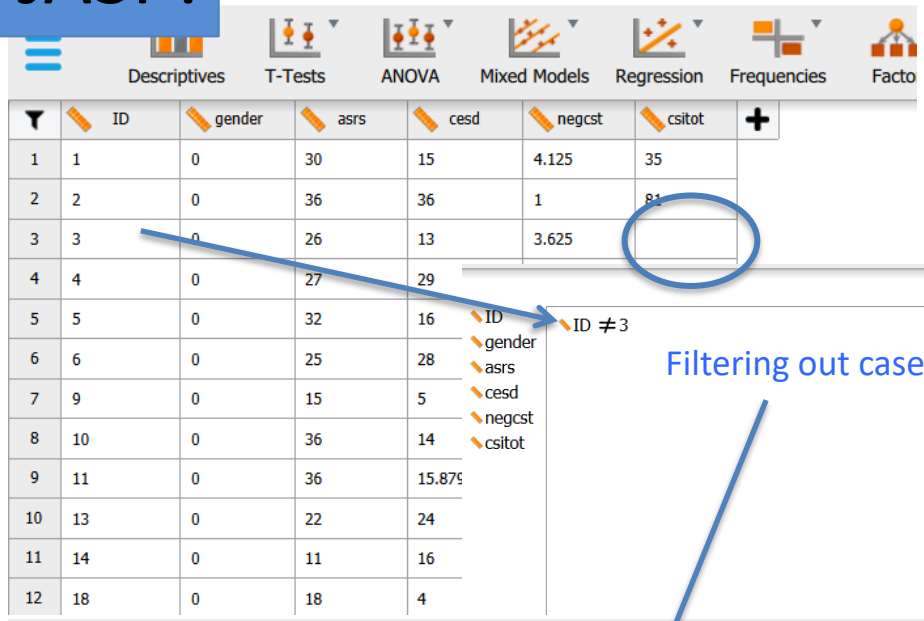
Through what mechanisms does ADHD influence relationship quality?

Method

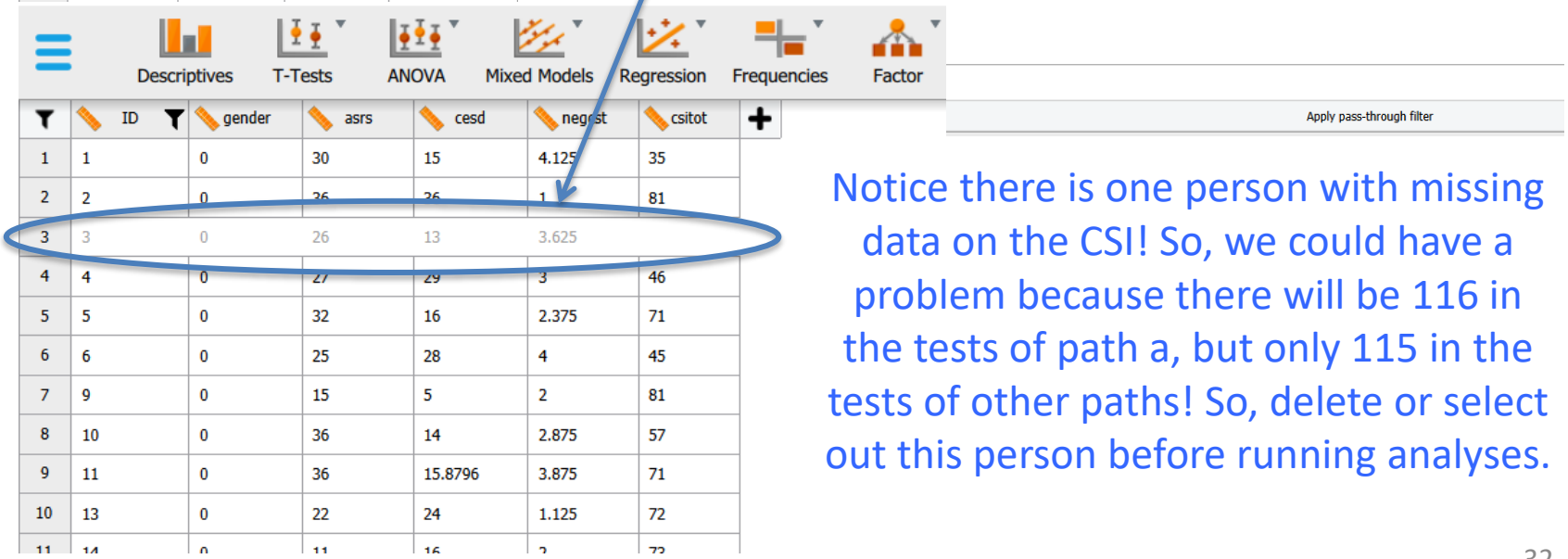
- Sample: 116 undergraduate women who received participation credit

- Completed self-report measures of ADHD symptoms (ASRS), relationship satisfaction (CSItot), and communication skills (negCST) via Survey Monkey
  - All three measures were Likert-type scales
  - Higher scores reflected more symptoms, satisfaction, and hostile communication.

In JASP:



Filtering out case #3 (the one with missing data)



Notice there is one person with missing data on the CSI! So, we could have a problem because there will be 116 in the tests of path a, but only 115 in the tests of other paths! So, delete or select out this person before running analyses.

In SPSS:

Correlations

		ADHD symptoms	Hostile Communication	Couple Satisfaction
ADHD symptoms	Pearson Correlation	1	.330**	-.232*
	Sig. (2-tailed)		.000	.013
	N	116	116	115
Hostile Communication	Pearson Correlation	.330**	1	-.694**
	Sig. (2-tailed)	.000		.000
	N	116	116	115
Couple Satisfaction	Pearson Correlation	-.232*	-.694**	1
	Sig. (2-tailed)	.013	.000	
	N	115	115	115

**. Correlation is significant at the 0.01 level (2-tailed).

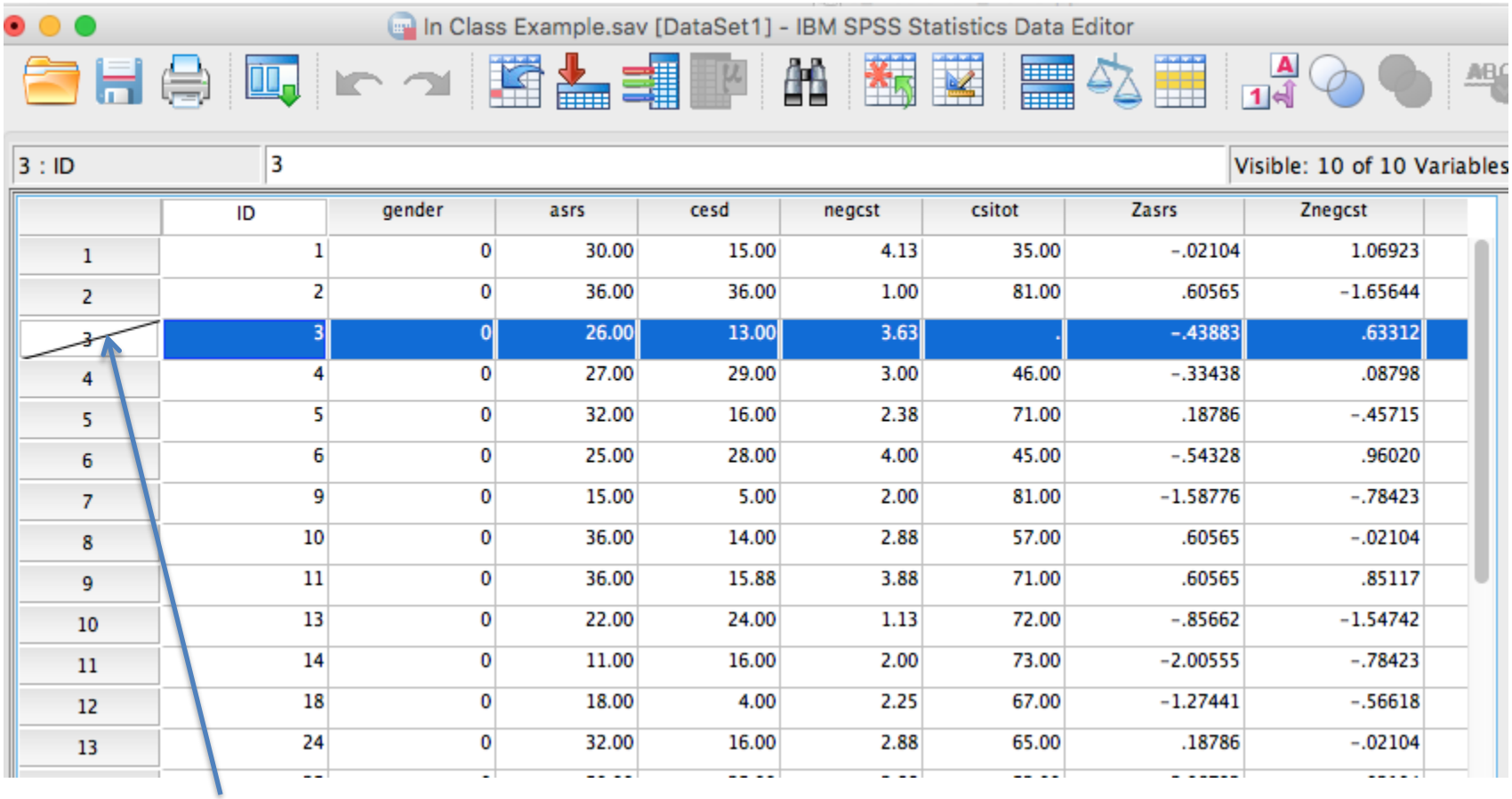
*. Correlation is significant at the 0.05 level (2-tailed).

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
ADHD symptoms	116	3.00	58.00	30.2014	9.57416
Hostile Communication	116	1.00	5.63	2.8991	1.14651
Couple Satisfaction	115	13.00	81.00	61.6946	16.13103
Valid N (listwise)	115				

Notice there is one person with missing data on the CSI! So, we could have a problem because there will be 116 in the tests of path a, but only 115 in the tests of other paths! So, delete or select out this person before running analyses.

In SPSS:



	ID	gender	asrs	cesd	negcst	csitot	Zasrs	Znegcst
1	1	0	30.00	15.00	4.13	35.00	-.02104	1.06923
2	2	0	36.00	36.00	1.00	81.00	.60565	-1.65644
3	3	0	26.00	13.00	3.63	.	-.43883	.63312
4	4	0	27.00	29.00	3.00	46.00	-.33438	.08798
5	5	0	32.00	16.00	2.38	71.00	.18786	-.45715
6	6	0	25.00	28.00	4.00	45.00	-.54328	.96020
7	9	0	15.00	5.00	2.00	81.00	-1.58776	-.78423
8	10	0	36.00	14.00	2.88	57.00	.60565	-.02104
9	11	0	36.00	15.88	3.88	71.00	.60565	.85117
10	13	0	22.00	24.00	1.13	72.00	-.85662	-1.54742
11	14	0	11.00	16.00	2.00	73.00	-2.00555	-.78423
12	18	0	18.00	4.00	2.25	67.00	-1.27441	-.56618
13	24	0	32.00	16.00	2.88	65.00	.18786	-.02104

Filtering out case #3 (the one with missing data)

In R:

```
rcorr(as.matrix(dat))
```

```
##           ID gender  asrs  cesd negcst csitot
## ID      1.00    NaN  0.02 -0.17   0.04  -0.01
## gender   NaN      1   NaN   NaN    NaN    NaN
## asrs    0.02    NaN  1.00  0.45   0.33  -0.23
## cesd   -0.17    NaN  0.45  1.00   0.24  -0.25
## negcst  0.04    NaN  0.33  0.24   1.00  -0.69
## csitot -0.01    NaN -0.23 -0.25  -0.69   1.00
##
## n
##          ID gender asrs cesd negcst csitot
## ID      116    116  116  116    116    115
## gender  116    116  116  116    116    115
## asrs    116    116  116  116    116    115
## cesd    116    116  116  116    116    115
## negcst  116    116  116  116    116    115
## csitot  115    115  115  115    115    115
##
## P
##         ID     gender asrs   cesd   negcst csitot
## ID                    0.8559 0.0682 0.6878 0.8794
## gender
## asrs    0.8559               0.0000 0.0003 0.0128
## cesd    0.0682        0.0000        0.0085 0.0072
## negcst  0.6878        0.0003 0.0085        0.0000
## csitot  0.8794        0.0128 0.0072 0.0000
```

Notice there is one person with missing data on the CSI! So, we could have a problem because there will be 116 in the tests of path a, but only 115 in the tests of other paths! So, delete or select out this person before running analyses.

In R:

```
#filter out case with missing data
dat.1 = filter(dat, ID !=3)
head(dat.1)
```

	ID <int>	gender <int>	asrs <dbl>	cesd <dbl>	negcst <dbl>	csitot <dbl>
1	1	0	30	15	4.125	35
2	2	0	36	36	1.000	81
3	4	0	27	29	3.000	46
4	5	0	32	16	2.375	71
5	6	0	25	28	4.000	45
6	9	0	15	5	2.000	81

6 rows

```
basicStats(dat.1)
```

	ID <dbl>	gender <dbl>	asrs <dbl>	cesd <dbl>	negcst <dbl>	csitot <dbl>
nobs	115.000000	115	115.000000	115.000000	115.000000	115.000000
NAs	0.000000	0	0.000000	0.000000	0.000000	0.000000
Minimum	1.000000	0	3.000000	1.000000	1.000000	13.000000
Maximum	200.000000	0	58.000000	47.000000	5.625000	81.000000
1. Quartile	69.000000	0	24.000000	9.000000	2.000000	52.000000
3. Quartile	158.500000	0	36.000000	21.500000	3.812500	74.500000
Mean	112.843478	0	30.237979	16.070721	2.892813	61.694594
Median	122.000000	0	30.000000	15.000000	2.875000	65.000000
Sum	12977.000000	0	3477.367530	1848.132958	332.673489	7094.878354
SE Mean	5.429453	0	0.895945	0.872771	0.107191	1.504227

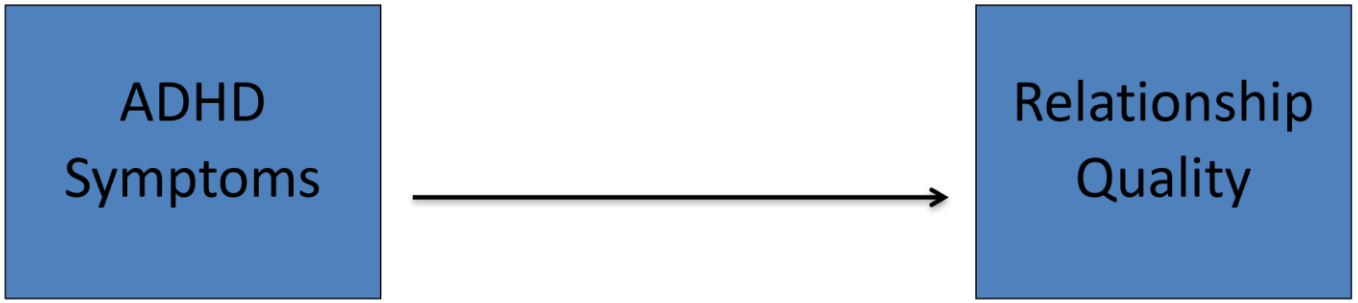
1-10 of 16 rows Previous 1 2 Next

Analyses and Results

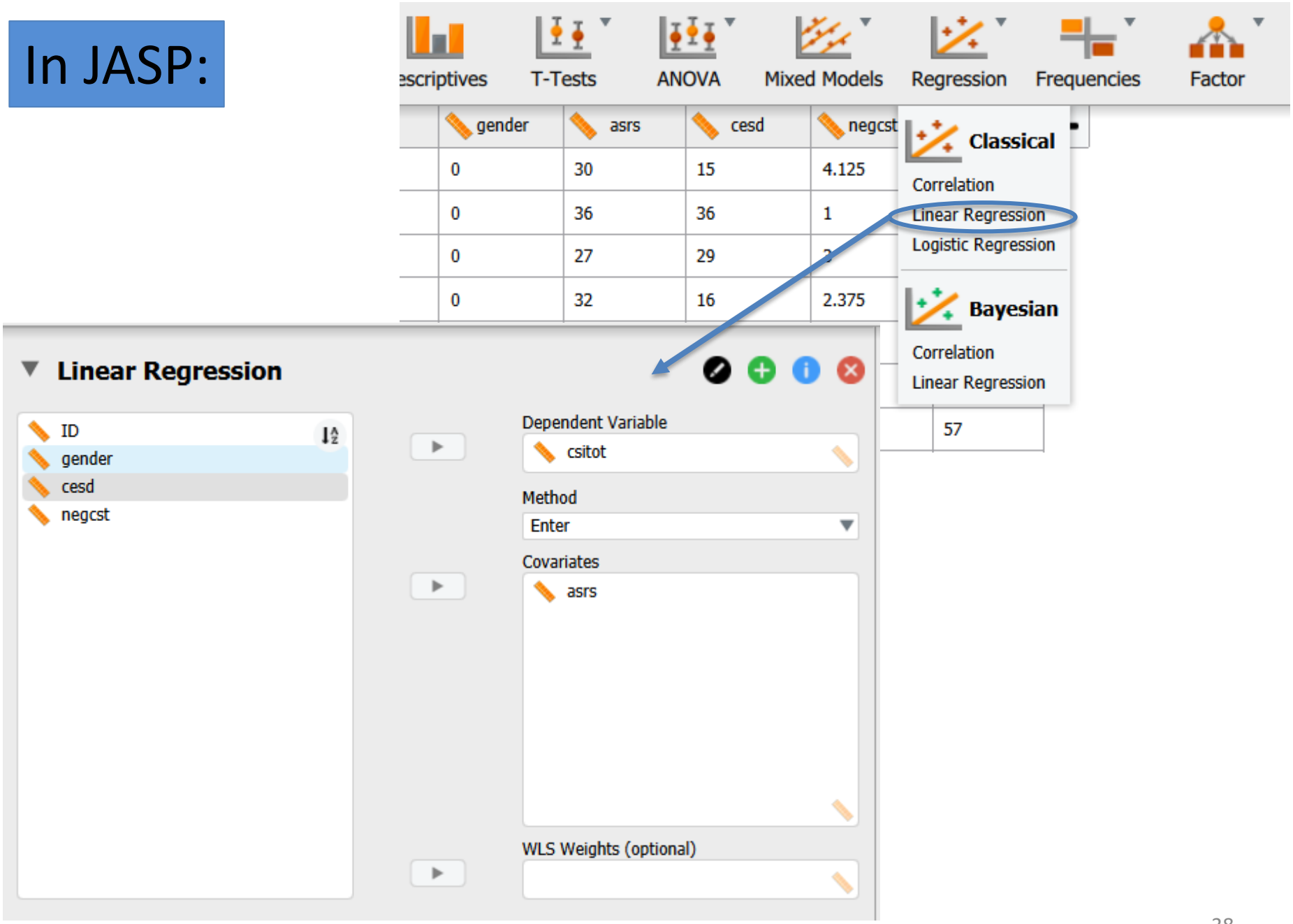
Step 1) Test whether the independent variable is associated with the dependent variable.

--- Path c (total effect)

--- Regress Y on X in a bivariate regression.



In JASP:


escriptives
T-Tests
ANOVA
Mixed Models
Regression
Frequencies
Factor
gender
asrs
cesd
negcst
0 30 15 4.125
0 36 36 1
0 27 29
0 32 16 2.375
Classical
Correlation
Linear Regression
Logistic Regression
Bayesian
Correlation
Linear Regression
57
Linear Regression
ID
gender
cesd
negcst
Dependent Variable
csitot
Method
Enter
Covariates
asrs
WLS Weights (optional)

In JASP:

Linear Regression

Model Summary - csitot

Model	R	R^2	Adjusted R^2	RMSE
H_0	0.000	0.000	0.000	16.131
H_1	0.232	0.054	0.045	15.762

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H_1	Regression	1590.266	1	1590.266	6.401	0.013
	Residual	28073.697	113	248.440		
	Total	29663.963	114			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
H_0	(Intercept)	61.695	1.504		41.014	< .001
H_1	(Intercept)	73.449	4.873		15.073	< .001
	asrs	−0.389	0.154	−0.232	−2.530	0.013

Descriptives

	N	Mean	SD	SE
csitot	115	61.695	16.131	1.504
asrs	115	30.238	9.608	0.896

In SPSS:

Variables Entered/Removed[b]

Model	Variables Entered	Variables Removed	Method
1	ADHD symptoms[a]	.	Enter

a. All requested variables entered.

b. Dependent Variable: Couple Satisfaction

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.232[a]	.054	.045	15.76197

a. Predictors: (Constant), ADHD symptoms

ANOVA[b]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1590.266	1	1590.266	6.401	.013[a]
	Residual	28073.697	113	248.440		
	Total	29663.963	114			

a. Predictors: (Constant), ADHD symptoms

b. Dependent Variable: Couple Satisfaction

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	73.449	4.873		15.073	.000
	ADHD symptoms	-.389	.154	-.232	-2.530	.013

a. Dependent Variable: Couple Satisfaction

In R:

```
m.1.1=lm(csitot ~ asrs,data=dat.1)
summary(m.1.1)
```

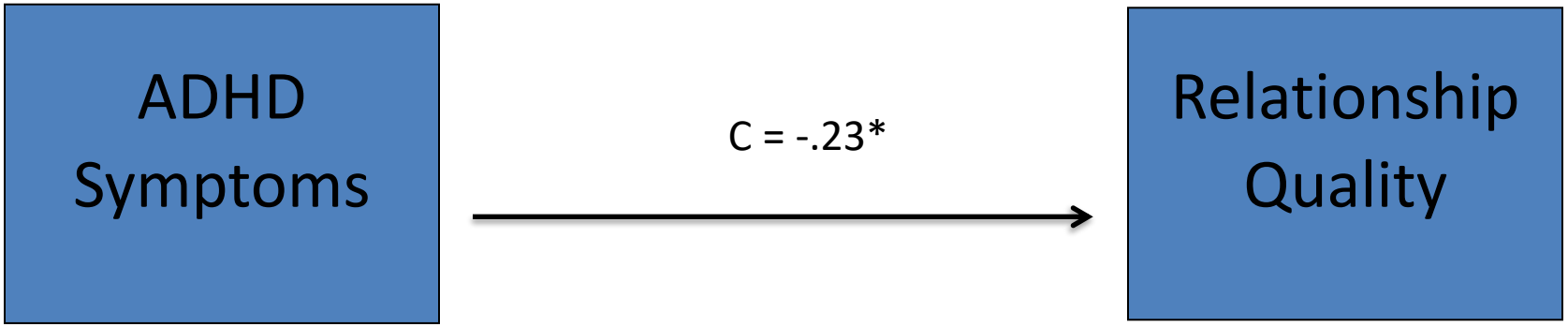
```
##
## Call:
## lm(formula = csitot ~ asrs, data = dat.1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -51.508  -6.981   3.602  11.545  29.097
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  73.4491     4.8730   15.07   <2e-16 ***
## asrs         -0.3887     0.1536   -2.53   0.0128 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.76 on 113 degrees of freedom
## Multiple R-squared:  0.05361,    Adjusted R-squared:  0.04523
## F-statistic: 6.401 on 1 and 113 DF,  p-value: 0.01278
```

In R:

```
m.1.2=lm(scale(csitot)~ scale(asrs),data=dat.1)
summary(m.1.2)
```

```
##
## Call:
## lm(formula = scale(csitot) ~ scale(asrs), data = dat.1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.1931 -0.4328  0.2233  0.7157  1.8038
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.220e-17  9.112e-02    0.00   1.0000
## scale(asrs) -2.315e-01  9.152e-02   -2.53   0.0128 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9771 on 113 degrees of freedom
## Multiple R-squared:  0.05361,    Adjusted R-squared:  0.04523
## F-statistic: 6.401 on 1 and 113 DF,  p-value: 0.01278
```

**note- for this example, in figures, will use standardized betas to aid interpretation. But, in tests of the significance of the indirect or mediated effect, use unstandardized coefficients.

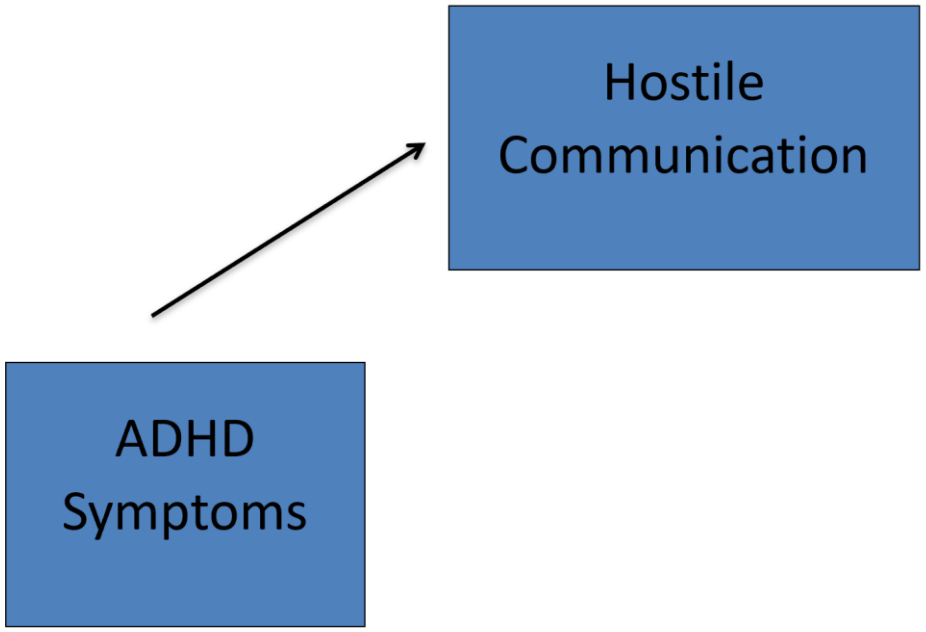


Analyses and Results

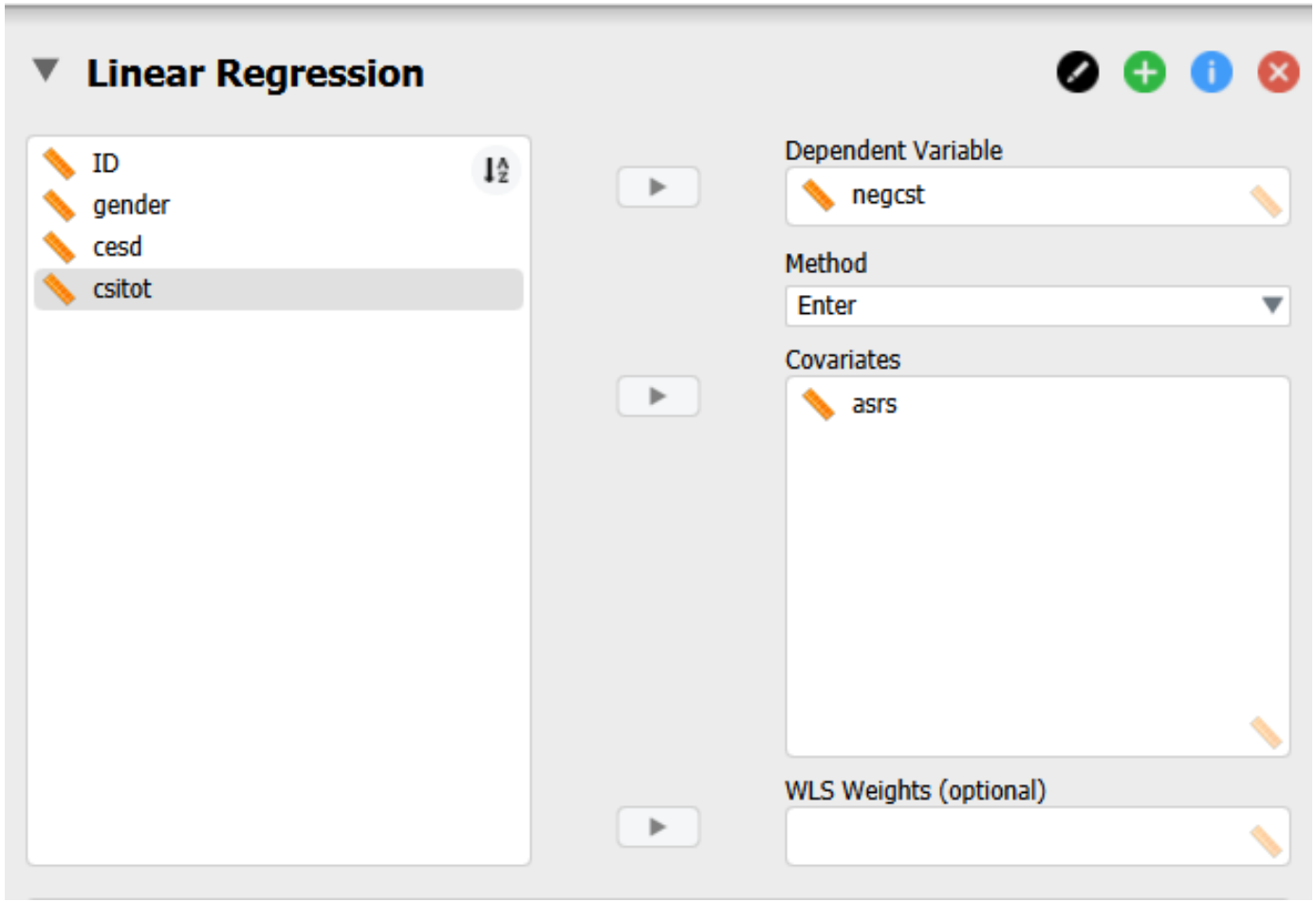
Step 2) Test whether the independent (or predictor) variable is correlated with the mediator

--- Path a

--- Regress M on X in a bivariate regression.



In JASP:


Linear Regression
ID
gender
cesd
csitot
Dependent Variable
negcst
Method
Enter
Covariates
asrs
WLS Weights (optional)

In JASP:

Linear Regression

Model Summary - negcst

Model	R	R^2	Adjusted R^2	RMSE
H_0	0.000	0.000	0.000	1.149
H_1	0.334	0.111	0.103	1.088

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H_1	Regression	16.771	1	16.771	14.158	< .001
	Residual	133.862	113	1.185		
	Total	150.634	114			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
H_0	(Intercept)	2.893	0.107		26.987	< .001
H_1	(Intercept)	1.686	0.336		5.010	< .001
	asrs	0.040	0.011	0.334	3.763	< .001

Descriptives

	N	Mean	SD	SE
negcst	115	2.893	1.149	0.107
asrs	115	30.238	9.608	0.896

In SPSS:

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	ADHD symptoms[b]	.	Enter

a. Dependent Variable: Hostile Communication

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.334[a]	.111	.103	1.08840

a. Predictors: (Constant), ADHD symptoms

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	16.771	1	16.771	14.158	.000[b]
	Residual	133.862	113	1.185		
	Total	150.634	114			

a. Dependent Variable: Hostile Communication

b. Predictors: (Constant), ADHD symptoms

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.686	.336		5.010	.000
	ADHD symptoms	.040	.011	.334	3.763	.000

a. Dependent Variable: Hostile Communication

In R:

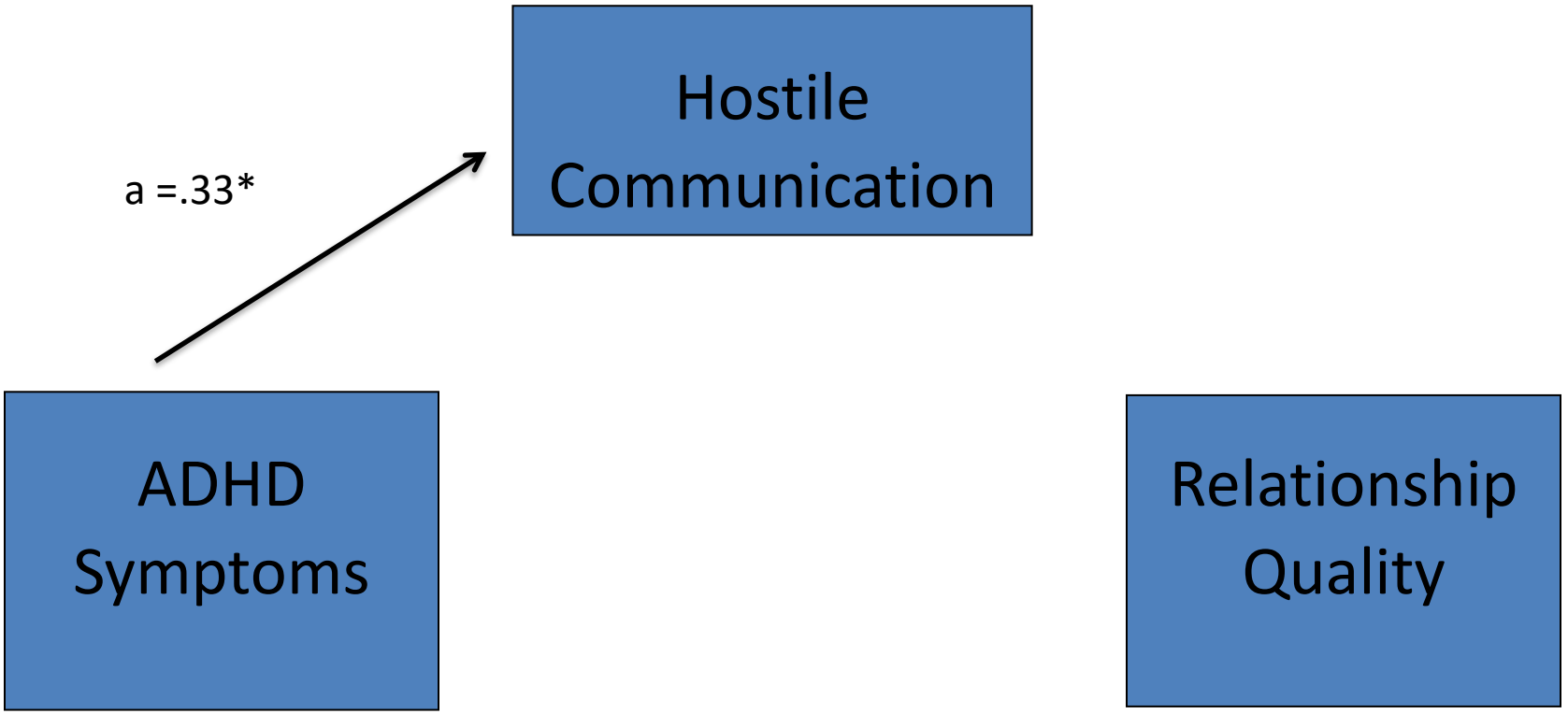
```
m.2.1=lm(negcst ~ asrs,data=dat.1)
summary(m.2.1)
```

```
##
## Call:
## lm(formula = negcst ~ asrs, data = dat.1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.00110 -0.82387 -0.01879  0.72696  2.69455
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.68568    0.33649   5.010 2.03e-06 ***
## asrs         0.03992    0.01061   3.763 0.000268 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.088 on 113 degrees of freedom
## Multiple R-squared:  0.1113, Adjusted R-squared:  0.1035
## F-statistic: 14.16 on 1 and 113 DF,  p-value: 0.0002682
```

In R:

```
m.2.2=lm(scale(negcst)~ scale(asrs),data=dat.1)
summary(m.2.2)
```

```
##
## Call:
## lm(formula = scale(negcst) ~ scale(asrs), data = dat.1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.61079 -0.71672 -0.01634  0.63242  2.34411
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.775e-17  8.829e-02   0.000 1.000000
## scale(asrs) 3.337e-01  8.868e-02   3.763 0.000268 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9469 on 113 degrees of freedom
## Multiple R-squared:  0.1113, Adjusted R-squared:  0.1035
## F-statistic: 14.16 on 1 and 113 DF,  p-value: 0.0002682
```

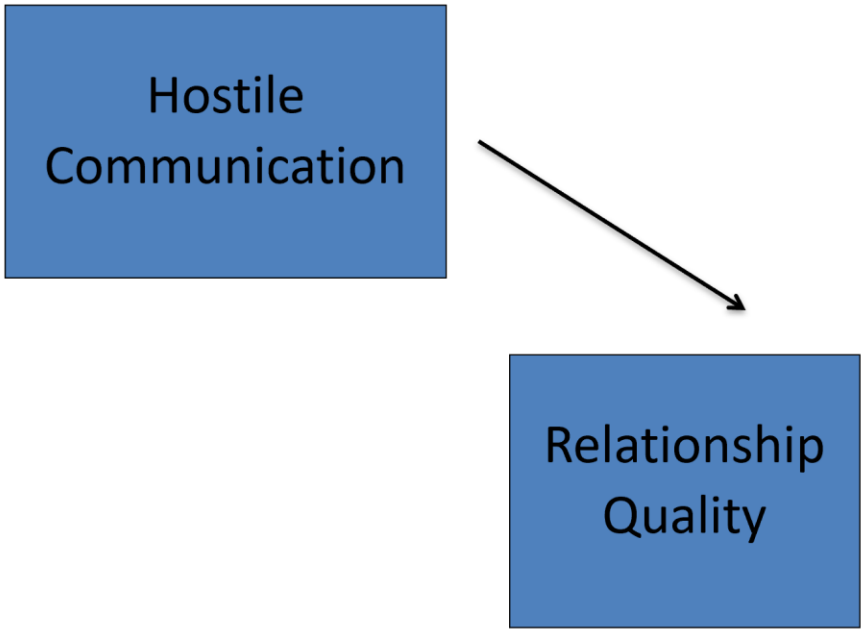

Hostile Communication
a =.33*
ADHD Symptoms
Relationship Quality

Analyses and Results

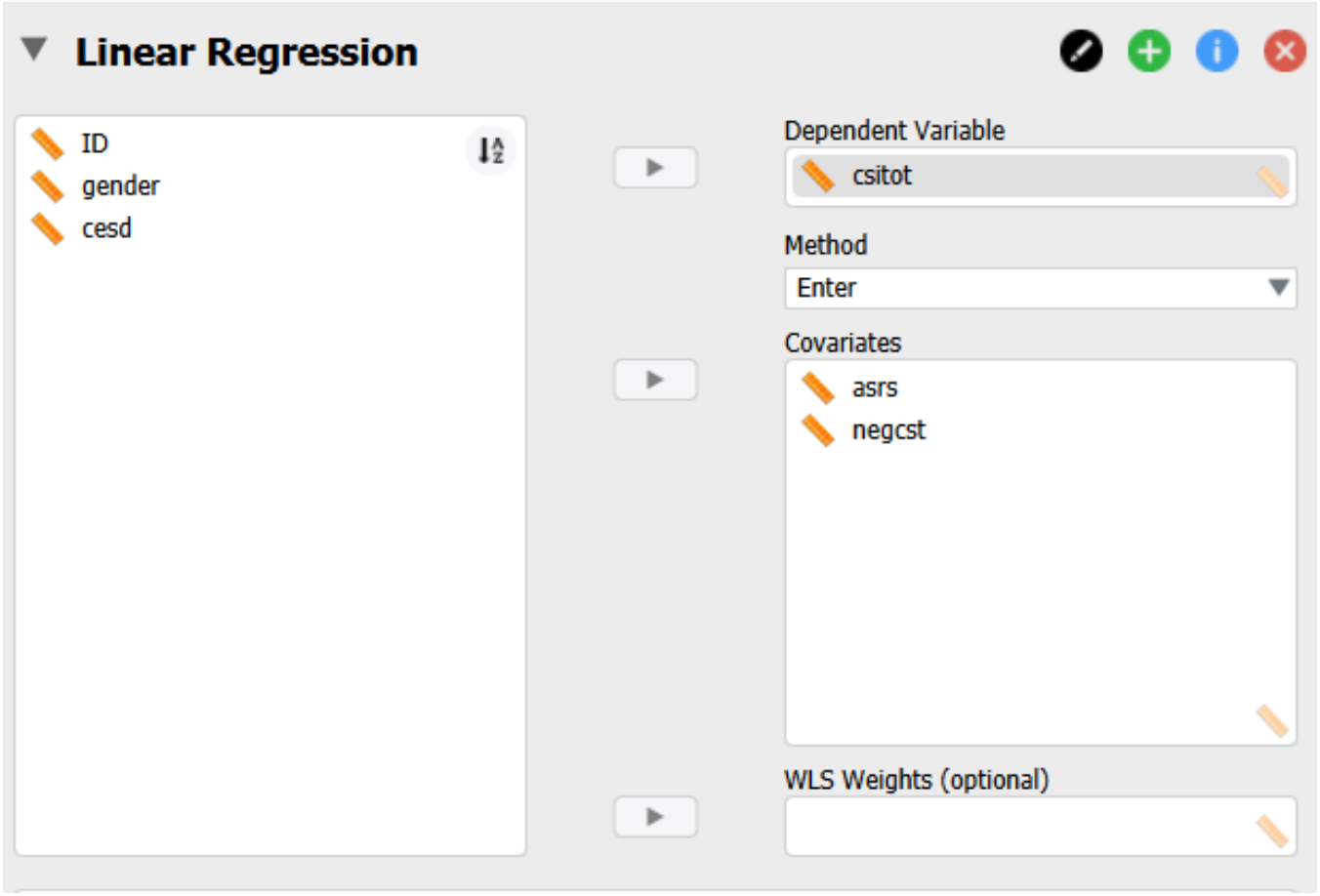
Step 3) Test whether the mediator affects the outcome variable (estimate and test path b)

--- Path b

--- Regress Y on X & M; check coefficient for M.



In JASP:


Linear Regression
ID
gender
cesd
Dependent Variable
csitot
Method
Enter
Covariates
asrs
negcst
WLS Weights (optional)

In JASP:

Linear Regression

Model Summary - csitot

Model	R	R^2	Adjusted R^2	RMSE
H_0	0.000	0.000	0.000	16.131
H_1	0.694	0.481	0.472	11.721

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H_1	Regression	14277.334	2	7138.667	51.963	< .001
	Residual	15386.629	112	137.381		
	Total	29663.963	114			

Note. The intercept model is omitted, as no meaningful information can be shown.

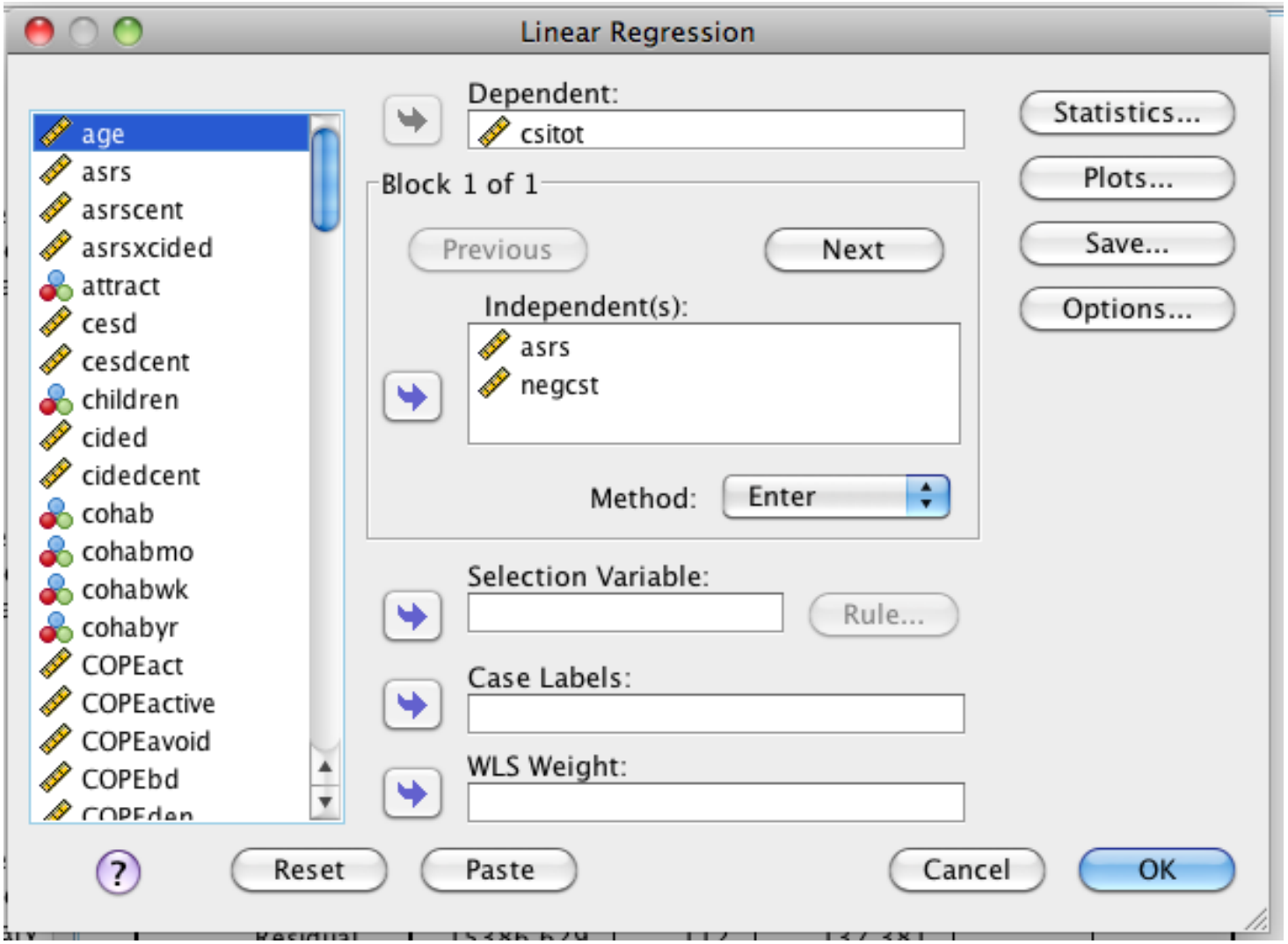
Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
H_0	(Intercept)	61.695	1.504		41.014	< .001
H_1	(Intercept)	89.860	4.006		22.432	< .001
	asrs	−8.924e −5	0.121	−5.315e −5	−7.363e −4	0.999
	negcst	−9.735	1.013	−0.694	−9.610	< .001

Descriptives

	N	Mean	SD	SE
csitot	115	61.695	16.131	1.504
asrs	115	30.238	9.608	0.896
negcst	115	2.893	1.149	0.107

In SPSS:


Linear Regression
age
asrs
asrscent
asrsxcided
attract
cesd
cesdcent
children
cided
cidedcent
cohab
cohabmo
cohabwk
cohabyr
COPEact
COPEactive
COPEavoid
COPEbd
Dependent:
csitot
Block 1 of 1
Previous
Next
Independent(s):
asrs
negcst
Method:
Enter
Selection Variable:
Rule...
Case Labels:
WLS Weight:
Statistics...
Plots...
Save...
Options...
Reset
Paste
Cancel
OK

In SPSS:

Variables Entered/Removed[b]

Model	Variables Entered	Variables Removed	Method
1	Hostile Communication, ADHD symptoms[a]	.	Enter

a. All requested variables entered.

b. Dependent Variable: Couple Satisfaction

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.694[a]	.481	.472	11.72095

a. Predictors: (Constant), Hostile Communication, ADHD symptoms

ANOVA[b]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	14277.334	2	7138.667	51.963	.000[a]
	Residual	15386.629	112	137.381		
	Total	29663.963	114			

a. Predictors: (Constant), Hostile Communication, ADHD symptoms

b. Dependent Variable: Couple Satisfaction

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	89.860	4.006		22.432	.000
	ADHD symptoms	-8.924E-5	.121	.000	-.001	.999
	Hostile Communication	-9.735	1.013	-.694	-9.610	.000

a. Dependent Variable: Couple Satisfaction

In R:

```
m.3.1=lm(csitot ~ asrs + negcst,data=dat.1)
summary(m.3.1)
```

```
##
## Call:
## lm(formula = csitot ~ asrs + negcst, data = dat.1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -41.267  -5.259   0.915   7.500  25.651
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.986e+01  4.006e+00  22.432  < 2e-16 ***
## asrs        -8.924e-05  1.212e-01  -0.001    0.999
## negcst      -9.735e+00  1.013e+00  -9.610 2.63e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.72 on 112 degrees of freedom
## Multiple R-squared:  0.4813, Adjusted R-squared:  0.472
## F-statistic: 51.96 on 2 and 112 DF,  p-value: < 2.2e-16
```

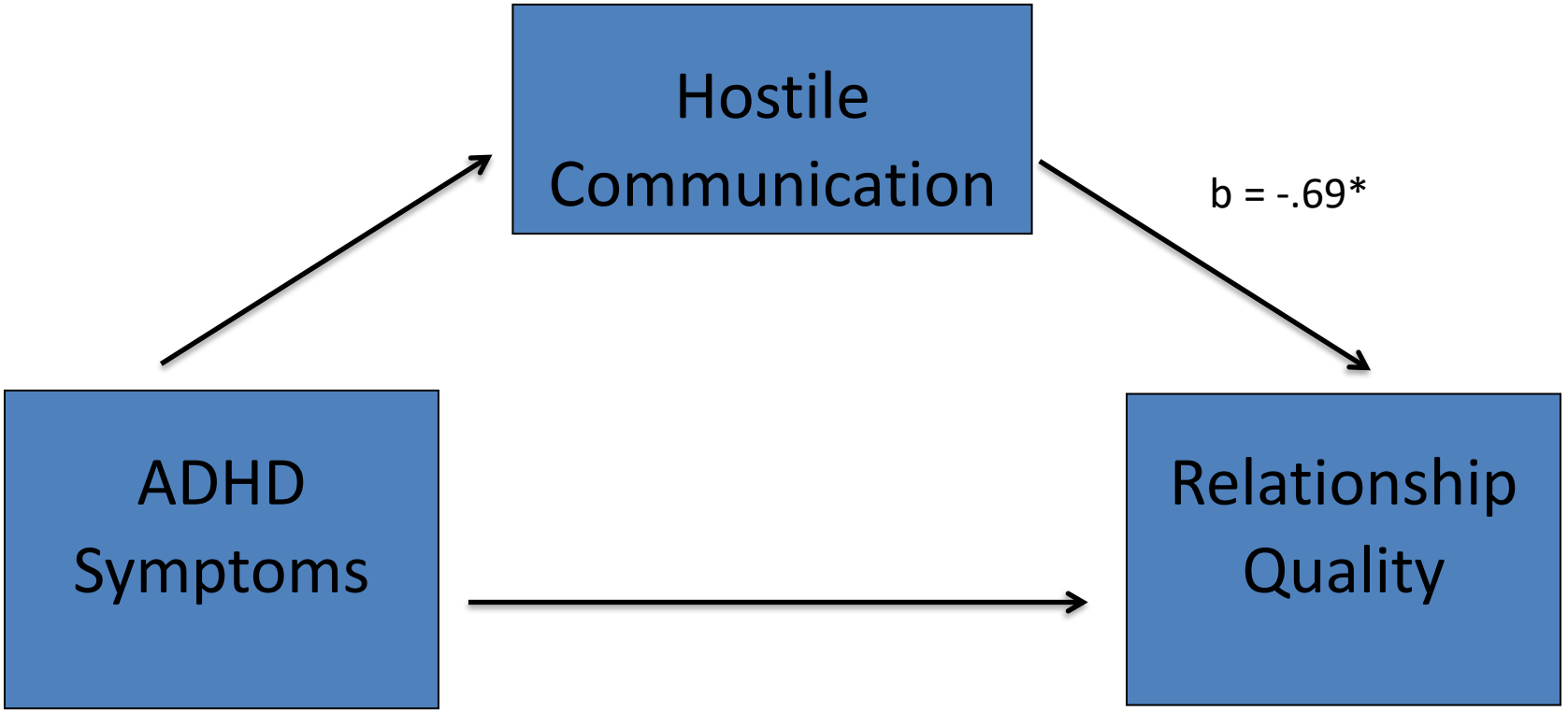


In R:

```
m.3.2=lm(scale(csitot)~ scale(asrs)+scale(negcst),data=dat.1)
summary(m.3.2)
```

```
##
## Call:
## lm(formula = scale(csitot) ~ scale(asrs) + scale(negcst), data = dat.1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.55822 -0.32604  0.05669  0.46492  1.59019
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    8.451e-17  6.776e-02   0.000    1.000
## scale(asrs)   -5.315e-05  7.219e-02  -0.001    0.999
## scale(negcst) -6.937e-01  7.219e-02  -9.610 2.63e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7266 on 112 degrees of freedom
## Multiple R-squared:  0.4813, Adjusted R-squared:  0.472
## F-statistic: 51.96 on 2 and 112 DF,  p-value: < 2.2e-16
```



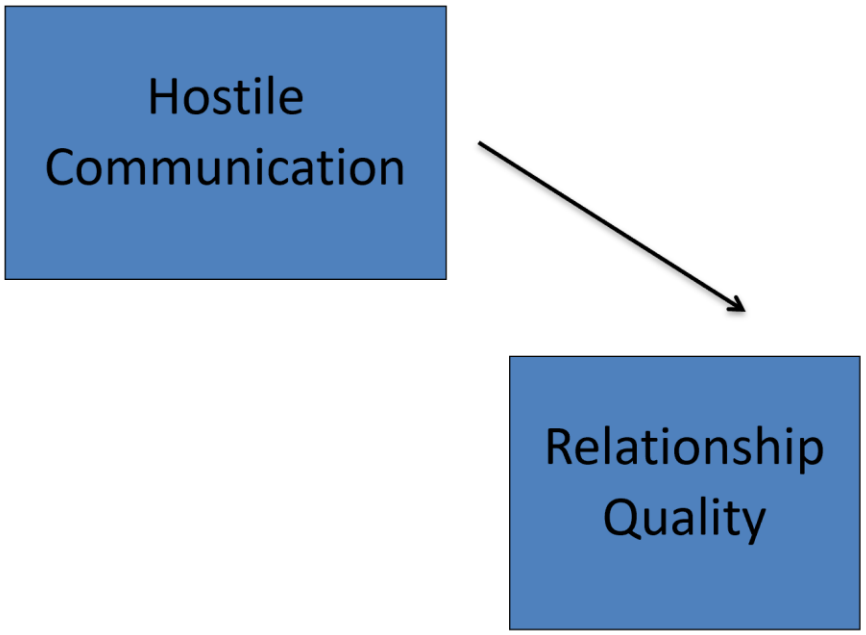

Hostile Communication
b = -.69*
ADHD Symptoms
Relationship Quality

Analyses and Results

Step 4) Test whether the effect of X on Y controlling for M is equal to zero

--- Path c'

--- Use Step 3 regression; check coefficient for X.



In JASP:

Model Summary - csitot

Model	R	R^2	Adjusted R^2	RMSE
H_0	0.000	0.000	0.000	16.131
H_1	0.694	0.481	0.472	11.721

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H_1	Regression	14277.334	2	7138.667	51.963	< .001
	Residual	15386.629	112	137.381		
	Total	29663.963	114			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
H_0	(Intercept)	61.695	1.504		41.014	< .001
H_1	(Intercept)	89.868	4.006		22.432	< .001
	asrs	−8.924e −5	0.121	−5.315e −5	−7.363e −4	0.999
	negcst	−9.735	1.013	−0.694	−9.610	< .001

In SPSS:

Variables Entered/Removed[b]

Model	Variables Entered	Variables Removed	Method
1	Hostile Communication, ADHD symptoms[a]	.	Enter

a. All requested variables entered.

b. Dependent Variable: Couple Satisfaction

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.694[a]	.481	.472	11.72095

a. Predictors: (Constant), Hostile Communication, ADHD symptoms

ANOVA[b]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	14277.334	2	7138.667	51.963	.000[a]
	Residual	15386.629	112	137.381		
	Total	29663.963	114			

a. Predictors: (Constant), Hostile Communication, ADHD symptoms

b. Dependent Variable: Couple Satisfaction

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	89.860	4.006		22.432	.000
	ADHD symptoms	-8.924E-5	.121	.000	-.001	.999
	Hostile Communication	-9.735	1.013	-.694	-9.610	.000

a. Dependent Variable: Couple Satisfaction

In R:

```
m.3.1=lm(csitot ~ asrs + negcst,data=dat.1)
summary(m.3.1)
```

```
##
## Call:
## lm(formula = csitot ~ asrs + negcst, data = dat.1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -41.267  -5.259   0.915   7.500  25.651
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.986e+01  4.006e+00  22.432  < 2e-16 ***
## asrs        -8.924e-05  1.212e-01  -0.001    0.999
## negcst      -9.735e+00  1.013e+00  -9.610 2.63e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.72 on 112 degrees of freedom
## Multiple R-squared:  0.4813, Adjusted R-squared:  0.472
## F-statistic: 51.96 on 2 and 112 DF,  p-value: < 2.2e-16
```

In R:

```
m.3.2=lm(scale(csitot)~ scale(asrs)+scale(negcst),data=dat.1)
summary(m.3.2)
```

```
##
## Call:
## lm(formula = scale(csitot) ~ scale(asrs) + scale(negcst), data = dat.1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.55822 -0.32604  0.05669  0.46492  1.59019
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    8.451e-17  6.776e-02   0.000    1.000
## scale(asrs)   -5.315e-05  7.219e-02  -0.001    0.999
## scale(negcst) -6.937e-01  7.219e-02  -9.610 2.63e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7266 on 112 degrees of freedom
## Multiple R-squared:  0.4813, Adjusted R-squared:  0.472
## F-statistic: 51.96 on 2 and 112 DF,  p-value: < 2.2e-16
```



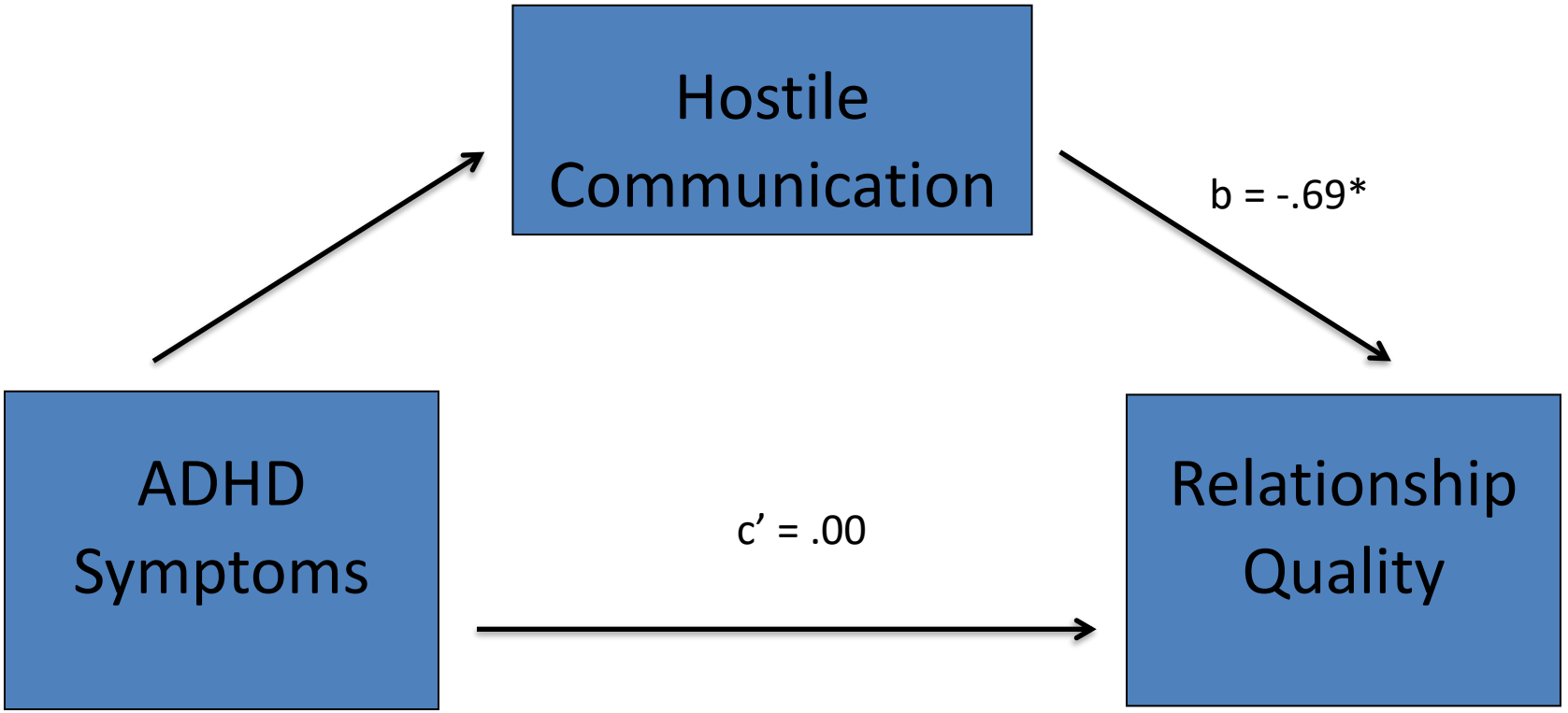
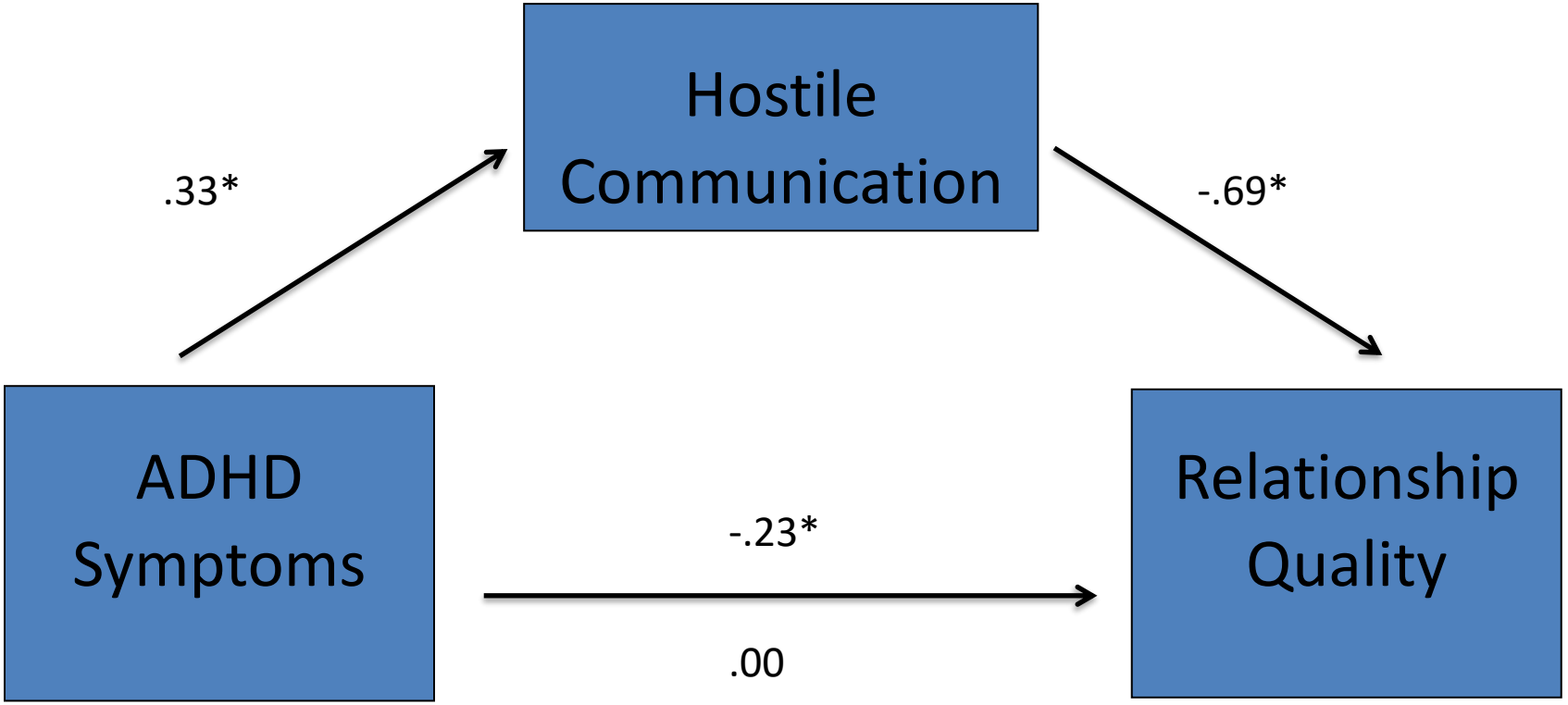

Hostile Communication
b = -.69*
ADHD Symptoms
c' = .00
Relationship Quality

Figure presented in Manuscript



Note. Coefficients above the arrow are estimates of the total effect; those below the arrow are estimates of the direct effect.

Test of the Indirect effect: #1: The Sobel Test

- Using the online calculator, insert:
  - the unstandardized coefficients for a and b
  - and their standard errors,
  - to calculate the Sobel test:

$$Z = \frac{a * b}{\sqrt{b^2 s_a^2 + a^2 s_b^2}}$$

CALCULATION FOR THE SOBEL TEST

An interactive calculation tool for mediation tests

- Curriculum vitae
- Selected publications
- Supplemental material for publications
- PSYC 2101: Introduction to Statistical Analysis
- PSYC 319: Factor Analysis
- Friends and colleagues
- Organizations
- Online utilities
- Mediation & moderation material
- Vanderbilt Dept. of Psychology & Human Development
- Vanderbilt Quantitative Methods (QM) program
- Contact me

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Details can be found in Baron and Kenny (1986), Sobel (1982), Goodman (1960), and MacKinnon, Warsi, and Dwyer (1995). Insert the *a*, *b*, s_a, and s_b into the cells below and this program will calculate the critical ratio as a test of whether the indirect effect of the IV on the DV via the mediator is significantly different from zero.

Input:			Test statistic:	Std. Error:	*p*-value:
a	.04	Sobel test:	-3.40102673	0.11449484	0.00067133
b	-9.735	Aroian test:	-3.38503334	0.11503579	0.0007117
s_a	.011	Goodman test:	-3.41724898	0.11395131	0.00063257
s_b	1.013	Reset all	Calculate		

Alternatively, you can insert t_a and t_b into the cells below, where t_a and t_b are the *t*-test statistics for the difference between the *a* and *b* coefficients and zero. Results should be identical to the first test, except for error due to rounding.

Input:			Test statistic:	*p*-value:
t_a		Sobel test:		
t_b		Aroian test:		
		Goodman test:		
		Reset all	Calculate	

The reported *p*-values (rounded to 8 decimal places) are drawn from the unit normal distribution

In R:

```
#sobel test variables
med = dat.1$negcst
iv = dat.1$asrs
dv = dat.1$csitot

sobel(iv,med,dv)
```

```
## $`Mod1: Y~X`
##               Estimate Std. Error    t value     Pr(>|t|)
## (Intercept) 73.4491179  4.8729673 15.072770 8.260900e-29
## pred        -0.3887338  0.1536484 -2.530022 1.278281e-02
##
## $`Mod2: Y~X+M`
##                   Estimate Std. Error       t value     Pr(>|t|)
## (Intercept)  8.985982e+01  4.0058722 22.4320241624 3.094714e-43
## pred        -8.923737e-05  0.1212026 -0.0007362662 9.994139e-01
## med         -9.735344e+00  1.0130559 -9.6098786196 2.627740e-16
##
## $`Mod3: M~X`
##               Estimate Std. Error  t value     Pr(>|t|)
## (Intercept) 1.68568306 0.33649066 5.009598 2.031411e-06
## pred        0.03992099 0.01060981 3.762650 2.682396e-04
##
## $Indirect.Effect
## [1] -0.3886445
##
## $SE
## [1] 0.1109253
##
## $z.value
## [1] -3.50366
##
## $N
## [1] 115
```

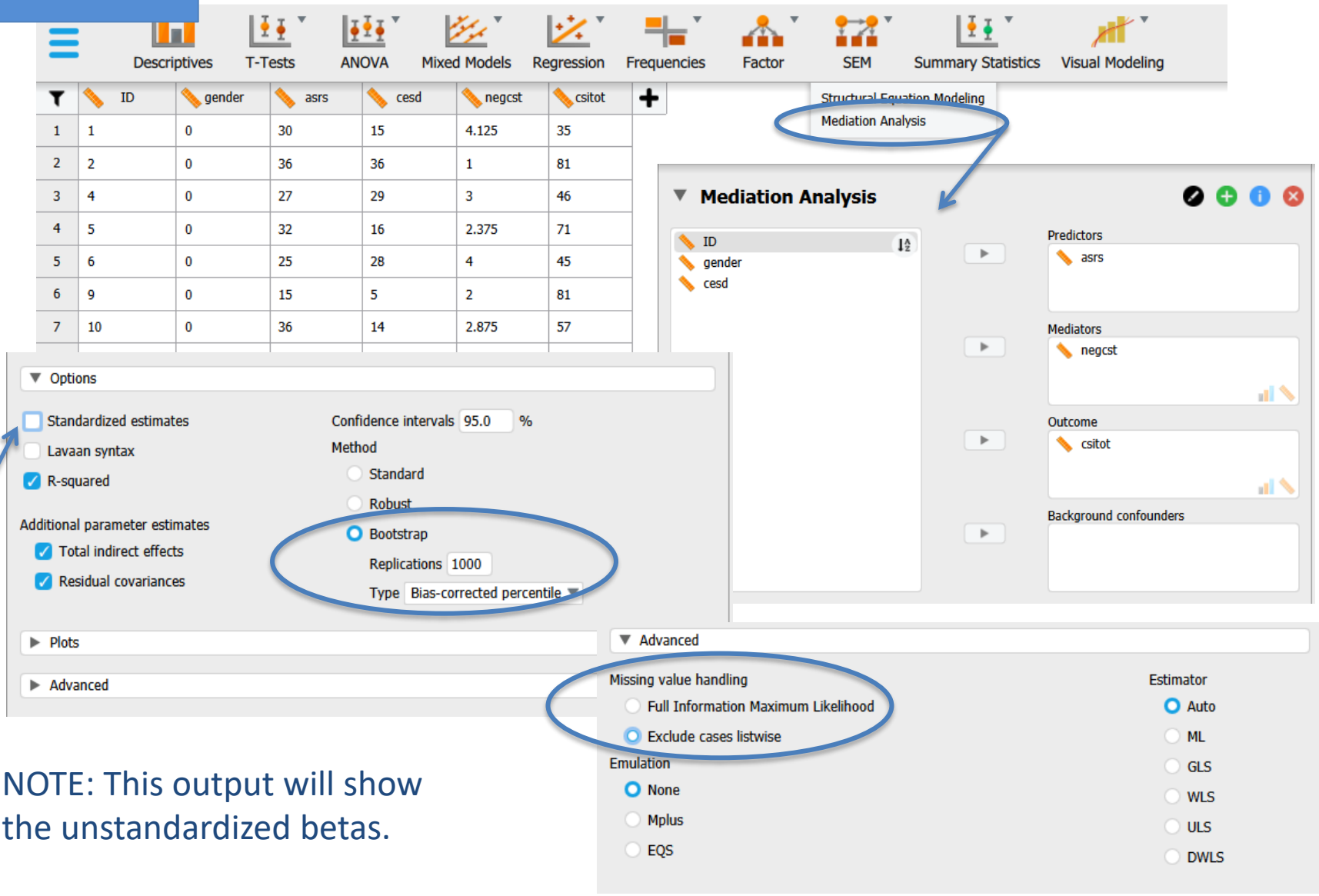
Example Report, using Sobel Test:

We used Baron and Kenny's (1986) procedure for estimating mediational effects using a series of regression analyses. As shown in Figure 1, there was a negative simple association between ADHD symptoms and relationship satisfaction (β =-.23, *p* =.01). Once communication was included in the model, the effect of ADHD symptom level on relationship satisfaction was reduced to zero (β = .00, *p* =.99), indicating full mediation. The Sobel test was used to determine if the reduction in prediction was statistically significant (*alternately, to determine if the indirect effect was statistically significant)*. Hostile communication significantly mediated the effect of ADHD symptoms on satisfaction, z' = -3.40, *p* <.001.

Test of the Indirect effect: #2: Bootstrapping Options

1. SEM (Structural Equation Modeling) Menu in JASP
2. Using the SPSS PROCESS macro from Andrew Hayes: http://www.processmacro.org/download.html
   - Same one you used for moderation
   - USE MODEL 4! (This is the one for mediation)
     - Often is the default setting in SPSS

In JASP:



	ID	gender	asrs	cesd	negcst	csitot
1	1	0	30	15	4.125	35
2	2	0	36	36	1	81
3	4	0	27	29	3	46
4	5	0	32	16	2.375	71
5	6	0	25	28	4	45
6	9	0	15	5	2	81
7	10	0	36	14	2.875	57

NOTE: This output will show the unstandardized betas.

In JASP:

Results

Mediation Analysis

NOTE: This output shows the unstandardized betas.

Parameter estimates

Direct effects

							95% Confidence Interval	
			Estimate	Std. Error	z-value	p	Lower	Upper
asrs	→	csitot	−8.923e −5	0.120	−7.460e −4	0.999	−0.270	0.227

Note. Delta method standard errors, bias-corrected percentile bootstrap confidence intervals, ML estimator.

Indirect effects

								95% Confidence Interval	
					Estimate	Std. Error	z-value	p	Lower	Upper
asrs	→	negcst	→	csitot	−0.389	0.110	−3.537	< .001	−0.667	−0.113

Note. Delta method standard errors, bias-corrected percentile bootstrap confidence intervals, ML estimator.

Total effects

							95% Confidence Interval	
			Estimate	Std. Error	z-value	p	Lower	Upper
asrs	→	csitot	−0.389	0.152	−2.552	0.011	−0.773	0.063

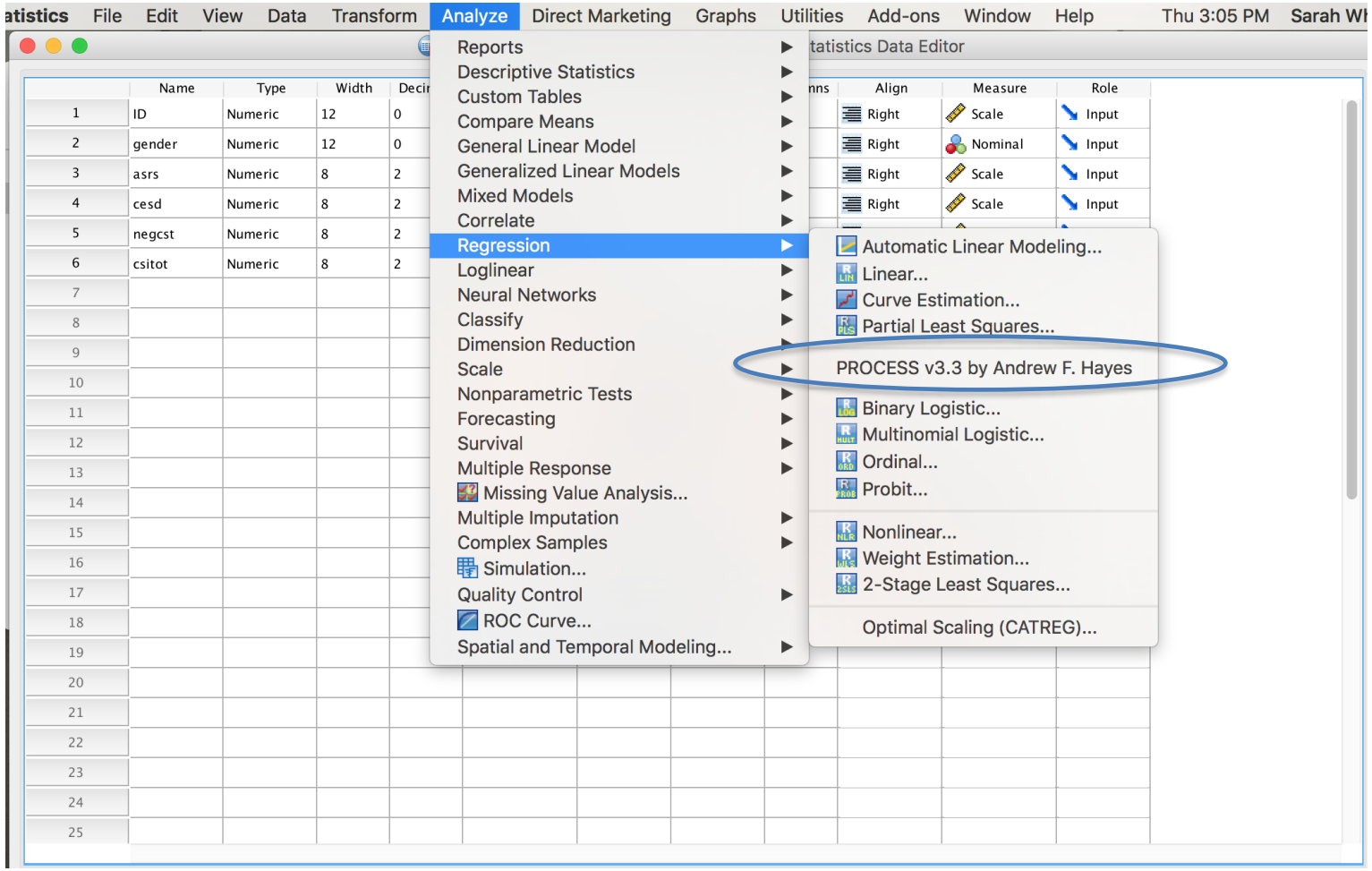
Note. Delta method standard errors, bias-corrected percentile bootstrap confidence intervals, ML estimator.

R-Squared

	R^2
csitot	0.481
negcst	0.111

In SPSS:

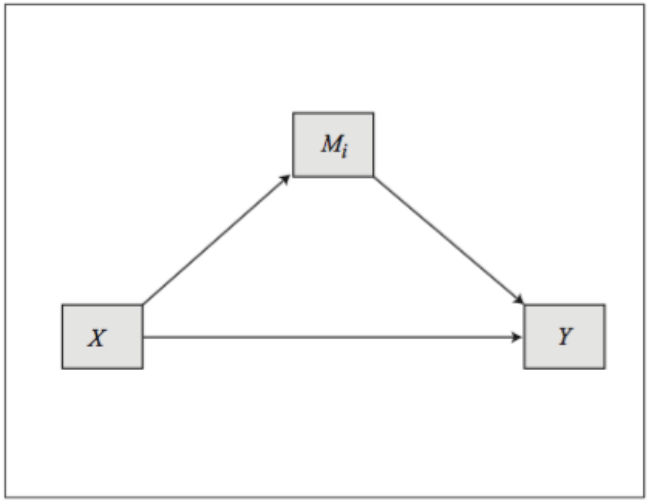
You can point and click to assess the significance of direct and indirect effects in your mediation model, using bootstrapping techniques in PROCESS:



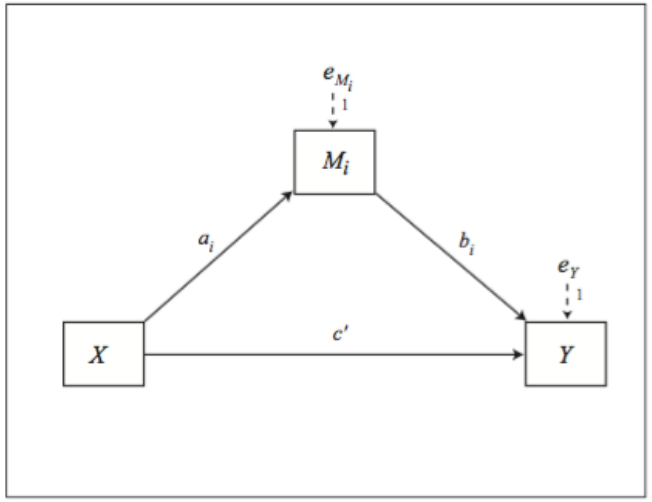
In SPSS:

Model 4

Conceptual Diagram



Statistical Diagram

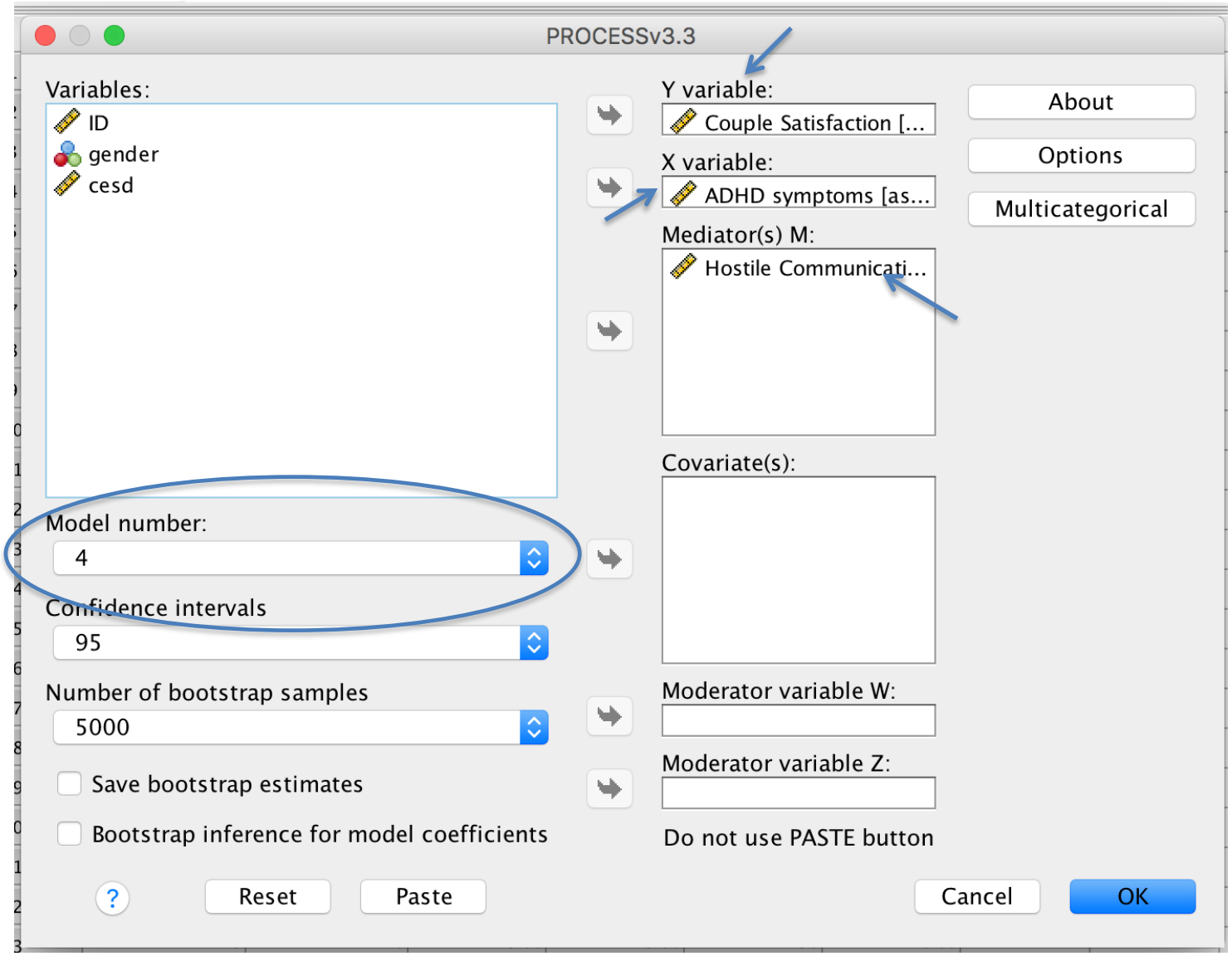


Indirect effect of X on Y through $M_i = a_i\, b_i$

Direct effect of X on $Y = c'$

Note: Model 4 allows up to 10 mediators operating in parallel.

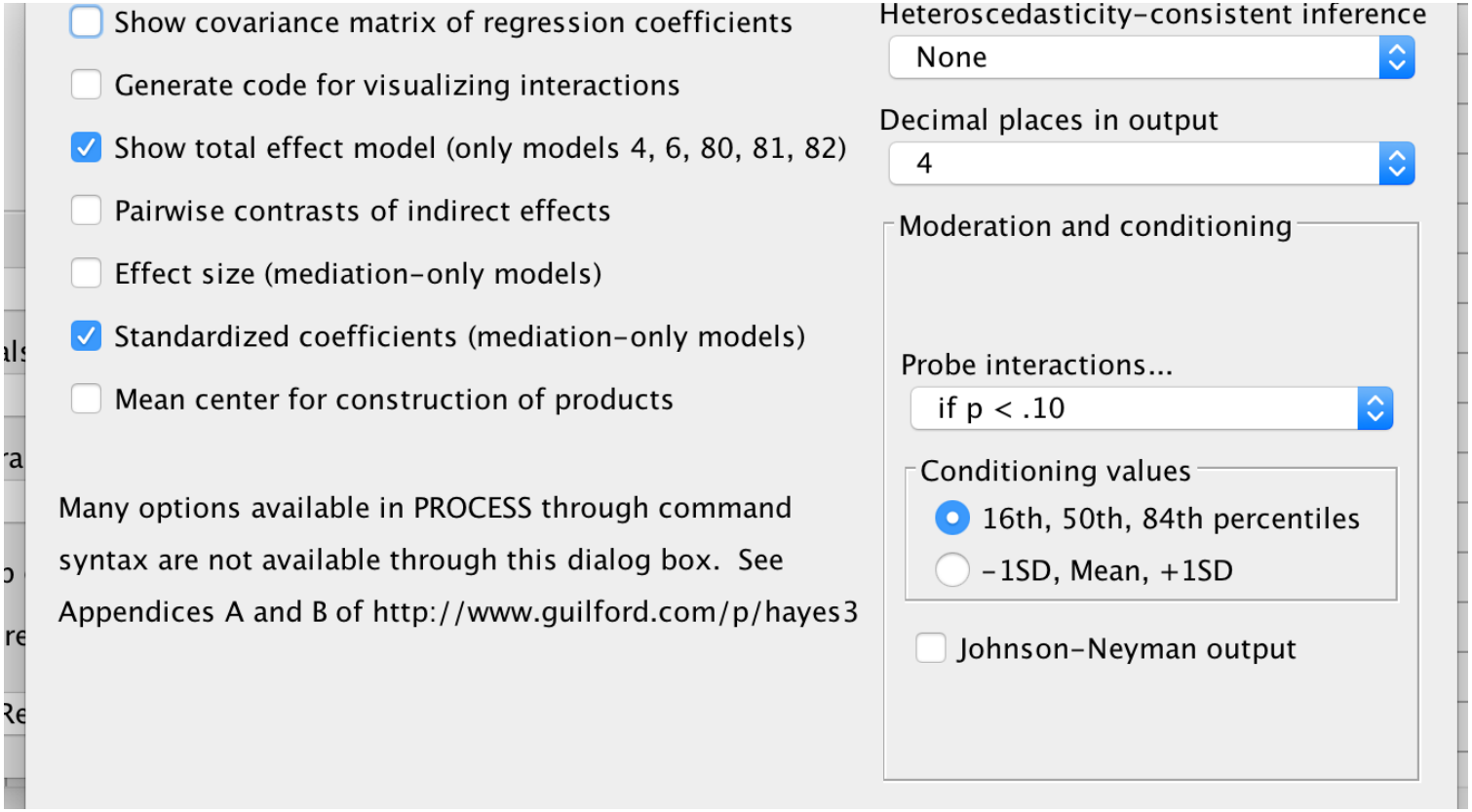
In SPSS:


PROCESSv3.3
Variables:
ID
gender
cesd
Y variable:
Couple Satisfaction [...
X variable:
ADHD symptoms [as...
Mediator(s) M:
Hostile Communicati...
Covariate(s):
Model number:
4
Confidence intervals
95
Number of bootstrap samples
5000
Moderator variable W:
Moderator variable Z:
Save bootstrap estimates
Bootstrap inference for model coefficients
Do not use PASTE button
About
Options
Multicategorical
Reset
Paste
Cancel
OK

In SPSS:

Under Options can select:

- Total effect model (to get estimate of path c)
- Standardized coefficients (this is new, Hayes used to not allow it because he views them as problematic)



In SPSS:


```
*************** PROCESS Procedure for SPSS Version 3.3 *******************

          Written by Andrew F. Hayes, Ph.D.       www.afhayes.com
    Documentation available in Hayes (2018). www.guilford.com/p/hayes3

**************************************************************************
Model  : 4
    Y  : csitot
    X  : asrs
    M  : negcst

Sample
Size:  115

**************************************************************************
OUTCOME VARIABLE:
 negcst

Model Summary
          R       R-sq        MSE          F        df1        df2          p
      .3337      .1113     1.1846    14.1575     1.0000   113.0000      .0003

Model
              coeff         se          t          p       LLCI       ULCI
constant     1.6857      .3365     5.0096      .0000     1.0190     2.3523
asrs          .0399      .0106     3.7626      .0003      .0189      .0609

Standardized coefficients
         coeff
asrs     .3337
```

First gives you M regressed on X.

Unstandardized beta for path a

Standardized beta for path a

In SPSS:

OUTPUT (p.2 of 3)

```
**************************************************************************
OUTCOME VARIABLE:
 csitot

Model Summary
          R       R-sq        MSE          F        df1        df2          p
      .6938      .4813   137.3806    51.9627     2.0000   112.0000      .0000

Model
              coeff         se          t          p       LLCI       ULCI
constant    89.8598     4.0059    22.4320     .0000    81.9227    97.7970
asrs         -.0001      .1212     -.0007     .9994     -.2402      .2401
negcst      -9.7353     1.0131    -9.6099     .0000   -11.7426    -7.7281

Standardized coefficients
            coeff
asrs       -.0001
negcst     -.6937

*************************** TOTAL EFFECT MODEL ****************************
OUTCOME VARIABLE:
 csitot

Model Summary
          R       R-sq        MSE          F        df1        df2          p
      .2315      .0536   248.4398     6.4010     1.0000   113.0000      .0128

Model
              coeff         se          t          p       LLCI       ULCI
constant    73.4491     4.8730    15.0728     .0000    63.7949    83.1034
asrs         -.3887      .1536    -2.5300     .0128     -.6931     -.0843

Standardized coefficients
           coeff
asrs      -.2315
```

Unstandardized b's for paths c' and b

Standardized betas for paths c' and b

Unstandardized beta for path c

Standardized beta for path c

In SPSS:

OUTPUT (p.3 of 3)

```
************** TOTAL, DIRECT, AND INDIRECT EFFECTS OF X ON Y **************

Total effect of X on Y
     Effect         se          t          p       LLCI       ULCI       c_ps       c_cs
     -.3887      .1536    -2.5300      .0128     -.6931     -.0843     -.0241     -.2315

Direct effect of X on Y
     Effect         se          t          p       LLCI       ULCI      c'_ps      c'_cs
     -.0001      .1212     -.0007      .9994     -.2402      .2401      .0000     -.0001

Indirect effect(s) of X on Y:
           Effect     BootSE   BootLLCI   BootULCI
negcst     -.3886      .1359     -.6592     -.1249

Partially standardized indirect effect(s) of X on Y:
           Effect     BootSE   BootLLCI   BootULCI
negcst     -.0241      .0084     -.0409     -.0076

Completely standardized indirect effect(s) of X on Y:
           Effect     BootSE   BootLLCI   BootULCI
negcst     -.2315      .0735     -.3663     -.0771

*********************** ANALYSIS NOTES AND ERRORS ************************

Level of confidence for all confidence intervals in output:
  95.0000

Number of bootstrap samples for percentile bootstrap confidence intervals:
  5000

------ END MATRIX -----
```

Estimated indirect effect = -.39 (SE=.14)

- Since CI does not include 0, it is statistically significant

In SPSS:

OUTPUT (p.3 of 3)

```
************** TOTAL, DIRECT, AND INDIRECT EFFECTS OF X ON Y **************

Total effect of X on Y
     Effect         se          t          p       LLCI       ULCI       c_ps       c_cs
     -.3887      .1536    -2.5300      .0128     -.6931     -.0843     -.0241     -.2315

Direct effect of X on Y
     Effect         se          t          p       LLCI       ULCI      c'_ps      c'_cs
     -.0001      .1212     -.0007      .9994     -.2402      .2401      .0000     -.0001

Indirect effect(s) of X on Y:
           Effect     BootSE   BootLLCI   BootULCI
negcst     -.3886      .1359     -.6592     -.1249

Partially standardized indirect effect(s) of X on Y:
           Effect     BootSE   BootLLCI   BootULCI
negcst     -.0241      .0084     -.0409     -.0076

Completely standardized indirect effect(s) of X on Y:
           Effect     BootSE   BootLLCI   BootULCI
negcst     -.2315      .0735     -.3663     -.0771

*********************** ANALYSIS NOTES AND ERRORS ************************

Level of confidence for all confidence intervals in output:
  95.0000

Number of bootstrap samples for percentile bootstrap confidence intervals:
  5000

------ END MATRIX -----
```

Partially standardized indirect effect = a*b/(SD of the DV)

- Doesn't take into account variation in IV and mediator, so it is called "partially standardized"

In SPSS:

OUTPUT (p.3 of 3)

```
************** TOTAL, DIRECT, AND INDIRECT EFFECTS OF X ON Y **************

Total effect of X on Y
     Effect         se          t          p       LLCI       ULCI       c_ps       c_cs
     -.3887      .1536    -2.5300      .0128     -.6931     -.0843     -.0241     -.2315

Direct effect of X on Y
     Effect         se          t          p       LLCI       ULCI      c'_ps      c'_cs
     -.0001      .1212     -.0007      .9994     -.2402      .2401      .0000     -.0001

Indirect effect(s) of X on Y:
           Effect     BootSE   BootLLCI   BootULCI
negcst     -.3886      .1359     -.6592     -.1249

Partially standardized indirect effect(s) of X on Y:
           Effect     BootSE   BootLLCI   BootULCI
negcst     -.0241      .0084     -.0409     -.0076

Completely standardized indirect effect(s) of X on Y:
           Effect     BootSE   BootLLCI   BootULCI
negcst     -.2315      .0735     -.3663     -.0771

*********************** ANALYSIS NOTES AND ERRORS ************************

Level of confidence for all confidence intervals in output:
  95.0000

Number of bootstrap samples for percentile bootstrap confidence intervals:
  5000

------ END MATRIX -----
```

Completely standardized indirect effect =
(a*b/SD of the DV) * SD of IV

- This fully standardizes it
- Aka “Index of Mediation”

In R:

Install the mediation package:

```
pacman::p_load("mediation")
library(mediation)
```

2. Use the Mediation calls to create lms models and calculate the effects.

```
#setup lm models based on paths c, a, and the mediation:
model.C <- lm(csitot ~ asrs, dat.1) #total effect of adhd symptoms on relationship satisfaction
model.A <- lm(negcst ~ asrs, dat.1) #path a effect of adhd symptoms on hostile communication
model.Y <- lm(csitot ~ asrs + negcst, dat.1) #mediated path's effect of adhd symptoms and hostile on relationship
 satisfaction
med.results <- mediate(model.A, model.Y, treat='asrs', mediator='negcst',
                   boot=TRUE, sims=100) #mediation calculation with 100 bootstrapped samples
```

In R:

```
summary(med.results)
```

```
##
## Causal Mediation Analysis
##
## Nonparametric Bootstrap Confidence Intervals with the Percentile Method
##
##                 Estimate 95% CI Lower 95% CI Upper p-value
## ACME           -3.89e-01    -6.30e-01        -0.14  <2e-16 ***
## ADE            -8.92e-05    -2.25e-01         0.26    0.98
## Total Effect   -3.89e-01    -7.44e-01         0.00    0.06 .
## Prop. Mediated  1.00e+00    -2.16e-01         2.39    0.06 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Sample Size Used: 115
##
##
## Simulations: 100
```

ACME: Test of the indirect (mediated) effect

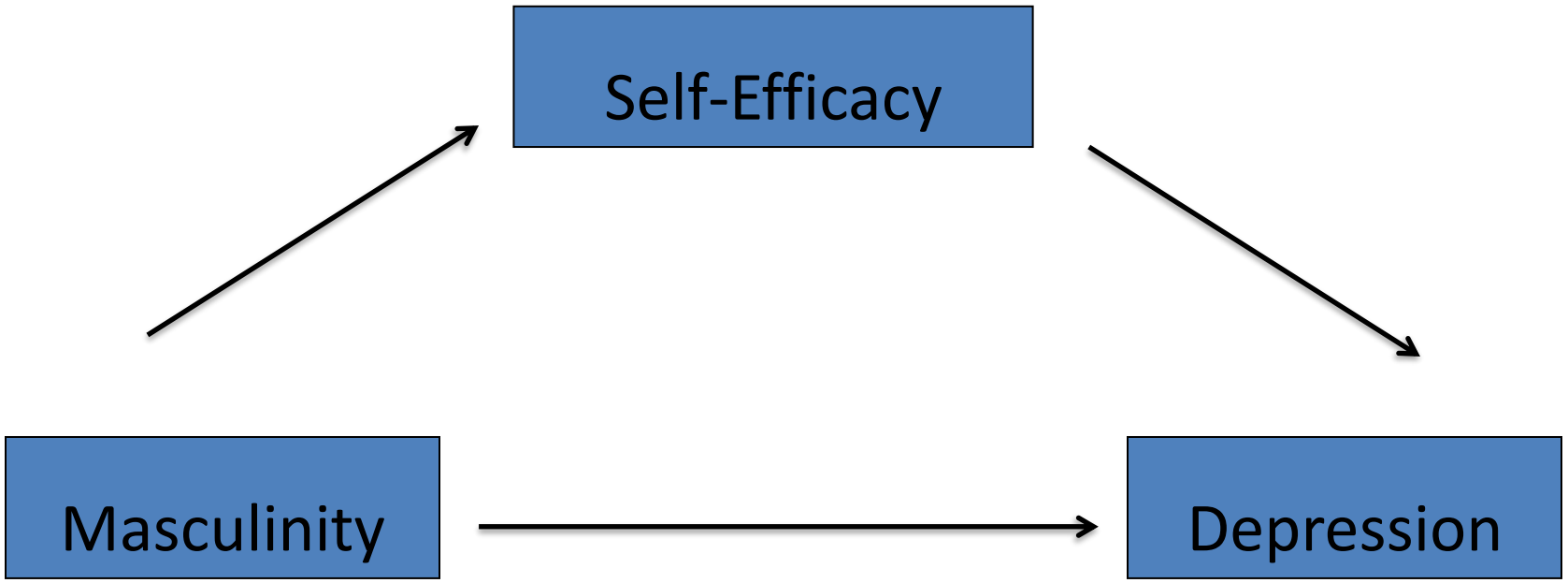
ADE: Direct effect (effect of X on Y after M is controlled)

Example Report using Mediation Bootstrapping:

The direct and indirect (i.e., mediated) effects were estimated with the PROCESS software (Hayes, 2013)/JASP Structural Equation Modeling and bootstrapping techniques. As shown in Figure 1, the indirect effect of ADHD symptoms on relationship satisfaction through hostile communication was *b* = -.39 (SE = .14; 95% CI = -.66 to -.12). In the mediated model, the direct effect of ADHD symptoms on relationship satisfaction was essentially zero, *b* = -.0001 (SE = .12; 95% CI = -.24 to .24; *p* =.99), suggesting full mediation.

5. In class Problem

- Background/Hypothesis:
  - Psychological masculinity is negatively associated with depressive symptom levels.
  - It is possible that individuals higher in masculinity have higher self-efficacy (perception that they can complete tasks or achieve goals successfully).
    - Counter to hopelessness and helplessness of depression


Self-Efficacy
Masculinity
Depression

In class Problem

- Sample: 467 undergrads who received participation credit
- Data: self-report measures of stereotypical masculinity, self-efficacy, and depressive symptoms via Survey Monkey
  - All three measures were Likert-type scales
  - Higher scores reflected more masculinity, self-efficacy, and depressive symptoms.
- Problem: Test the hypothesis that self-efficacy mediates the negative association between masculinity and depressive symptoms
  - Baron & Kenny method with Sobel test
  - Mediation package method

In class Problem

Solution

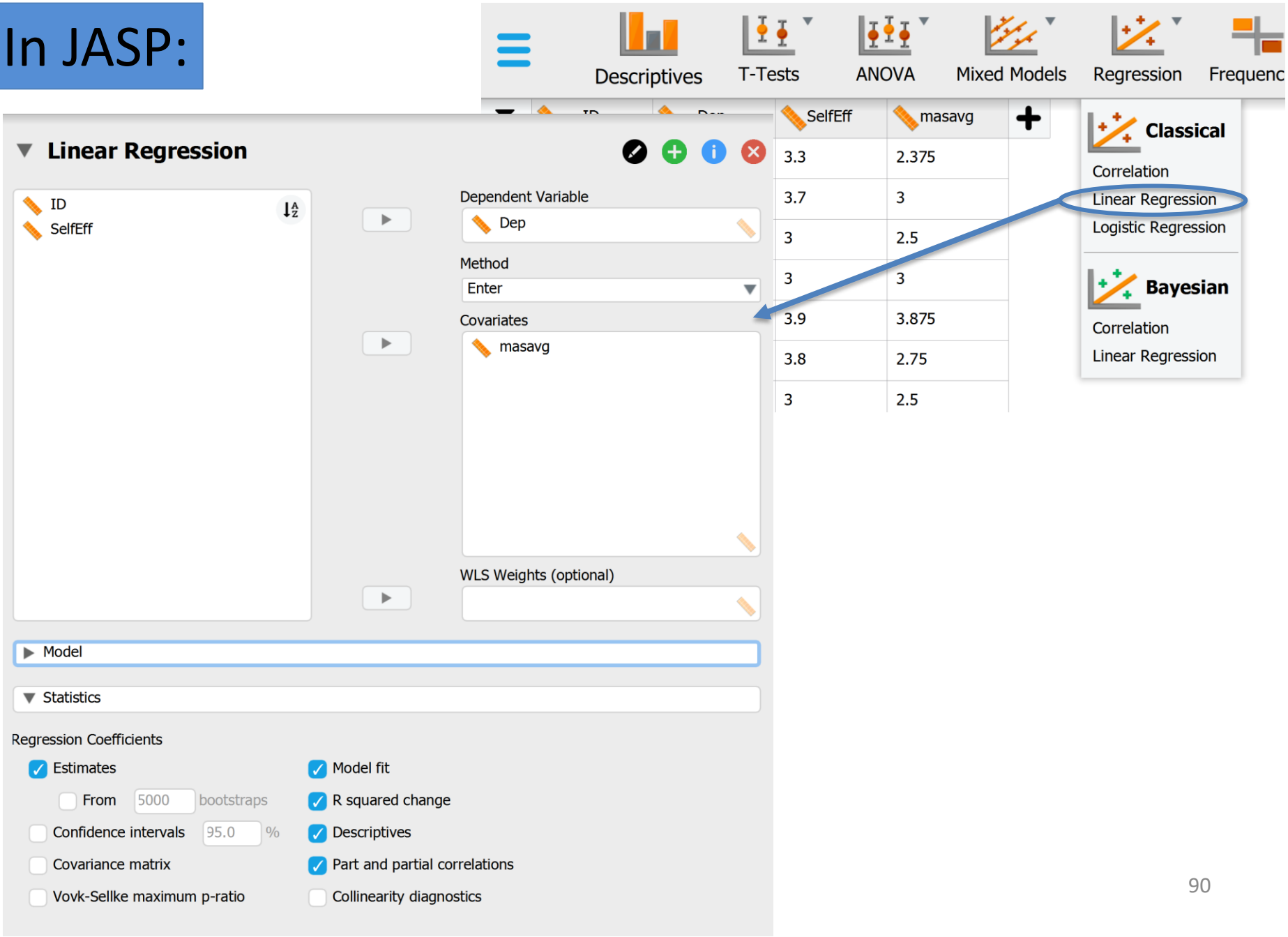
Analyses and Results

Step 1) Test whether the independent variable is correlated with the dependent variable.

- i.e., estimate path c
- by regressing the IV on the DV



In JASP:


Descriptives
T-Tests
ANOVA
Mixed Models
Regression
Frequenc
Linear Regression
ID
SelfEff
Dependent Variable
Dep
Method
Enter
Covariates
masavg
WLS Weights (optional)
Model
Statistics
Regression Coefficients
Estimates
From
5000
bootstraps
Confidence intervals
95.0
%
Covariance matrix
Vovk-Sellke maximum p-ratio
Model fit
R squared change
Descriptives
Part and partial correlations
Collinearity diagnostics
SelfEff
masavg
3.3
2.375
3.7
3
3
2.5
3
3
3.9
3.875
3.8
2.75
3
2.5
Classical
Correlation
Linear Regression
Logistic Regression
Bayesian
Correlation
Linear Regression

In JASP:

Results

Linear Regression

Model Summary - Dep

Model	R	R^2	Adjusted R^2	RMSE	R^2 Change	F Change	df1	df2	p
H_0	0.000	0.000	0.000	10.038	0.000		0	466	
H_1	0.312	0.097	0.096	9.546	0.097	50.229	1	465	< .001

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H_1	Regression	4577.286	1	4577.286	50.229	< .001
	Residual	42375.077	465	91.129		
	Total	46952.363	466			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
H_0	(Intercept)	15.424	0.464		33.206	< .001
H_1	(Intercept)	28.270	1.866		15.150	< .001
	masavg	−5.156	0.728	−0.312	−7.087	< .001

Descriptives

	N	Mean	SD	SE
Dep	467	15.424	10.038	0.464
masavg	467	2.492	0.608	0.028

In SPSS:

Regression

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.312[a]	.097	.096	9.54616

a. Predictors: (Constant), Masculinity Mean

ANOVA[b]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	4577.286	1	4577.286	50.229	.000[a]
	Residual	42375.077	465	91.129		
	Total	46952.363	466			

a. Predictors: (Constant), Masculinity Mean

b. Dependent Variable: Depressive Symptoms

Coefficients[a]

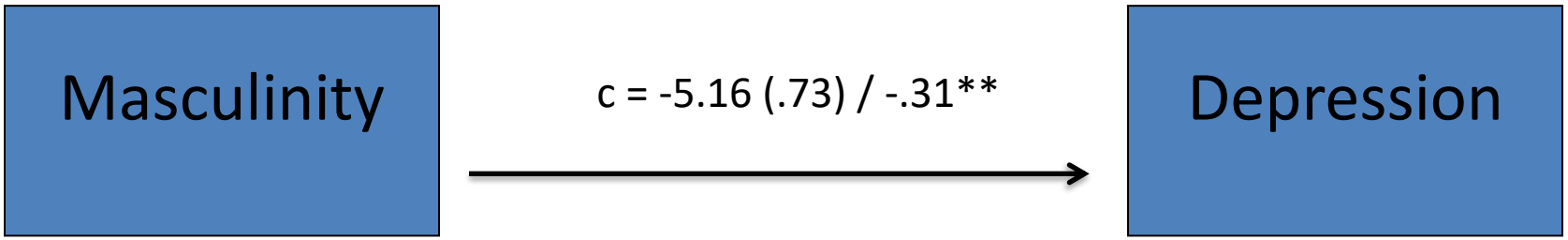
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	28.276	1.866		15.150	.000
	Masculinity Mean	-5.156	.728	-.312	-7.087	.000

a. Dependent Variable: Depressive Symptoms

In R:

```
m.1.2=lm(scale(Dep)~ scale(masavg),data=dat.1)
summary(m.1.2)
```

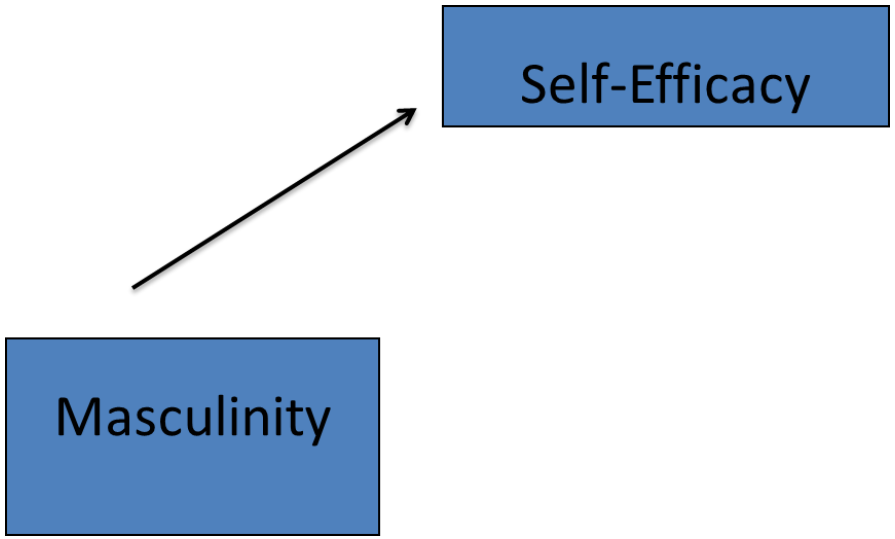
```
##
## Call:
## lm(formula = scale(Dep) ~ scale(masavg), data = dat.1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.7253 -0.6472 -0.2045  0.4853  3.8824
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -4.447e-17  4.401e-02   0.000        1
## scale(masavg) -3.122e-01  4.406e-02  -7.087 5.11e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.951 on 465 degrees of freedom
## Multiple R-squared:  0.09749,    Adjusted R-squared:  0.09555
## F-statistic: 50.23 on 1 and 465 DF,  p-value: 5.106e-12
```


Masculinity
c = -5.16 (.73) / -.31**
Depression

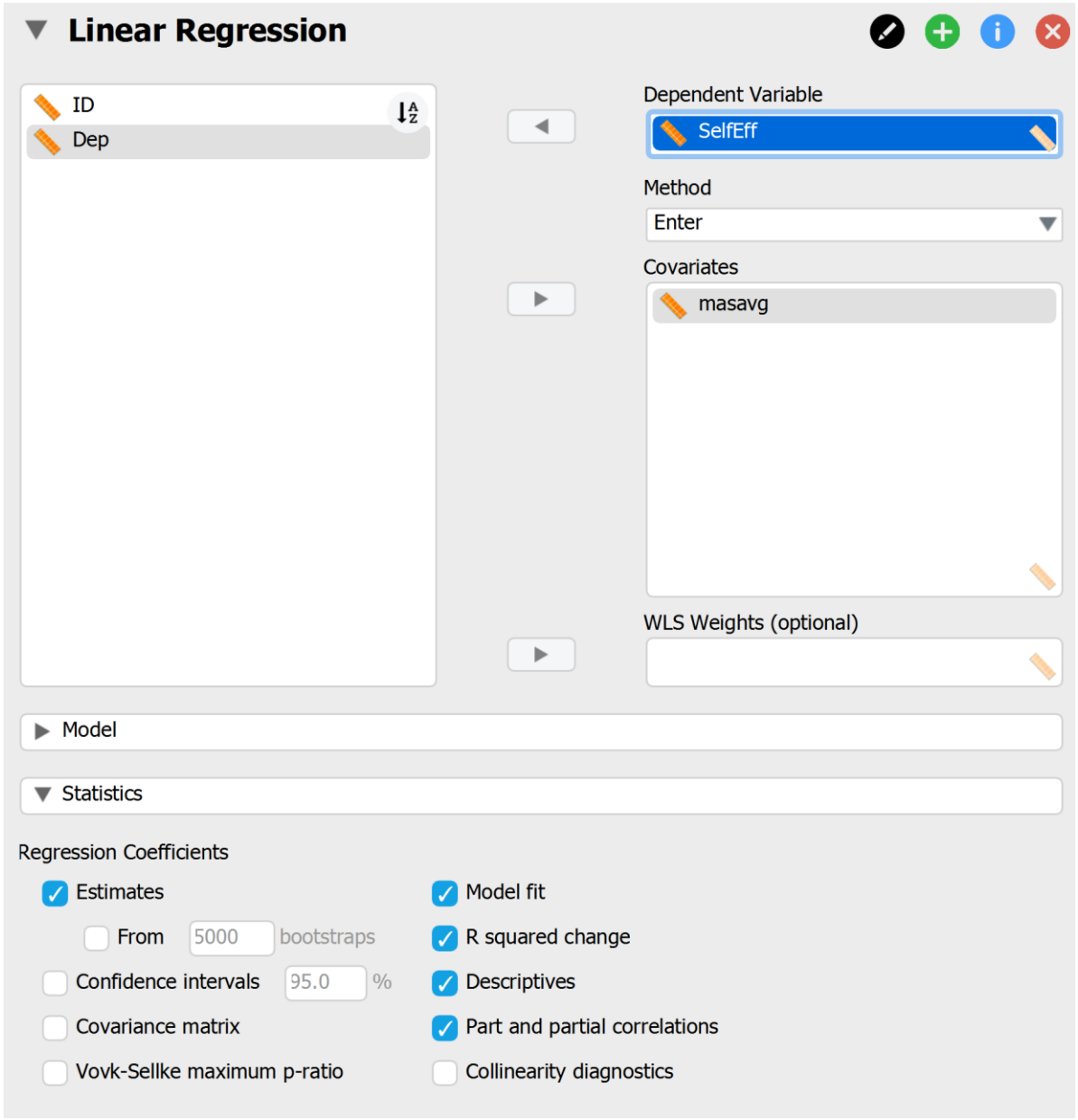
Analyses and Results

Step 2) Test whether the independent (or predictor) variable is correlated with the mediator.

- estimate and test path a



In JASP:


Linear Regression
ID
Dep
Dependent Variable
SelfEff
Method
Enter
Covariates
masavg
WLS Weights (optional)
Model
Statistics
Regression Coefficients
Estimates
From
5000
bootstraps
Confidence intervals
95.0
%
Covariance matrix
Vovk-Sellke maximum p-ratio
Model fit
R squared change
Descriptives
Part and partial correlations
Collinearity diagnostics

In JASP:

Linear Regression

Model Summary - SelfEff

Model	R	R^2	Adjusted R^2	RMSE	R^2 Change	F Change	df1	df2	p
H_0	0.000	0.000	0.000	0.446	0.000		0	466	
H_1	0.455	0.207	0.205	0.398	0.207	121.171	1	465	< .001

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H_1	Regression	19.155	1	19.155	121.171	< .001
	Residual	73.508	465	0.158		
	Total	92.663	466			

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
H_0	(Intercept)	3.126	0.021		151.498	< .001
H_1	(Intercept)	2.295	0.078		29.521	< .001
	masavg	0.334	0.030	0.455	11.008	< .001

Descriptives

	N	Mean	SD	SE
SelfEff	467	3.126	0.446	0.021
masavg	467	2.492	0.608	0.028

In SPSS:

Variables Entered/Removed[b]

Model	Variables Entered	Variables Removed	Method
1	Masculinity Mean[a]	.	Enter

a. All requested variables entered.

b. Dependent Variable: Self-Efficacy

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.455[a]	.207	.205	.39759

a. Predictors: (Constant), Masculinity Mean

ANOVA[b]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	19.155	1	19.155	121.171	.000[a]
	Residual	73.508	465	.158		
	Total	92.663	466			

a. Predictors: (Constant), Masculinity Mean

b. Dependent Variable: Self-Efficacy

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	2.295	.078		29.521	.000
	Masculinity Mean	.334	.030	.455	11.008	.000

a. Dependent Variable: Self-Efficacy

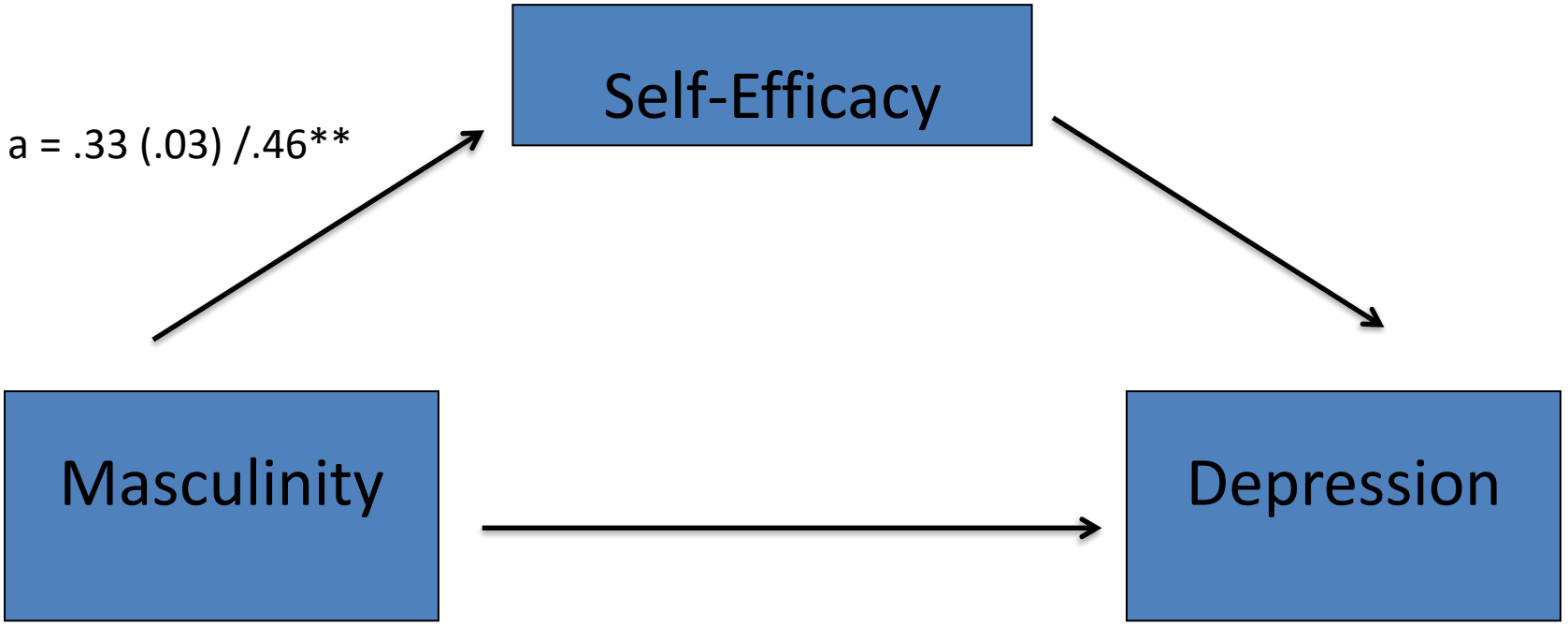
In R:

```
m.2.1=lm(SelfEff ~ masavg,data=dat.1)
summary(m.2.1)
```

```
##
## Call:
## lm(formula = SelfEff ~ masavg, data = dat.1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.92056 -0.22462 -0.01208  0.25471  1.03810
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.29477    0.07773   29.52   <2e-16 ***
## masavg       0.33356    0.03030   11.01   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3976 on 465 degrees of freedom
## Multiple R-squared:  0.2067, Adjusted R-squared:  0.205
## F-statistic: 121.2 on 1 and 465 DF,  p-value: < 2.2e-16
```

```
m.2.2=lm(scale(SelfEff)~ scale(masavg),data=dat.1)
summary(m.2.2)
```

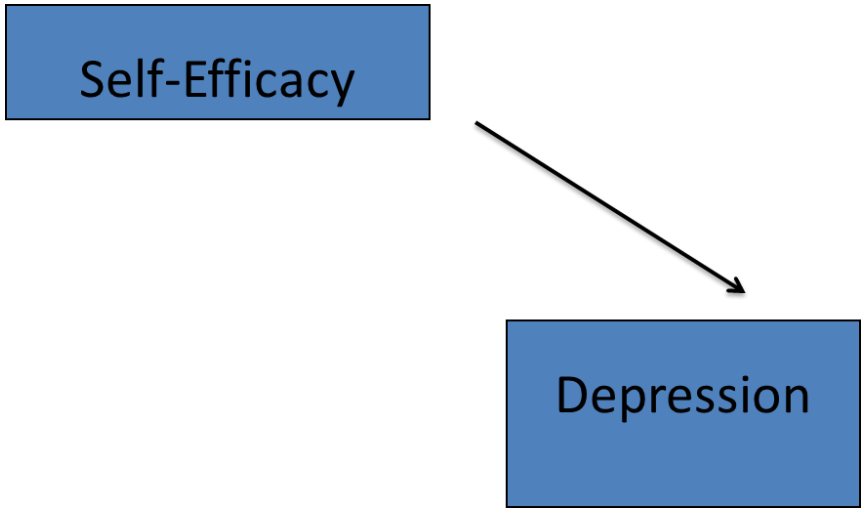
```
##
## Call:
## lm(formula = scale(SelfEff) ~ scale(masavg), data = dat.1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.3069 -0.5037 -0.0271  0.5712  2.3280
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.344e-16  4.126e-02    0.00        1
## scale(masavg) 4.547e-01  4.130e-02   11.01   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8916 on 465 degrees of freedom
## Multiple R-squared:  0.2067, Adjusted R-squared:  0.205
## F-statistic: 121.2 on 1 and 465 DF,  p-value: < 2.2e-16
```


a = .33 (.03) /.46**
Self-Efficacy
Masculinity
Depression

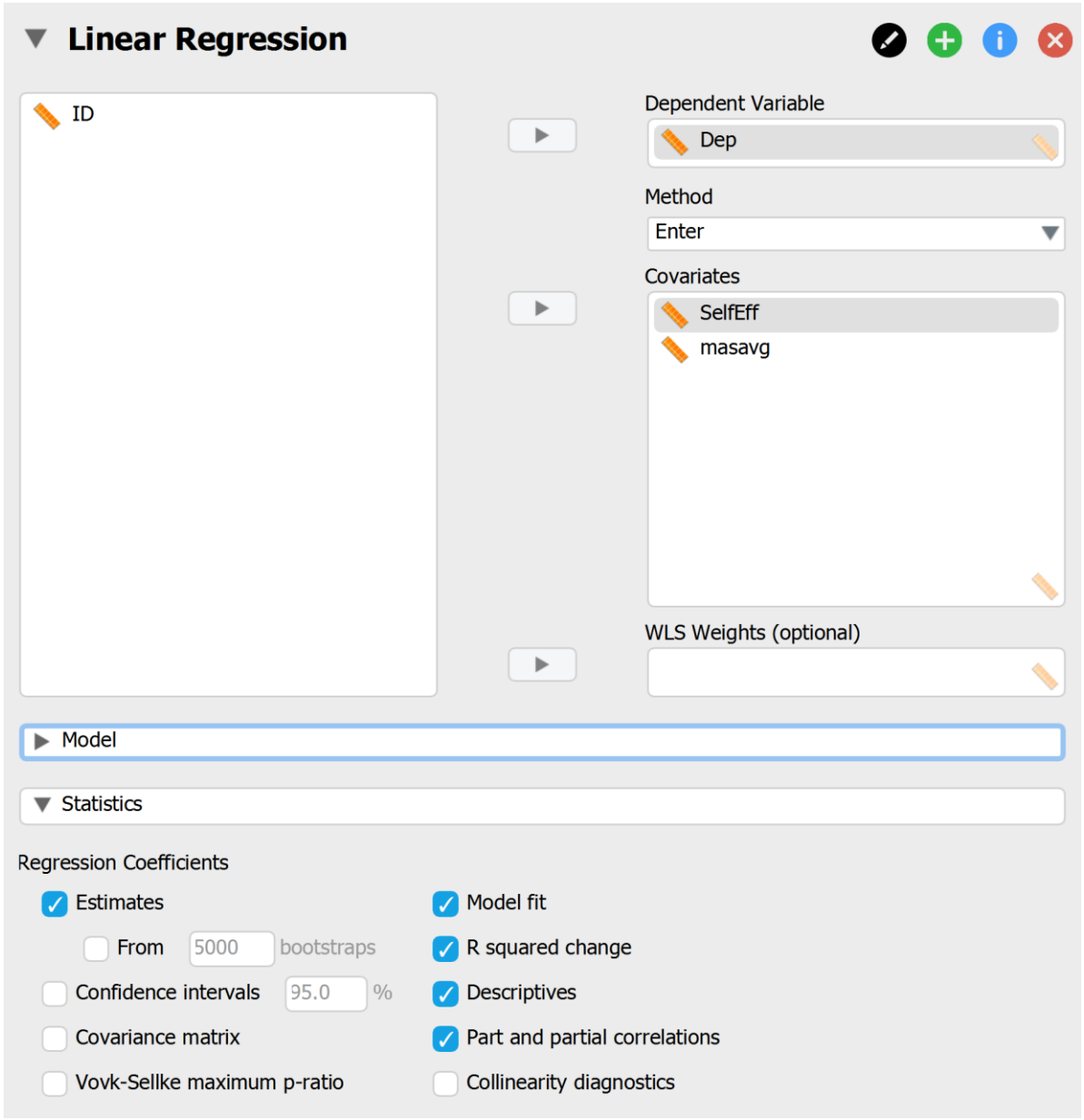
Analyses and Results

Step 3) Test whether the mediator affects the outcome variable.

- estimate and test path b
- To do so, regress Y on both X and M simultaneously
  - Look at the coefficient for M



In JASP:


Linear Regression
ID
Dependent Variable
Dep
Method
Enter
Covariates
SelfEff
masavg
WLS Weights (optional)
Model
Statistics
Regression Coefficients
Estimates
From
5000
bootstraps
Confidence intervals
95.0
%
Covariance matrix
Vovk-Sellke maximum p-ratio
Model fit
R squared change
Descriptives
Part and partial correlations
Collinearity diagnostics

In JASP:

Results

Linear Regression

Model Summary - Dep

Model	R	R^2	Adjusted R^2	RMSE	R^2 Change	F Change	df1	df2	p
H_0	0.404	0.163	0.159	9.203	0.163	45.173	2	464	< .001

Note. Null model includes SelfEff, masavg

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H_0	Regression	7652.173	2	3826.087	45.173	< .001
	Residual	39300.190	464	84.699		
	Total	46952.363	466			

Note. Null model includes SelfEff, masavg

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
H_0	(Intercept)	43.117	3.050		14.135	< .001
	SelfEff	−6.468	1.073	−0.287	−6.025	< .001
	masavg	2.000	0.738	0.182	3.808	< .001

Descriptives

	N	Mean	SD	SE
Dep	467	15.424	10.038	0.464
SelfEff	467	3.126	0.446	0.021
masavg	467	2.492	0.608	0.028

In SPSS:

Variables Entered/Removed[b]

Model	Variables Entered	Variables Removed	Method
1	Self-Efficacy, Masculinity Mean	.	Enter

b. Dependent Variable: Depressive Symptoms

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.404[a]	.163	.159	9.20319

a. Predictors: (Constant), Self-Efficacy, Masculinity Mean

ANOVA[b]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	7652.173	2	3826.087	45.173	.000[a]
	Residual	39300.190	464	84.699		
	Total	46952.363	466			

a. Predictors: (Constant), Self-Efficacy, Masculinity Mean

b. Dependent Variable: Depressive Symptoms

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	43.117	3.050		14.135	.000
	Masculinity Mean	-2.999	.788	-.182	-3.808	.000
	Self-Efficacy	-6.468	1.073	-.287	-6.025	.000

a. Dependent Variable: Depressive Symptoms

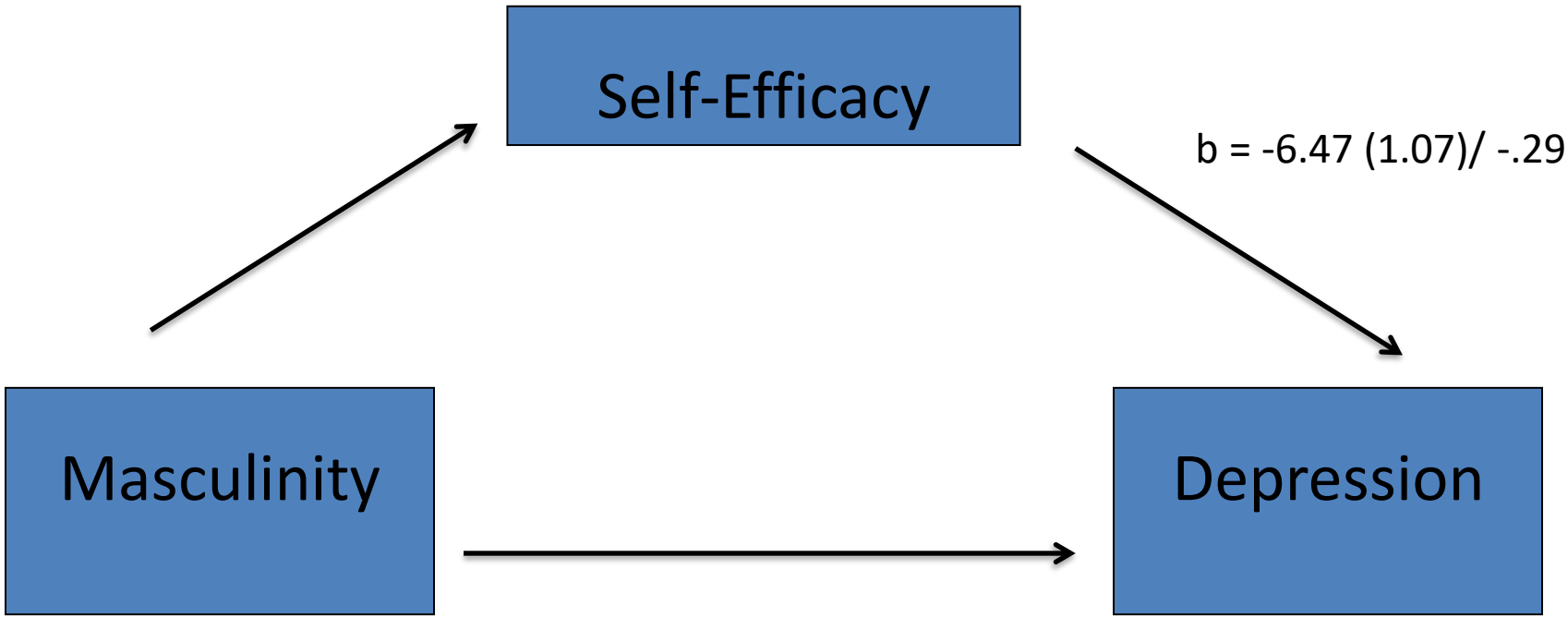
In R:

```
m.3.1=lm(Dep ~ masavg + SelfEff,data=dat.1)
summary(m.3.1)
```

```
##
## Call:
## lm(formula = Dep ~ masavg + SelfEff, data = dat.1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -17.494  -6.143  -1.635   5.071  39.055
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  43.1173     3.0505  14.135  < 2e-16 ***
## masavg       -2.9990     0.7875  -3.808 0.000159 ***
## SelfEff      -6.4677     1.0734  -6.025 3.44e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.203 on 464 degrees of freedom
## Multiple R-squared:  0.163,  Adjusted R-squared:  0.1594
## F-statistic: 45.17 on 2 and 464 DF,  p-value: < 2.2e-16
```

```
m.3.2=lm(scale(Dep)~ scale(masavg)+scale(SelfEff),data=dat.1)
summary(m.3.2)
```

```
##
## Call:
## lm(formula = scale(Dep) ~ scale(masavg) + scale(SelfEff), data = dat.1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.7428 -0.6120 -0.1629  0.5052  3.8908
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -5.845e-18  4.243e-02   0.000 1.000000
## scale(masavg)  -1.816e-01  4.769e-02  -3.808 0.000159 ***
## scale(SelfEff) -2.873e-01  4.769e-02  -6.025 3.44e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9169 on 464 degrees of freedom
## Multiple R-squared:  0.163,  Adjusted R-squared:  0.1594
## F-statistic: 45.17 on 2 and 464 DF,  p-value: < 2.2e-16
```


Self-Efficacy
b = -6.47 (1.07)/ -.29**
Masculinity
Depression

Analyses and Results

Step 4) Test whether the effect of X on Y controlling for M is equal to zero.

- estimate and test path c'

- Use same regression equation as in Step 3.



In JASP:

Results

Linear Regression

Model Summary - Dep

Model	R	R^2	Adjusted R^2	RMSE	R^2 Change	F Change	df1	df2	p
H_0	0.404	0.163	0.159	9.203	0.163	45.173	2	464	< .001

Note. Null model includes SelfEff, masavg

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
H_0	Regression	7652.173	2	3826.087	45.173	< .001
	Residual	39300.190	464	84.699		
	Total	46952.363	466			

Note. Null model includes SelfEff, masavg

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
H_0	(Intercept)	43.117	3.050		14.135	< .001
	SelfEff	−6.468	1.073	0.287	−6.025	< .001
	masavg	−2.999	0.788	−0.182	−3.808	< .001

Descriptives

	N	Mean	SD	SE
Dep	467	15.424	10.038	0.464
SelfEff	467	3.126	0.446	0.021
masavg	467	2.492	0.608	0.028

In SPSS:

Variables Entered/Removed[b]

Model	Variables Entered	Variables Removed	Method
1	Self-Efficacy, Masculinity Mean	.	Enter

b. Dependent Variable: Depressive Symptoms

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.404[a]	.163	.159	9.20319

a. Predictors: (Constant), Self-Efficacy, Masculinity Mean

ANOVA[b]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	7652.173	2	3826.087	45.173	.000[a]
	Residual	39300.190	464	84.699		
	Total	46952.363	466			

a. Predictors: (Constant), Self-Efficacy, Masculinity Mean

b. Dependent Variable: Depressive Symptoms

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	43.117	3.050		14.135	.000
	Masculinity Mean	-2.999	.788	-.182	-3.808	.000
	Self-Efficacy	-6.468	1.073	-.287	-6.025	.000

a. Dependent Variable: Depressive Symptoms

In R:

```
m.3.1=lm(Dep ~ masavg + SelfEff,data=dat.1)
summary(m.3.1)
```

```
##
## Call:
## lm(formula = Dep ~ masavg + SelfEff, data = dat.1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -17.494  -6.143  -1.635   5.071  39.055
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  43.1173     3.0505  14.135  < 2e-16 ***
## masavg       -2.9990     0.7875  -3.808 0.000159 ***
## SelfEff      -6.4677     1.0734  -6.025 3.44e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.203 on 464 degrees of freedom
## Multiple R-squared:  0.163,  Adjusted R-squared:  0.1594
## F-statistic: 45.17 on 2 and 464 DF,  p-value: < 2.2e-16
```

```
m.3.2=lm(scale(Dep)~ scale(masavg)+scale(SelfEff),data=dat.1)
summary(m.3.2)
```

```
##
## Call:
## lm(formula = scale(Dep) ~ scale(masavg) + scale(SelfEff), data = dat.1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.7428 -0.6120 -0.1629  0.5052  3.8908
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -5.845e-18  4.243e-02   0.000 1.000000
## scale(masavg)  -1.816e-01  4.769e-02  -3.808 0.000159 ***
## scale(SelfEff) -2.873e-01  4.769e-02  -6.025 3.44e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9169 on 464 degrees of freedom
## Multiple R-squared:  0.163,  Adjusted R-squared:  0.1594
## F-statistic: 45.17 on 2 and 464 DF,  p-value: < 2.2e-16
```

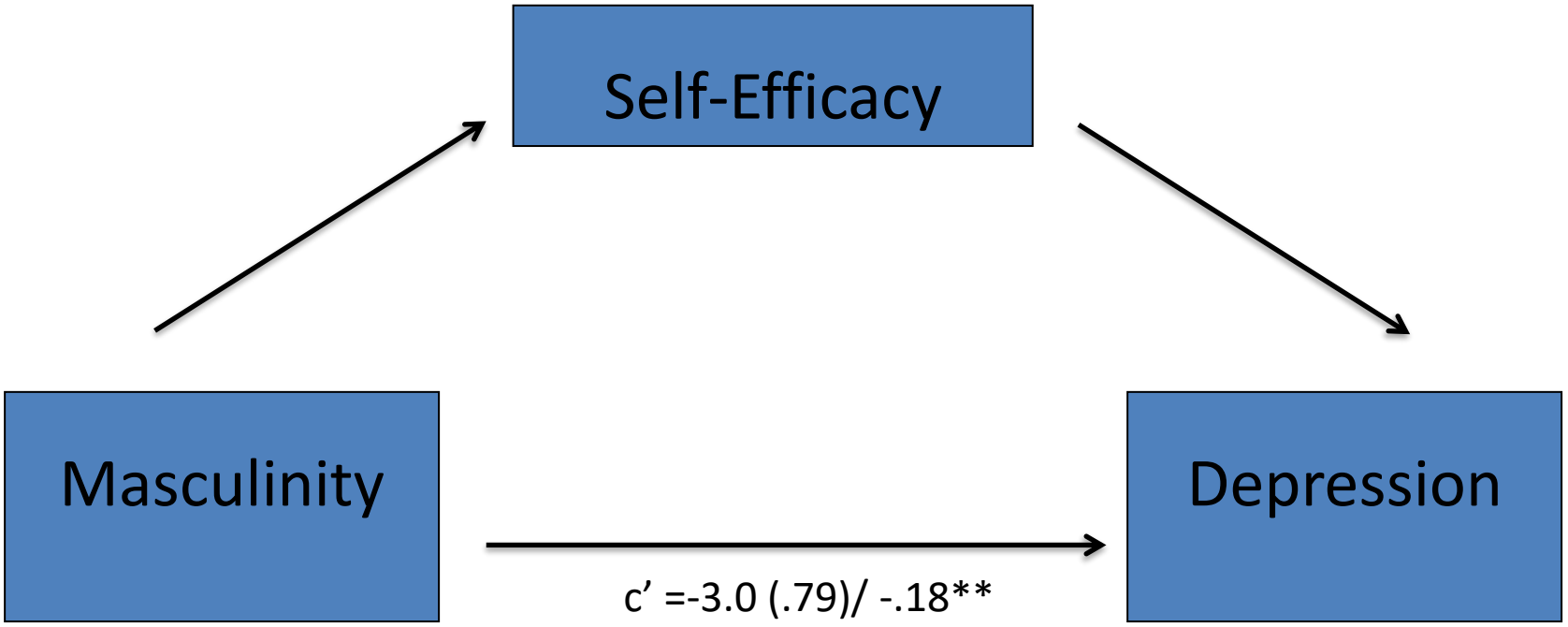
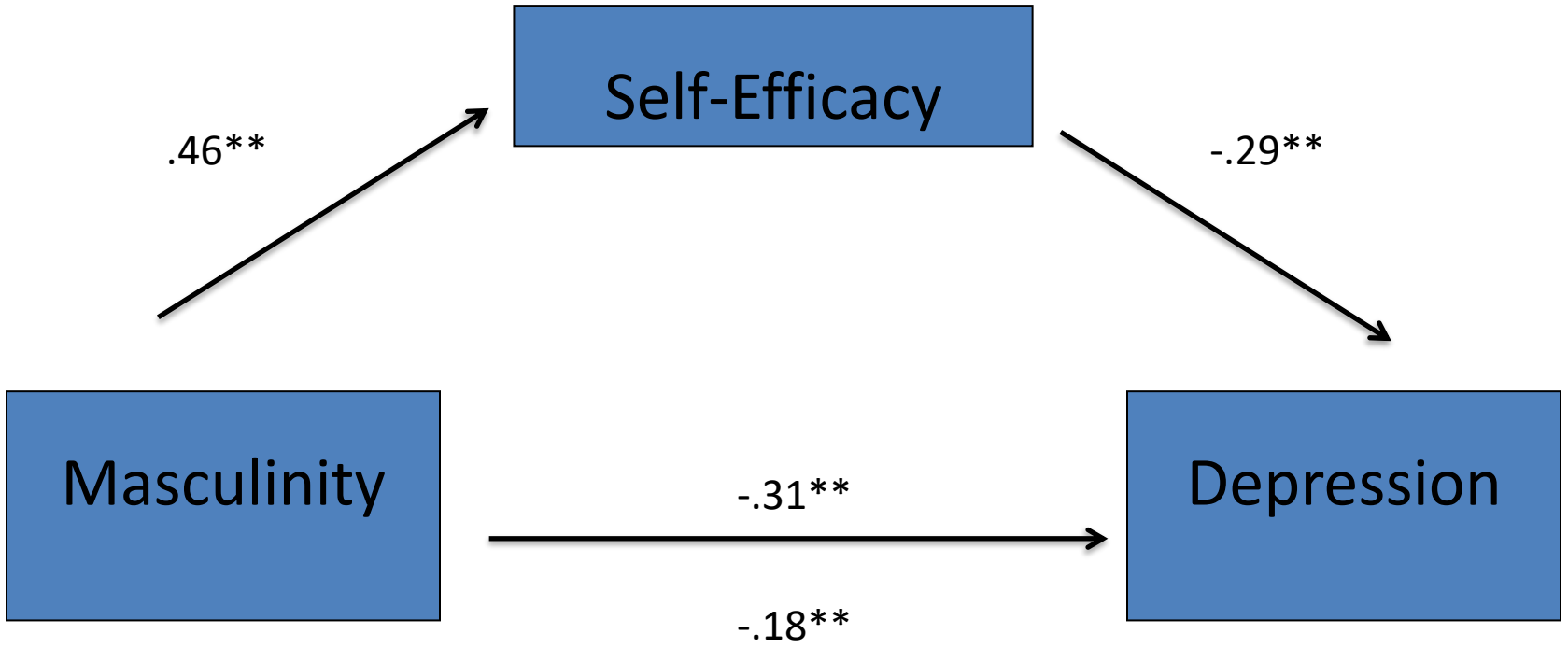
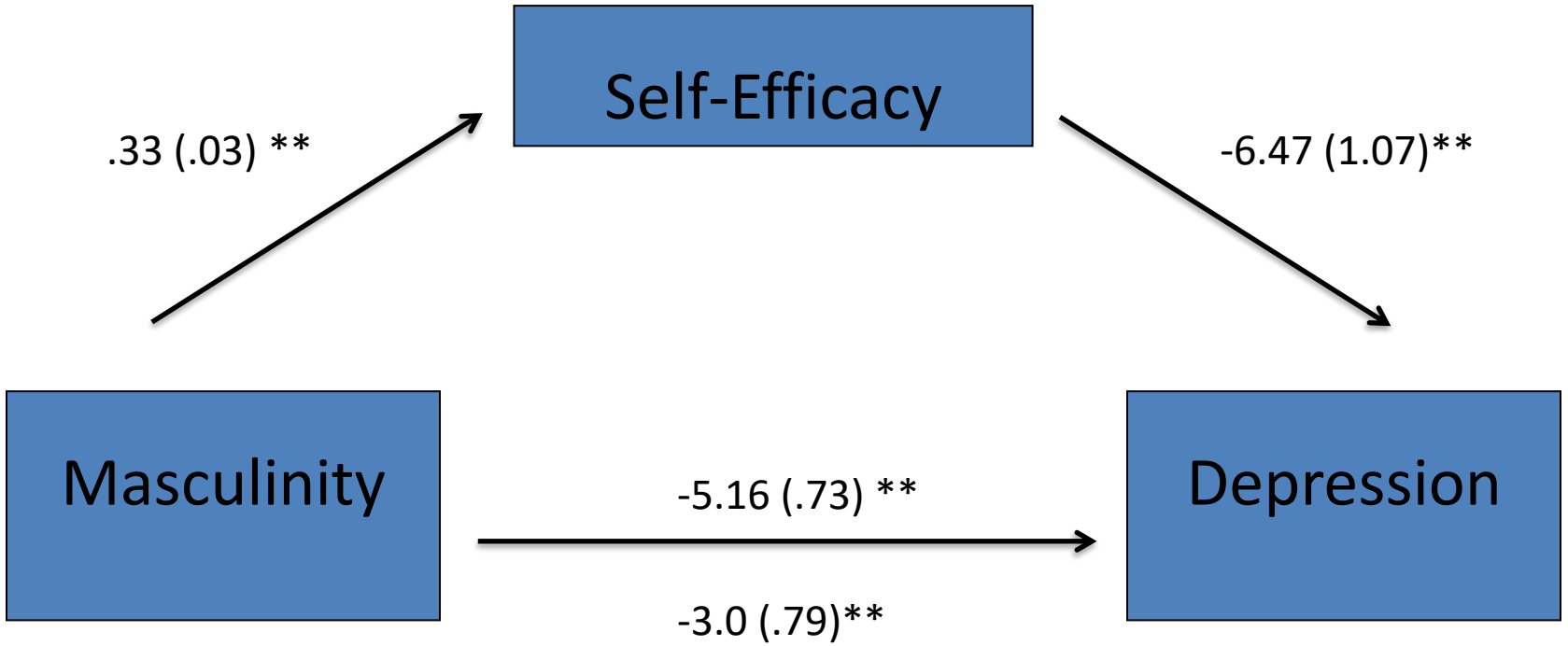

Self-Efficacy
Masculinity
Depression
c' =-3.0 (.79)/ -.18**

Figure to present (with standardized betas)



Note. Unmediated path (total effect) displayed above the arrow; mediated path (direct effect) below.

Figure to present (with unstandardized betas)



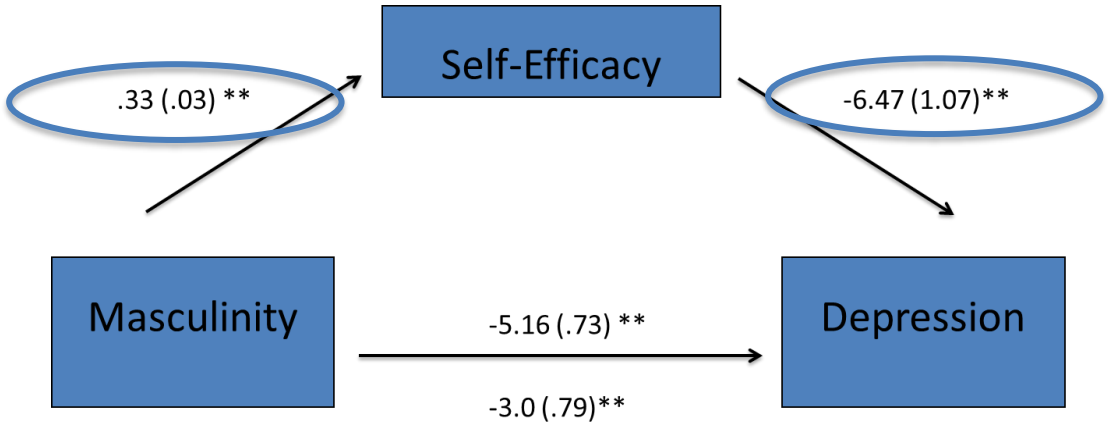
Note. Unmediated path (total effect) displayed above the arrow; mediated path (direct effect) below.

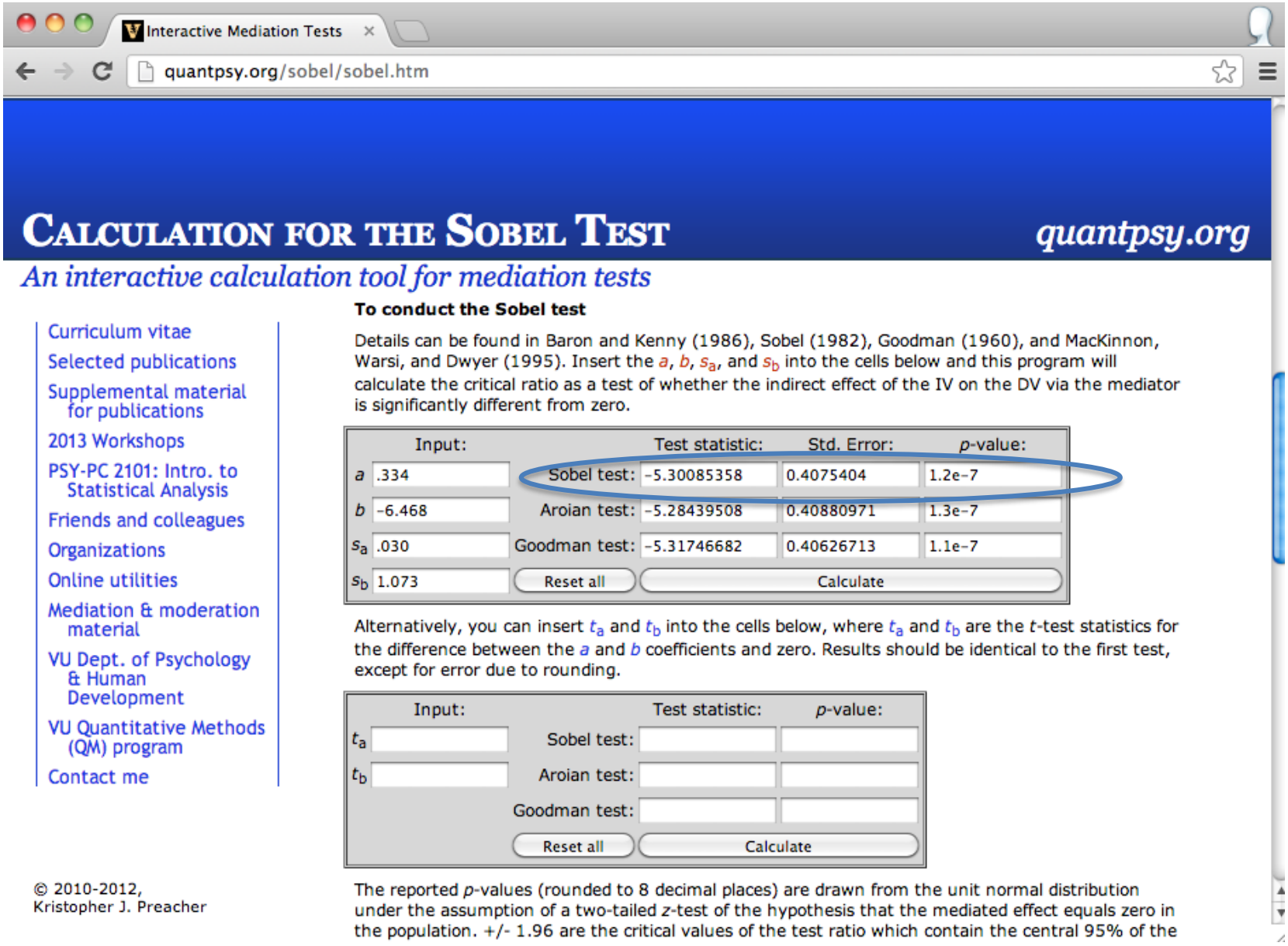
Test of the Indirect effect

#1 Using Sobel Method:

- Using the online calculator, and the unstandardized coefficients and their standard errors, calculate the Sobel test:

$$Z = \frac{a * b}{\sqrt{b^2 s_a^2 + a^2 s_b^2}}$$




Interactive Mediation Tests
quantpsy.org/sobel/sobel.htm
CALCULATION FOR THE SOBEL TEST
quantpsy.org
An interactive calculation tool for mediation tests
Curriculum vitae
Selected publications
Supplemental material for publications
2013 Workshops
PSY-PC 2101: Intro. to Statistical Analysis
Friends and colleagues
Organizations
Online utilities
Mediation & moderation material
VU Dept. of Psychology & Human Development
VU Quantitative Methods (QM) program
Contact me
© 2010-2012, Kristopher J. Preacher
To conduct the Sobel test
Details can be found in Baron and Kenny (1986), Sobel (1982), Goodman (1960), and MacKinnon, Warsi, and Dwyer (1995). Insert the a, b, sa, and sb into the cells below and this program will calculate the critical ratio as a test of whether the indirect effect of the IV on the DV via the mediator is significantly different from zero.
Input:
Test statistic:
Std. Error:
p-value:
a .334
b -6.468
sa .030
sb 1.073
Sobel test: -5.30085358 0.4075404 1.2e-7
Aroian test: -5.28439508 0.40880971 1.3e-7
Goodman test: -5.31746682 0.40626713 1.1e-7
Reset all
Calculate
Alternatively, you can insert ta and tb into the cells below, where ta and tb are the t-test statistics for the difference between the a and b coefficients and zero. Results should be identical to the first test, except for error due to rounding.
Input:
Test statistic:
p-value:
ta
tb
Sobel test:
Aroian test:
Goodman test:
Reset all
Calculate
The reported p-values (rounded to 8 decimal places) are drawn from the unit normal distribution under the assumption of a two-tailed z-test of the hypothesis that the mediated effect equals zero in the population. +/- 1.96 are the critical values of the test ratio which contain the central 95% of the

In R:

```
#sobel test variables
med = dat.1$SelfEff
iv = dat.1$masavg
dv = dat.1$Dep

sobel(iv,med,dv)
```

```
## $`Mod1: Y~X`
##              Estimate Std. Error   t value     Pr(>|t|)
## (Intercept) 28.275506  1.8663973 15.149778 1.987332e-42
## pred        -5.156381  0.7275615 -7.087209 5.106419e-12
##
## $`Mod2: Y~X+M`
##              Estimate Std. Error   t value     Pr(>|t|)
## (Intercept) 43.117332  3.0504597 14.134700 5.599176e-38
## pred        -2.998995  0.7875259 -3.808123 1.588364e-04
## med         -6.467664  1.0734237 -6.025266 3.442066e-09
##
## $`Mod3: M~X`
##              Estimate Std. Error  t value      Pr(>|t|)
## (Intercept) 2.2947740 0.07773489 29.52051 1.143767e-108
## pred        0.3335649 0.03030272 11.00775  3.342026e-25
##
## $Indirect.Effect
## [1] -2.157386
##
## $SE
## [1] 0.4081858
##
## $z.value
## [1] -5.285303
##
## $N
## [1] 467
```

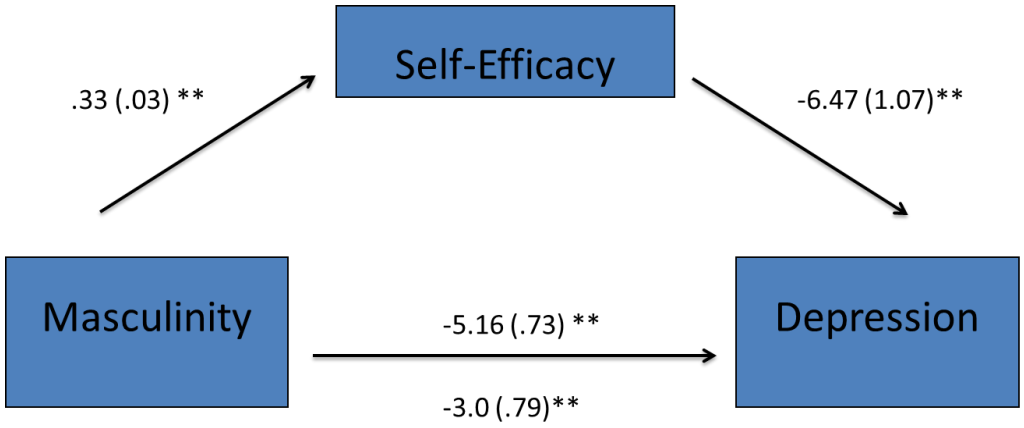
Write Up using B&K Method & Sobel Test

We used Baron and Kenny's (1986) procedure for estimating mediational effects using a series of regression analyses, and the Sobel test to assess for the presence of a significant indirect, or mediating, effect. As shown in Figure 1, there was a negative association masculinity and depressive symptoms (β =-.31, p <.001). When self-efficacy was included in the model, the association between masculinity and depressive symptoms was reduced, but remained significantly different from zero (β = -.18, p <.001), indicating a small, negative direct effect and partial mediation. The Sobel test indicated a statistically significant indirect effect of masculinity on depression via self-efficacy, z' = -5.30, p <.05.

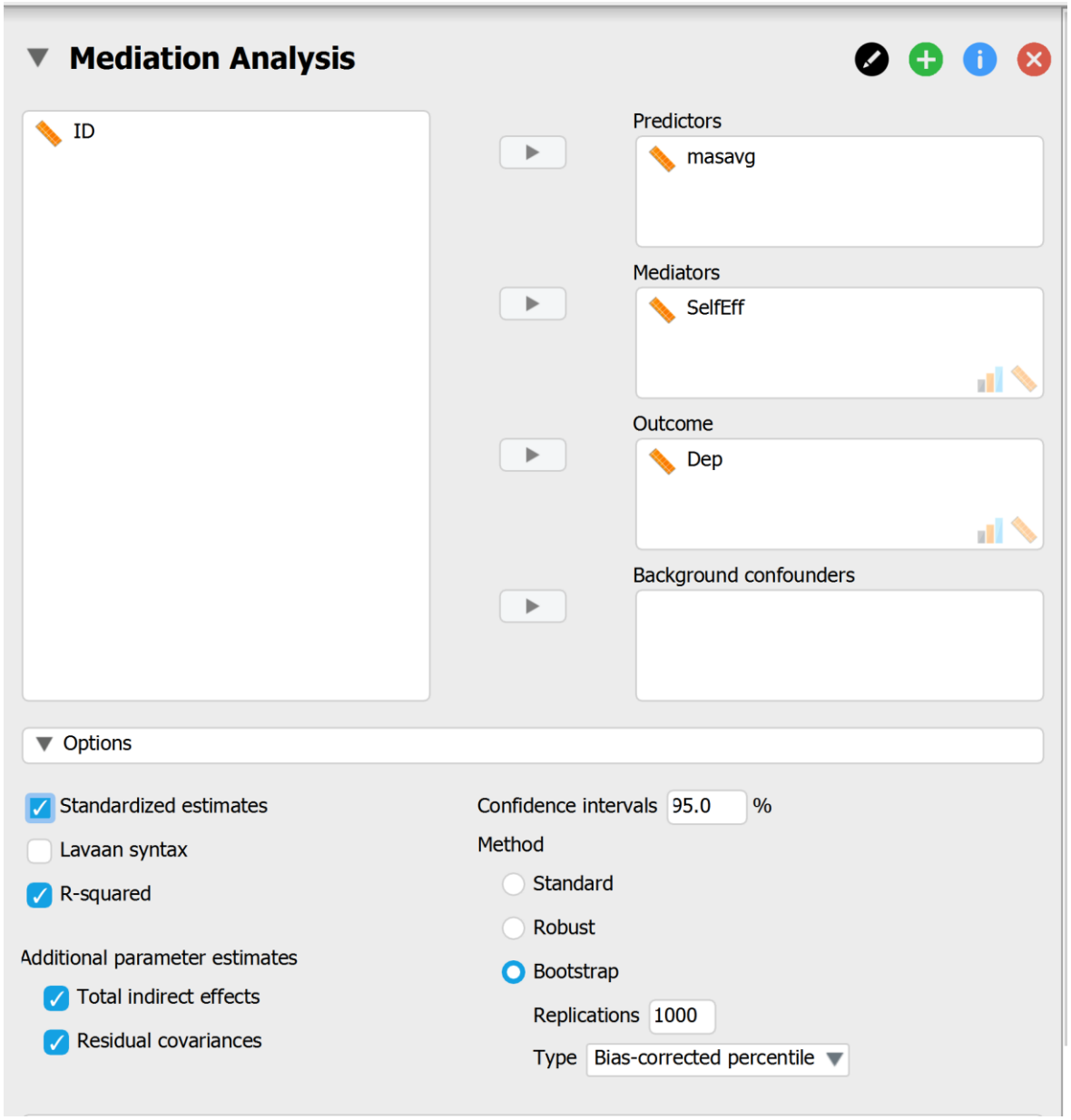
Test of the Indirect effect

#2 Using Bootstrapping via one of the following:

- Structural Equation Modeling via JASP
- PROCESS Macro via SPSS



In JASP:


Mediation Analysis
ID
Predictors
masavg
Mediators
SelfEff
Outcome
Dep
Background confounders
Options
Standardized estimates
Lavaan syntax
R-squared
Additional parameter estimates
Total indirect effects
Residual covariances
Confidence intervals 95.0 %
Method
Standard
Robust
Bootstrap
Replications 1000
Type Bias-corrected percentile

In JASP:

In Class Week 9 - Mediation Analysis ▼

NOTE: This output shows the standardized betas.

Parameter estimates

Direct effects

						95% Confidence Interval	
			Estimate	Std. Error	z-value	p	Lower	Upper
masavg	→	Dep	−0.299	0.078	−3.820	< .001	−0.458	−0.129

Note. Delta method standard errors, bias-corrected percentile bootstrap confidence intervals, ML estimator.

Indirect effects

								95% Confidence Interval	
					Estimate	Std. Error	z-value	p	Lower	Upper
masavg	→	SelfEff	→	Dep	−0.215	0.041	−5.301	< .001	−0.295	−0.135

Note. Delta method standard errors, bias-corrected percentile bootstrap confidence intervals, ML estimator.

Total effects

							95% Confidence Interval	
			Estimate	Std. Error	z-value	p	Lower	Upper
masavg	→	Dep	−0.514	0.072	−7.102	< .001	−0.643	−0.365

Note. Delta method standard errors, bias-corrected percentile bootstrap confidence intervals, ML estimator.

R-Squared

	R^2
Dep	0.163
SelfEff	0.207

In JASP:

In Class Week 9 - Mediation Analysis

NOTE: This output shows the unstandardized betas.

Parameter estimates

Direct effects

							95% Confidence Interval	
			Estimate	Std. Error	z-value	p	Lower	Upper
masavg	→	Dep	−2.999	0.785	−3.820	< .001	−4.624	−1.301

Note. Delta method standard errors, bias-corrected percentile bootstrap confidence intervals, ML estimator.

Indirect effects

								95% Confidence Interval	
					Estimate	Std. Error	z-value	p	Lower	Upper
masavg	→	SelfEff	→	Dep	−2.157	0.407	−5.301	< .001	−3.033	−1.358

Note. Delta method standard errors, bias-corrected percentile bootstrap confidence intervals, ML estimator.

Total effects

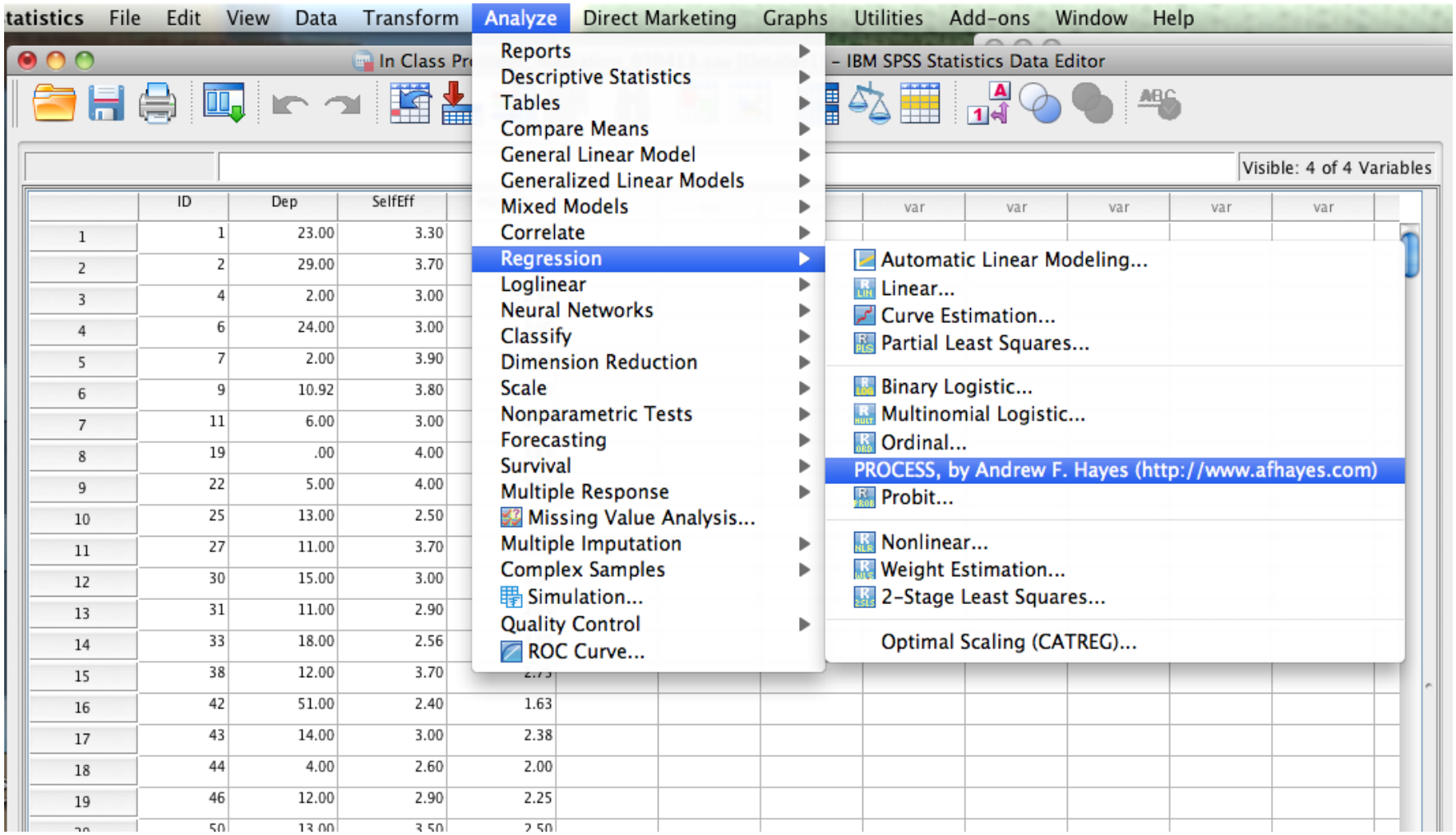
							95% Confidence Interval	
			Estimate	Std. Error	z-value	p	Lower	Upper
masavg	→	Dep	−5.156	0.726	−7.102	< .001	−6.622	−3.717

Note. Delta method standard errors, bias-corrected percentile bootstrap confidence intervals, ML estimator.

R-Squared

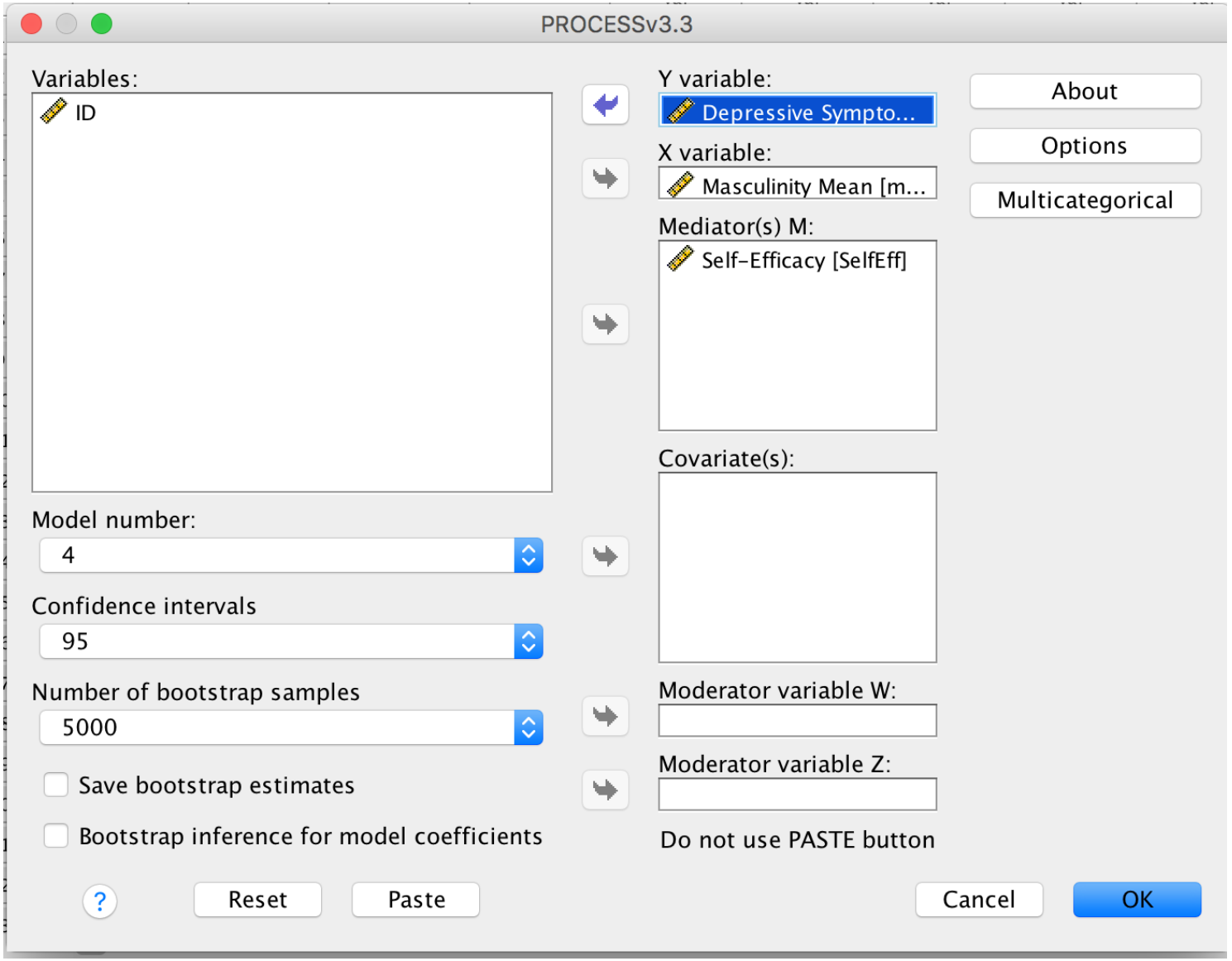
	R^2
Dep	0.163
SelfEff	0.207

In SPSS:

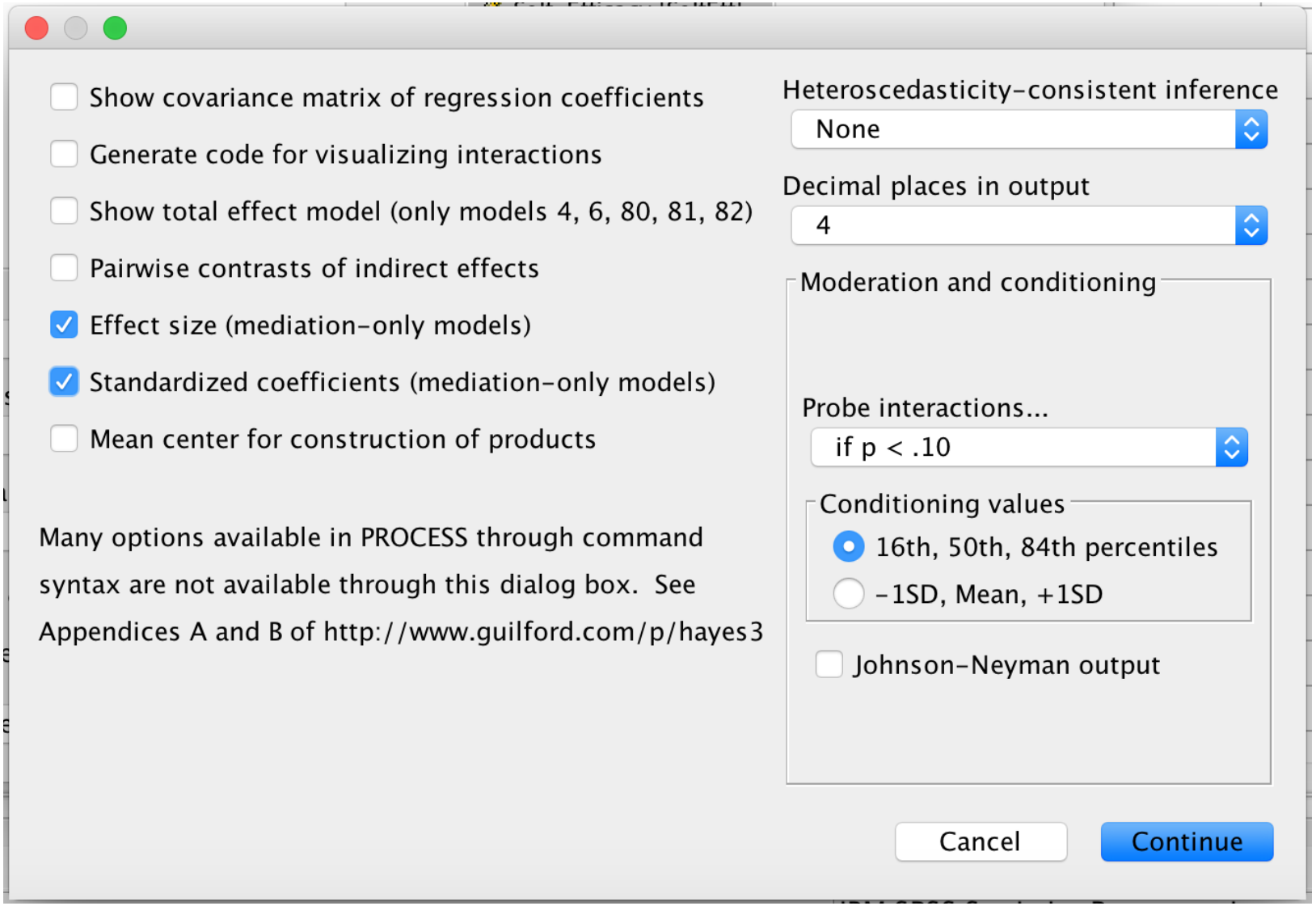


	ID	Dep	SelfEff	
1	1	23.00	3.30	
2	2	29.00	3.70	
3	4	2.00	3.00	
4	6	24.00	3.00	
5	7	2.00	3.90	
6	9	10.92	3.80	
7	11	6.00	3.00	
8	19	.00	4.00	
9	22	5.00	4.00	
10	25	13.00	2.50	
11	27	11.00	3.70	
12	30	15.00	3.00	
13	31	11.00	2.90	
14	33	18.00	2.56	
15	38	12.00	3.70	
16	42	51.00	2.40	1.63
17	43	14.00	3.00	2.38
18	44	4.00	2.60	2.00
19	46	12.00	2.90	2.25
	50	13.00	3.50	2.50

In SPSS:


PROCESSv3.3
Variables:
ID
Y variable:
Depressive Sympto...
X variable:
Masculinity Mean [m...
Mediator(s) M:
Self-Efficacy [SelfEff]
Covariate(s):
Moderator variable W:
Moderator variable Z:
Do not use PASTE button
About
Options
Multicategorical
Model number:
4
Confidence intervals
95
Number of bootstrap samples
5000
Save bootstrap estimates
Bootstrap inference for model coefficients
?
Reset
Paste
Cancel
OK

In SPSS:


Show covariance matrix of regression coefficients
Generate code for visualizing interactions
Show total effect model (only models 4, 6, 80, 81, 82)
Pairwise contrasts of indirect effects
Effect size (mediation-only models)
Standardized coefficients (mediation-only models)
Mean center for construction of products
Many options available in PROCESS through command syntax are not available through this dialog box. See Appendices A and B of http://www.guilford.com/p/hayes3
Heteroscedasticity-consistent inference
None
Decimal places in output
4
Moderation and conditioning
Probe interactions...
if p < .10
Conditioning values
16th, 50th, 84th percentiles
-1SD, Mean, +1SD
Johnson-Neyman output
Cancel
Continue

In SPSS:

```
********* PROCESS Procedure for SPSS Version 3.3 *********

          Written by Andrew F. Hayes, Ph.D.       www.afhayes.com
    Documentation available in Hayes (2018). www.guilford.com/p/hayes3

**************************************************************************
Model  : 4
    Y  : Dep
    X  : masavg
    M  : SelfEff

Sample
Size:  467

**************************************************************************
OUTCOME VARIABLE:
 SelfEff

Model Summary
          R       R-sq        MSE          F        df1        df2          p
      .4547      .2067      .1581   121.1707     1.0000   465.0000      .0000

Model
              coeff         se          t          p       LLCI       ULCI
constant     2.2948       .0777    29.5205       .0000     2.1420     2.4475
masavg        .3336       .0303    11.0078       .0000      .2740      .3931

Standardized coefficients
            coeff
masavg      .4547

**************************************************************************
```

Path a

In SPSS:

```
**************************************************************************
OUTCOME VARIABLE:
 Dep

Model Summary
          R       R-sq        MSE          F        df1        df2          p
      .4037      .1630    84.6987    45.1729     2.0000   464.0000      .0000

Model
              coeff         se          t          p       LLCI       ULCI
constant    43.1173     3.0505    14.1347      .0000    37.1229    49.1118
masavg      -2.9990      .7875    -3.8081      .0002    -4.5466    -1.4514
SelfEff     -6.4677     1.0734    -6.0253      .0000    -8.5770    -4.3583

Standardized coefficients
             coeff
masavg      -.1816
SelfEff     -.2873

****************** DIRECT AND INDIRECT EFFECTS OF X ON Y *****************

Direct effect of X on Y
    Effect         se          t          p       LLCI       ULCI      c'_ps      c'_cs
   -2.9990      .7875    -3.8081      .0002    -4.5466    -1.4514     -.2988     -.1816

Indirect effect(s) of X on Y:
            Effect     BootSE   BootLLCI   BootULCI
SelfEff    -2.1574      .4260    -2.9811    -1.3289

Partially standardized indirect effect(s) of X on Y:
            Effect     BootSE   BootLLCI   BootULCI
SelfEff     -.2149      .0408     -.2937     -.1345

Completely standardized indirect effect(s) of X on Y:
            Effect     BootSE   BootLLCI   BootULCI
SelfEff     -.1306      .0251     -.1794     -.0810

/*/*/*/*/*/*/*/*/*/*/*/*/*/*/*/*/*/*/*/*/ ANALYSIS NOTES AND ERRORS /*/*/*/*/*/*/*/*/*/*/*/*/*/*/*/*/*/*/*/*/
```

Path b

Path c’

Direct effect (effect of X on Y after M is controlled)

Indirect (mediated) effect

Path c (total effect): *b* of direct effect + *b* of indirect effect

In R:

```
#setup lm models based on paths c, a, and the mediation:
model.C <- lm(Dep ~ masavg, dat.1) #total effect of adhd symptoms on relationship satisfaction
model.A <- lm(SelfEff ~ masavg, dat.1) #path a effect of adhd symptoms on hostile communication
model.Y <- lm(Dep ~ masavg + SelfEff, dat.1) #mediated path's effect of adhd symptoms and hostile on relationship satisfacti
on
med.results <- mediate(model.A, model.Y, treat='masavg', mediator='SelfEff',
                   boot=TRUE, sims=100) #mediation calculation with 100 bootstrapped samples
```

```
## Running nonparametric bootstrap
```

```
summary(med.results)
```

```
##
## Causal Mediation Analysis
##
## Nonparametric Bootstrap Confidence Intervals with the Percentile Method
##
##                Estimate 95% CI Lower 95% CI Upper p-value
## ACME             -2.157       -2.961        -1.25  <2e-16 ***
## ADE              -2.999       -4.544        -1.52  <2e-16 ***
## Total Effect     -5.156       -6.383        -3.62  <2e-16 ***
## Prop. Mediated    0.418        0.248         0.63  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Sample Size Used: 467
##
##
## Simulations: 100
```

ACME: Test of the indirect (mediated) effect

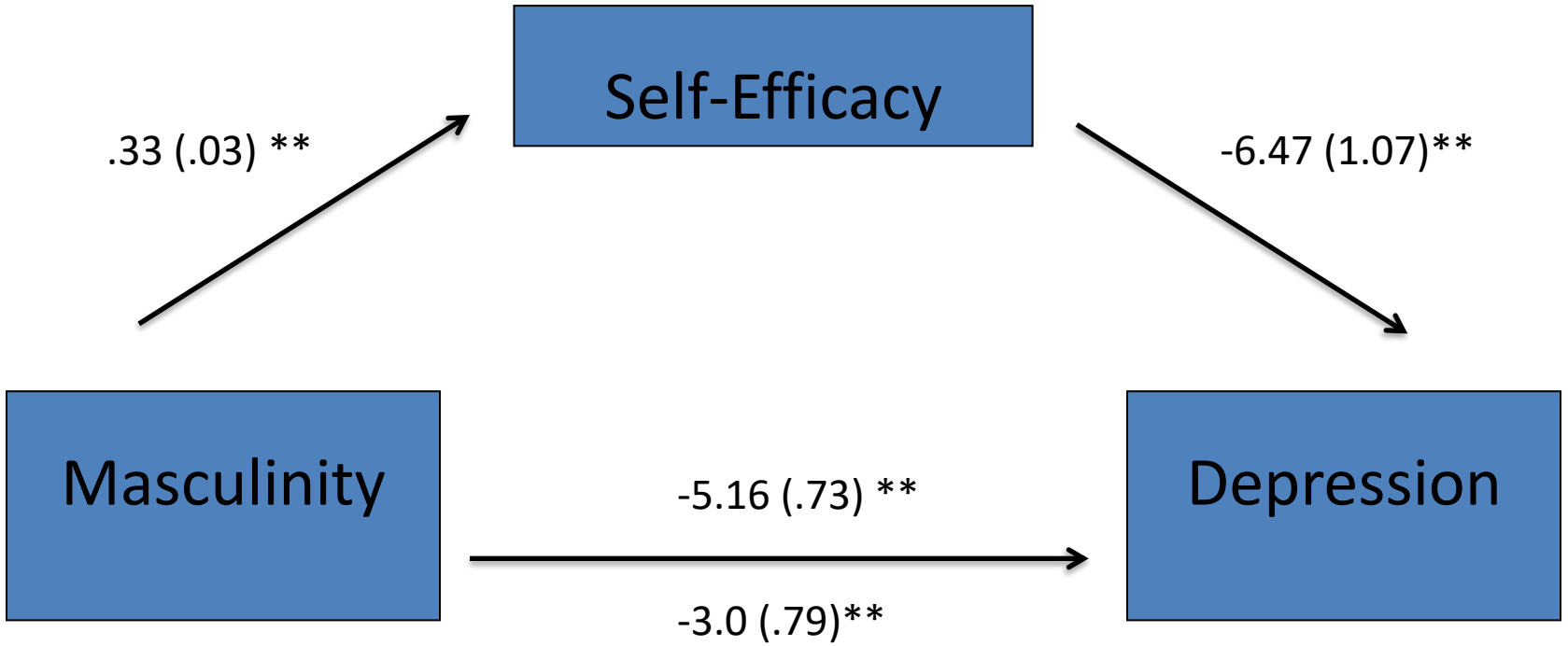
ADE: Direct effect (effect of X on Y after M is controlled)

Write Up using Bootstrapping

(uses unstandardized coefficients, you can use standardized throughout if you prefer)

The direct and indirect (i.e., mediated) effects were estimated with the PROCESS software (Hayes, 2013)/JASP Structural Equation Modeling and bootstrapping techniques. As shown in Figure 1, the indirect effect of masculinity on depressive symptoms via self-efficacy was b = -2.16 (SE = .43; 95% CI = -2.98 to -1.33, p <.001). In the mediated model, the direct effect of masculinity on depressive symptoms remained significant (b = -3.0, SE = .79; 95% CI = -4.55 to -1.45, p <.001), suggesting that self-efficacy does not entirely account for this association.

Figure to present (this uses unstandardized betas; you can use standardized throughout if you prefer)



Note. Unmediated path (total effect) displayed above the arrow; mediated path (direct effect) below.

6. Final Notes:

- Important point to remember:
  - Mediation models are hypothesized causal relations between variables
    - Based on theory, previous research
    - Results, especially from cross-sectional data, only support the predicted model, don't "prove" it
    - Stronger support for mediation comes from:
      - Temporal ordering (X measured before M before Y).
      - Controlling for possible confounding variables

        The mediation analyses work if controls are consistent across regression equations

Avoiding Specification Error (do you have the right model?)

Common Potential Errors:

1. The proposed IV and/or mediator may be caused by the outcome

Solutions:

- Does your model make sense theoretically?
- Test the alternative direction model (run the model with switching M and Y and test mediation again)

** Study Design:

a) with appropriate temporal order (measure mediator before DV)

b) measure previous values of M and Y and control for them in models predicting later values of M and Y.

c) Experimentally manipulate IV (e.g., an intervention or condition, whose effect on the DV is proposed to be indirect via the mediator)

Avoiding Specification Error (do you have the right model?)

Common Potential Errors, cont.:

2. Measurement error in mediator
    - Solution: Measure with high reliability
3. Mediator and outcome share variance due to common method
    - Solution: Use different methods to measure mediator and DV
4. An omitted variable is causing both Mediator and DV.
    - Solution: Specify, measure and control for it

Mediation Decision Flowchart

